

Elementary Number Theory

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Introduction

Elementary number theory is a category in the topic of number theory. The word ‘elementary’ does not mean the things in this area are simple. This is only the name that we call, so as to distinguish with the other more advanced branches of number theory like algebraic number theory and analytic number theory. In Olympiad, we only need knowledge in elementary number theory.

In recent years, number theory problems in Olympiad usually cover only the very basic techniques and knowledge in this area. These include divisibilities and congruences introduced in section 1, and Diophantine equations introduced in section 2. Also, one needs quite a lot of algebraic tricks to solve these kinds of problems. We will cover many different examples in sections 1, 2 and 3, which are the main components of this note.

In section 4, we introduce a few more advanced theorems on congruences. These may not be so useful since we are not supposed to know them. On the other hand, the concepts introduced in sections 5 and 6, though being advanced in the area of elementary number theory, are more important as we sometimes need to apply them to solve some kind of MO problems. These concepts include the topic of quadratic residues, orders and primitive roots.

The last two sections, sections 7 and 8, cover yet more advanced theorems, which have a little use for us. They are included only for self-interest. Indeed, some of those results are far beyond the area of elementary number theory.

1 Divisibilities and Congruences

In this section, we revise the basic knowledge in elementary number theory. Divisibilities and congruences are the fundamental concepts that are needed in this area.

Those who are familiar with these basic topics may skip this introductory section, or move to the problems directly.

In this note, we use $\mathbb{N}$ and $\mathbb{Z}$ to denote the set of all positive integers and the set of all integers respectively. We also use the notation $\mathbb{N}_0$ to denote the set of all nonnegative integers, though this is not a common notation.

1.1 Divisibilities

We first begin with the concept of divisibilities.

Theorem 1.1. (**Division algorithm**(除法算則)) Let a and b be integers where $b \neq 0$. Then there exist unique integers q and r such that

$$a = qb + r \quad \text{and} \quad 0 \leq r < |b|.$$

This is simply a formal way to express the division of an integer by a nonzero integer. Here, q is called the *quotient*(商) while r is called the *remainder*(餘數) when a is divided by b .

Definition 1.1. An (nonzero) integer b is said to *divide*(整除) an integer a if $a = kb$ for some integer k . We use the notation $b \mid a$ to denote this. If there is no such integer k , then b does not divide a , and we use the notation $b \nmid a$.

If $b \mid a$, we also say that b is a *divisor*(因子) of a , and a is a *multiple*(倍數) of b , or a is *divisible by*(被...整除) b . Note that these definitions apply to negative integers as well. For example, we have $-2 \mid 4$ and $-3 \nmid -4$. By convention, we usually assume b is nonzero when we write $b \mid a$ or $b \nmid a$. However, some may take into account the case $b = 0$ in the above definition. This means $0 \mid a$ if and only if(iff) $a = 0$. Yet this special case is usually rejected for our purpose.

The following gives a few useful properties of divisibility.

Proposition 1.1. Let a, b, c, d be integers. Then the following hold.

- (a) If $a \mid b$ and $b \mid c$, then $a \mid c$.
- (b) If $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ for any integers m and n .
- (c) If $a \mid b$ and $c \mid d$, then $ac \mid bd$.
- (d) If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.

Proof.

(a) Since $a \mid b$ and $b \mid c$, we have $b = ma$ and $c = nb$ for some $m, n \in \mathbb{Z}$. Hence, we get $c = nb = nma$. As $nm \in \mathbb{Z}$, this implies $a \mid c$.

(b) Since $a \mid b$ and $a \mid c$, we have $b = ka$ and $c = \ell a$ for some $k, \ell \in \mathbb{Z}$. Hence, we get

$$mb + nc = mka + n\ell a = (mk + n\ell)a.$$

As $mk + n\ell \in \mathbb{Z}$, this implies $a \mid mb + nc$.

(c) Since $a \mid b$ and $c \mid d$, we have $b = ma$ and $d = nc$ for some $m, n \in \mathbb{Z}$. Hence, we get $bd = manc = (mn)ac$. As $mn \in \mathbb{Z}$, this implies $ac \mid bd$.

(d) Since $a \mid b$, we have $b = ma$ for some $m \in \mathbb{Z}$. As $b \neq 0$, we have $m \neq 0$. Therefore, $|m| \geq 1$. Hence, we get $|b| = |m| |a| \geq |a|$ as desired.

□

From (b), we know in particular that $a \mid b \pm c$. The first 3 items are the basic operations that we can do for the divisibility symbol. Part (d) is a useful property that can convert a number theory problem into an algebra problem. We always need to use this simple fact to solve divisibility problems.

Definition 1.2. A positive integer $p > 1$ is called a *prime number* (質數) or simply a *prime* if its only positive divisors are 1 and p . A positive integer $n > 1$ which is not a prime number is called a *composite number* (合成數).

A prime number is usually denoted by p , sometimes also by q . The first few primes are 2, 3, 5, 7, 11, ... It is known that every positive integer greater than 1 can be expressed in the form $p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ uniquely, where $p_1 < p_2 < \cdots < p_s$ are primes and $\alpha_1, \alpha_2, \dots, \alpha_s$ are positive integers. This is called the *fundamental theorem of arithmetic* (算術基本定理) or the *unique factorization theorem* (唯一分解定理). For convenience, sometimes we may allow some of the exponents α_j 's to be zero, which clearly does not affect the value of the product. Note that 1 is neither a prime nor a composite number.

Proposition 1.2. There are infinitely many prime numbers.

This is a well-known fact, whose proof is also well-known.

Proof. Suppose on the contrary that there are only finitely many primes. We order all the primes as $p_1 < p_2 < \dots < p_m$. Consider $n = p_1 p_2 \dots p_m + 1$. Clearly, $n > 1$. Note that n is not divisible by any p_j since the remainder when n is divided by p_j is 1. By the unique factorization theorem, n should contain a prime divisor, which is different from $p_1, p_2, \dots, p_m$. This contradicts the assumption that $p_1, p_2, \dots, p_m$ are all the prime numbers. Hence, there are infinitely many primes. $\square$

Prime numbers are useful when solving divisibility problems because of the following properties.

Proposition 1.3. Let p be a prime number, let a, b be integers, and let n be a positive integer. Then the following hold.

- (a) If $a \mid p$, then $a = \pm 1, \pm p$.
- (b) If $p \mid ab$, then $p \mid a$ or $p \mid b$.
- (c) If $p \mid a^n$, then $p \mid a$.

Proof.

- (a) This follows from the definition of a prime number. One only needs to take into account the negative divisors of p .
- (b) Since $p \mid ab$, we have $ab = pc$ for some $c \in \mathbb{Z}$. Let

$$a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}, \quad b = q_1^{\beta_1} q_2^{\beta_2} \dots q_t^{\beta_t}, \quad c = r_1^{\gamma_1} r_2^{\gamma_2} \dots r_u^{\gamma_u}$$

where $p_1, \dots, p_s, q_1, \dots, q_t, r_1, \dots, r_u$ are prime numbers and $\alpha_i, \beta_j, \gamma_k \in \mathbb{N}$. Then

$$p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s} q_1^{\beta_1} q_2^{\beta_2} \dots q_t^{\beta_t} = p r_1^{\gamma_1} r_2^{\gamma_2} \dots r_u^{\gamma_u}.$$

Since the prime p appears on the right-hand side, by the uniqueness of factorization, p must also appear on the left-hand side. Without loss of generality (WLOG), assume $p_1 = p$. This implies $p \mid a$. (If $q_1 = p$, then $p \mid b$.)

- (c) This is a corollary of (b). Indeed, one can use (b) to show a more general result by induction. If $p \mid a_1 a_2 \dots a_n$ where $a_j \in \mathbb{Z}$, then p divides one of $a_1, a_2, \dots, a_n$. By taking $a_1 = a_2 = \dots = a_n = a$, we obtain the desired result.

$\square$

Note that (b) does not hold if p is not a prime number.

Definition 1.3. Let a and b be integers which are not both zero. A nonzero integer c is said to be a *common divisor*(公因數) of a and b if $c \mid a$ and $c \mid b$. In particular, there exists a unique integer $d > 0$, called the *greatest common divisor*(最大公因數) of a and b , such that

- (i) $d \mid a$ and $d \mid b$, and
- (ii) whenever $c \mid a$ and $c \mid b$, we have $c \leq d$.

The greatest common divisor of a and b is denoted by $d = \gcd(a, b)$ or simply $d = (a, b)$. In particular, if $(a, b) = 1$, we say that a and b are *relatively prime*(互質) or *coprime*.

We give a few properties of the greatest common divisor.

Proposition 1.4. Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ and $b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ be positive integers where $p_1, p_2, \dots, p_s$ are distinct primes and α_j, β_j are nonnegative integers. Then

$$(a, b) = p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s}$$

where $\gamma_j = \min \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$.

Proof. Note that $c \in \mathbb{N}$ divides a iff c has the form $p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s}$ where $\gamma_j \leq \alpha_j$ for $j = 1, 2, \dots, s$. Therefore, c is a common divisor of a and b iff $\gamma_j \leq \min \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$. The result follows readily by the definition of the greatest common divisor. $\square$

This gives us one way to find (a, b) .

Proposition 1.5. Let a, b, c be integers. Then the following hold.

- (a) $(a, b) = (a + kb, b)$ for any integer k .
- (b) $(ka, kb) = |k|(a, b)$ for any nonzero integer k .
- (c) If $(a, c) = 1$, then $(a, bc) = (a, b)$.
- (d) If $a \mid bc$, then $\frac{a}{(a, c)} \mid b$. In particular, if $a \mid bc$ and $(a, c) = 1$, then $a \mid b$.
- (e) (*Bézout's identity*(貝祖定理)) If $d = (a, b)$, then there exist integers m and n such that $ma + nb = d$.
- (f) If $c \mid a$ and $c \mid b$, then $c \mid (a, b)$.

Proof.

- (a) Let $d = (a, b)$. From $d \mid a$ and $d \mid b$, we have $d \mid a + kb$. This shows d is a common divisor of $a + kb$ and b . Consider any common divisor c of $a + kb$ and b . Then $c \mid a + kb$ and $c \mid b$ imply $c \mid a + kb - kb = a$. Hence, c is a common divisor of a and b . By the definition of the greatest common divisor, we get $c \leq d$. This shows $d = (a + kb, b)$ by definition.

- (b) Let $a = \pm p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$, $b = \pm p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ and $k = \pm p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s}$ where $p_1, p_2, \dots, p_s$ are distinct prime numbers and $\alpha_j, \beta_j, \gamma_j$ are nonnegative integers. By [proposition 1.4](#), we have

$$(a, b) = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s}$$

where $m_j = \min \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$, and

$$(ka, kb) = p_1^{n_1} p_2^{n_2} \cdots p_s^{n_s}$$

where $n_j = \min \{\alpha_j + \gamma_j, \beta_j + \gamma_j\} = \min \{\alpha_j, \beta_j\} + \gamma_j$ for $j = 1, 2, \dots, s$. It is clear that

$$(ka, kb) = p_1^{m_1 + \gamma_1} p_2^{m_2 + \gamma_2} \cdots p_s^{m_s + \gamma_s} = |k| (a, b).$$

- (c) Let $a = \pm p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ and $c = \pm q_1^{\gamma_1} q_2^{\gamma_2} \cdots q_t^{\gamma_t}$ be the prime factorizations of a and c . Since $(a, c) = 1$, all p_i, q_j 's are distinct.

Let $b = \pm p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s} q_1^{\delta_1} q_2^{\delta_2} \cdots q_t^{\delta_t} r_1^{\nu_1} r_2^{\nu_2} \cdots r_u^{\nu_u}$ where $r_1, r_2, \dots, r_u$ are distinct prime numbers different from p_i, q_j . The exponents β_i, δ_j are nonnegative, while ν_k are positive. By [proposition 1.4](#), we have

$$(a, b) = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s}$$

where $m_j = \min \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$, and

$$(a, bc) = p_1^{m_1} p_2^{m_2} \cdots p_s^{m_s} = (a, b).$$

- (d) Since $a \mid bc$, there exists $n \in \mathbb{Z}$ such that $bc = na$. We write $a = (a, c)a'$ and $c = (a, c)c'$. Note that $(a', c') = 1$ since otherwise

$$(a, c) = ((a, c)a', (a, c)c') = (a, c)(a', c') > (a, c)$$

by [\(b\)](#). From $bc = na$, we have $bc' = na'$. As a' and c' are relatively prime, if we consider the prime factorizations of both sides, we see that all prime divisors of a' must come from b . This implies $a' \mid b$ as desired.

- (e) Consider the set

$$S = \{ma + nb : m, n \in \mathbb{Z}, ma + nb > 0\}.$$

Since this is a set of positive integers, it contains a smallest number. Suppose $d' = ma + nb$ is the smallest among all numbers in S . We claim that d' is the

greatest common divisor of a and b . First of all, let $a = qd' + r$ where $0 \leq r \leq d' - 1$ using the division algorithm. Then

$$r = a - qd' = a - q(ma + nb) = (1 - qm)a + (-qn)b$$

belongs to S by definition unless $r = 0$. However, since $r < d'$, $r \notin S$ by the minimality of d' . Therefore, we must have $r = 0$ and hence $d' \mid a$. Similarly, $d' \mid b$.

Next, suppose c is any common divisor of a and b . Then $c \mid a$ and $c \mid b$ imply $c \mid ma + nb = d'$. As $d' > 0$, this gives $c \leq |c| \leq d'$ by [proposition 1.1\(d\)](#). Therefore, $d' = (a, b) = d$ by definition. From the construction, we notice that $d = ma + nb$ as desired.

(f) This follows from the proof of part (e). □

Parts (a), (b) and (c) are the basic operations that can be done to relate different greatest common divisors. Bézout's identity is an important property of the greatest common divisor. It is a useful way to express (a, b) if a and b are not integers given explicitly.

One can define analogously the greatest common divisor of more than 2 integers. We would not go through the details here.

Definition 1.4. Let a and b be nonzero integers. An integer c is said to be a *common multiple* (公倍数) of a and b if $a \mid c$ and $b \mid c$. In particular, there exists a unique integer $\ell > 0$, called the *least common multiple* (最小公倍数) of a and b , such that

- (i) $a \mid \ell$ and $b \mid \ell$, and
- (ii) whenever $a \mid c$ and $b \mid c$ for $c \neq 0$, we have $\ell \leq |c|$.

The least common multiple of a and b is denoted by $\ell = \text{lcm}(a, b)$ or simply $\ell = [a, b]$.

Analogous to the case of greatest common divisors, we have the following.

Proposition 1.6. Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ and $b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ be positive integers where $p_1, p_2, \dots, p_s$ are distinct primes and α_j, β_j are nonnegative integers. Then

$$[a, b] = p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s}$$

where $\gamma_j = \max \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$.

Proof. Note that $c \in \mathbb{N}$ is divisible by a iff c has the form $p_1^{\gamma_1} p_2^{\gamma_2} \cdots p_s^{\gamma_s}$ where $\gamma_j \geq \alpha_j$ for $j = 1, 2, \dots, s$. Therefore, c is a common multiple of a and b iff $\gamma_j \geq \max \{\alpha_j, \beta_j\}$ for $j = 1, 2, \dots, s$. The result follows readily by the definition of the least common multiple. □

Proposition 1.7. Let a, b, c be integers. Then the following hold.

- (a) $|a||b| = (a, b)[a, b]$
- (b) if $a \mid c$ and $b \mid c$, then $[a, b] \mid c$.

Proof.

- (a) Let $d = (a, b)$, $a = da'$ and $b = db'$ where $a', b' \in \mathbb{Z}$. Note that $(a', b') = 1$. We need to prove that $[a, b] = d|a'||b'|$.

Firstly, it is obvious that $a = da'$ divides $d|a'||b'|$ and $b = db'$ divides $d|a'||b'|$. If c is a nonzero common multiple of a and b , let $c = ac_1$ and $c = bc_2$. As there are $m, n \in \mathbb{Z}$ such that $d = ma + nb$ by Bézout's identity, we have

$$cd = c(ma + nb) = (bc_2)ma + (ac_1)nb = (mc_2 + nc_1)ab,$$

which is a multiple of ab . Therefore, $|cd| \geq |ab| = d^2|a'||b'|$ by [proposition 1.1\(d\)](#) as $c, d \neq 0$. Hence, $|c| \geq d|a'||b'|$. By definition, $d|a'||b'|$ is the least common multiple of a and b .

- (b) From the proof of part (a), we know that $|a||b| \mid cd$. Using the notation in that proof, this implies $d^2|a'||b'| \mid cd$. Hence, $[a, b] = d|a'||b'| \mid c$.

□

By part (a), in particular, if $(a, b) = 1$, then $[a, b] = |a||b|$. Similarly, by part (b), if $a \mid c$, $b \mid c$ and $(a, b) = 1$, then $ab \mid c$. Also, in view of (a), the least common multiple can be converted to the greatest common divisor easily. The former is less frequently used than the latter in number theory.

Problem 1.1. For each $n \in \mathbb{N}$, find the greatest common divisor of $n^3 + n^2 + 2n + 1$ and $n^2 + n + 1$.

Analysis. We can use [proposition 1.5\(a\)](#) to reduce the degrees of the polynomials when finding the greatest common divisor.

Solution. Since $n^3 + n^2 + 2n + 1 = n(n^2 + n + 1) + n + 1$, we have

$$(n^3 + n^2 + 2n + 1, n^2 + n + 1) = (n + 1, n^2 + n + 1).$$

Similarly, as $n^2 + n + 1 = n(n + 1) + 1$, we have

$$(n + 1, n^2 + n + 1) = (n + 1, 1).$$

Note that the only positive divisor of 1 is 1. Therefore, $(n + 1, 1)$ must be equal to 1. These show $(n^3 + n^2 + 2n + 1, n^2 + n + 1) = 1$. □

Remarks. A similar idea can be used to compute the greatest common divisor of two given integers. Indeed, given two integers a, b with $|a| > |b|$, we can write $a = qb + r$ for some $r = 0, 1, \dots, |b| - 1$. Then we have $(a, b) = (qb + r, b) = (r, b)$. The size of $\max\{|a|, |b|\}$ is reduced in every such step. Eventually, we can obtain a pair of small integers whose gcd can be easily found. This method is called the *Euclidean algorithm* (輾轉相除法).

Problem 1.2. Let $a, b, n \in \mathbb{N}$. Prove that $a \mid b$ iff $n^a - 1 \mid n^b - 1$.

Analysis. The only if part is rather easy to prove. For the converse, we can write $b = qa + r$ using the division algorithm and try to show that $r = 0$. This is one common trick in proving $a \mid b$.

Solution. If $a \mid b$, then $b = ka$ for some $k \in \mathbb{N}$. Then

$$n^b - 1 = n^{ka} - 1 = (n^a - 1)(n^{(k-1)a} + n^{(k-2)a} + \dots + 1)$$

is divisible by $n^a - 1$.

Conversely, if $n^a - 1 \mid n^b - 1$, we write $b = qa + r$ where $0 \leq r \leq a - 1$ using the division algorithm. This gives $n^a - 1 \mid n^{qa+r} - 1$. Note that $n^a - 1 \mid n^{qa} - 1$ from above. Therefore,

$$\begin{aligned} & n^a - 1 \mid n^{qa+r} - 1 \\ \Rightarrow & n^a - 1 \mid (n^{qa} - 1)n^r + n^r - 1 \\ \Rightarrow & n^a - 1 \mid n^r - 1. \end{aligned}$$

As $r \leq a - 1$, we must have $n^a - 1 > n^r - 1$. In view of [proposition 1.1\(d\)](#), we need $n^r - 1 = 0$, i.e. $r = 0$. Hence, $a \mid b$.

The equivalence is established. □

Problem 1.3. ([Frobenius coin problem](#) (弗羅貝尼烏斯硬幣問題)) Let $m, n \in \mathbb{N}$ be relatively prime. Find the smallest positive integer N such that for every integer $k > N$, we can write $k = am + bn$ for some $a, b \in \mathbb{N}_0$.

Analysis. This is a modification of Bézout's identity where we restrict a, b to be nonnegative. We can use that fact to solve this problem. The bound N arises naturally if we follow the arguments in the solution.

Solution. By Bézout's identity, we can always find $a, b \in \mathbb{Z}$ such that $k = am + bn$. We pick the pair (a, b) for which $a \in \mathbb{N}_0$ is the smallest possible. We claim that $a \leq n - 1$. Indeed, if $a \geq n$, then we can write

$$k = am + bn = (a - n)m + (b + m)n.$$

This shows the pair $(a - n, b + m)$ has the first entry $a - n \in \mathbb{N}_0$ which is smaller than a . This contradicts the minimality of a . Now, if we can show that $a \leq n - 1$ implies

$b \geq 0$, then the equation is solvable in nonnegative integers. Suppose $b \leq -1$. Then we have

$$k = am + bn \leq (n-1)m + (-1)n = mn - m - n.$$

Therefore, if we define $N = mn - m - n$, then whenever $k > N$, we must have $b \in \mathbb{N}_0$. In other words, any $k > N$ can be represented in the desired form.

It remains to show that N cannot be represented in the desired form. Suppose

$$mn - m - n = am + bn$$

for some $a, b \in \mathbb{N}_0$. This can be rewritten as

$$(n-1-a)m = (b+1)n.$$

This implies $m \mid (b+1)n$. As $(m, n) = 1$, this yields $m \mid b+1$. Let $b = km - 1$ for some $k \in \mathbb{Z}$. Note that $k \geq 1$ since b is nonnegative. Therefore, $b \geq m - 1$. By symmetry, we have $a \geq n - 1$. Then

$$mn - m - n = am + bn \geq (n-1)m + (m-1)n = 2mn - m - n.$$

This is a contradiction. Hence, the answer is $N = mn - m - n$. $\square$

1.2 Arithmetic Functions

In this short subsection, we introduce a few useful arithmetic functions. An arithmetic function is simply a function defined on the set of positive integers.

Definition 1.5. Let α be a real number, and let n be a positive integer. We define a *divisor function* (除數函數) of n by

$$\sigma_\alpha(n) = \sum_{d \mid n} d^\alpha,$$

where the summation runs through all positive divisors d of n .

In particular, when $\alpha = 0$, $\sigma_0(n)$ is the number of positive divisors of n and is usually denoted by $d(n)$ or $\tau(n)$. When $\alpha = 1$, $\sigma_1(n)$ is the sum of all positive divisors of n and is usually denoted by $\sigma(n)$. Their formulae are given below.

Proposition 1.8. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be the prime factorization of $n > 1$. Then the following hold.

- (a) $d(n) = (\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1)$
- (b) $\sigma(n) = (1 + p_1 + \cdots + p_1^{\alpha_1})(1 + p_2 + \cdots + p_2^{\alpha_2}) \cdots (1 + p_s + \cdots + p_s^{\alpha_s})$

Proof.

- (a) Each divisor of n is of the form $p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$ where $0 \leq \beta_j \leq \alpha_j$ for $j = 1, 2, \dots, s$. Since there are $\alpha_j + 1$ choices for each β_j , there are exactly

$$(\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1)$$

positive divisors of n . By definition, $d(n) = (\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1)$.

- (b) Observe that there are exactly $d(n)$ terms in the expansion of

$$(1 + p_1 + \cdots + p_1^{\alpha_1})(1 + p_2 + \cdots + p_2^{\alpha_2}) \cdots (1 + p_s + \cdots + p_s^{\alpha_s}).$$

In addition, every term corresponds to a distinct positive divisor of n (simply of the form $p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}$ where $0 \leq \beta_j \leq \alpha_j$). In other words, this is just the sum of all positive divisors of n , and hence we obtain the desired formula for $\sigma(n)$. □

Note that we can use the sum of geometric sequence to get

$$\sigma(n) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \times \frac{p_2^{\alpha_2+1} - 1}{p_2 - 1} \times \cdots \times \frac{p_s^{\alpha_s+1} - 1}{p_s - 1}.$$

Definition 1.6. Let n be a positive integer. We define $\varphi(n)$ to be the number of positive integers from 1 to n which are coprime to n . The function φ is called *Euler's totient function* or simply the *Euler function*(歐拉函數).

Proposition 1.9. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be the prime factorization of $n > 1$. Then

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_s}\right).$$

Proof. Let N_m be the number of integers from 1 to n which are divisible by m . In particular, if $m \mid n$, then $N_m = \frac{n}{m}$. To find the number of integers between 1 and n which are coprime to n , we simply need to find those integers not divisible by any of $p_1, p_2, \dots, p_s$. By the inclusion-exclusion principle, this is equal to

$$\begin{aligned} & n - \sum_{i=1}^n N_{p_i} + \sum_{1 \leq i < j \leq n} N_{p_i p_j} - \cdots + (-1)^s N_{p_1 p_2 \cdots p_s} \\ &= n - \sum_{i=1}^n \frac{n}{p_i} + \sum_{1 \leq i < j \leq n} \frac{n}{p_i p_j} - \cdots + (-1)^s \frac{n}{p_1 p_2 \cdots p_s}. \end{aligned}$$

Note that this is just the expansion of $n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_s}\right)$. □

In particular, for any prime number p and any integer $k \geq 1$, we have $\varphi(p) = p - 1$ and $\varphi(p^k) = p^{k-1}(p - 1)$.

Note that the divisor functions and Euler's totient function are *multiplicative* (積性). This means for any coprime integers m and n , we have

$$\sigma_\alpha(mn) = \sigma_\alpha(m)\sigma_\alpha(n) \quad \text{and} \quad \varphi(mn) = \varphi(m)\varphi(n).$$

Problem 1.4. (IMO Shortlist 2000 N2) For a positive integer n , let $d(n)$ be the number of all positive divisors of n . Find all positive integers n such that $d(n)^3 = 4n$.

Analysis. We can use the formula given in [proposition 1.8\(a\)](#) to set up an equation. Since this formula is related to prime factorizations, we can use the uniqueness of prime factorization to compare the two sides. Note that $4n$ is in general larger than $d(n)^3$. Therefore, we can use some bounds to reject all but finitely many cases.

Solution. Clearly, $n \neq 1$. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ be its prime factorization. By [proposition 1.8\(a\)](#), the given equation becomes

$$(\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1)^3 = 4p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}.$$

Let

$$(\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_s + 1) = q_1^{\beta_1} q_2^{\beta_2} \cdots q_t^{\beta_t} \tag{1}$$

be its prime factorization. Then

$$q_1^{3\beta_1} q_2^{3\beta_2} \cdots q_t^{3\beta_t} = 4p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}. \tag{2}$$

Since 4 divides the right-hand side of (2), the prime 2 must appear on the left. WLOG assume $q_1 = 2$. As the exponent of 2 on the left is a multiple of 3, while $4 = 2^2$, we must have more powers of 2 on the right. WLOG assume $p_1 = 2$. By comparing the exponents, we have $3\beta_1 = \alpha_1 + 2$. Then we obtain

$$q_2^{3\beta_2} \cdots q_t^{3\beta_t} = p_2^{\alpha_2} \cdots p_s^{\alpha_s}.$$

By the uniqueness of prime factorization, by permuting the indices if necessary, we must have $t = s$, $p_j = q_j$ and $\alpha_j = 3\beta_j$ for $j = 2, \dots, s$. Therefore, from (1), we get

$$(3\beta_1 - 1)(3\beta_2 + 1) \cdots (3\beta_s + 1) = 2^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}. \tag{3}$$

Note that none of the p_j 's can be 3 as the left-hand side is not a multiple of 3. We shall show that the right-hand side of (3) is in general larger than the left-hand side. Firstly, for $\beta_1 \geq 4$, by the binomial theorem, we have

$$2^{\beta_1} = (1 + 1)^{\beta_1} > 1 + \beta_1 + \binom{\beta_1}{2} = \frac{\beta_1^2 + \beta_1 + 2}{2} \geq 3\beta_1 - 1$$

since the last inequality is equivalent to $(\beta_1 - 1)(\beta_1 - 4) \geq 0$. Secondly, for $p \geq 5$ and $\beta \geq 1$, we have

$$p^\beta \geq (1 + 4)^\beta \geq 1 + 4\beta > 3\beta + 1.$$

These help us eliminate many cases.

- If $\beta_1 \geq 4$, then the right-hand side of (3) is larger than the left-hand side.
- If $\beta_1 = 1$ or $\beta_1 = 3$, then $3\beta_1 - 1 = 2^{\beta_1}$. In view of the above bound for $p \geq 5$, in order that (3) holds, we cannot have primes other than 2. Therefore, $s = 1$, in which case (3) is true. We get two solutions $n = 2^1 = 2$ and $n = 2^7 = 128$.
- If $\beta_1 = 2$, then $\frac{3\beta_1 - 1}{2^{\beta_1}} = \frac{5}{4}$. This shows 5 divides the left-hand side of (3). WLOG assume $p_2 = 5$. If $\beta_2 \geq 2$, then

$$5^{\beta_2} = (1 + 4)^{\beta_2} \geq 1 + 4\beta_2 + 16\binom{\beta_2}{2} = 8\beta_2^2 - 4\beta_2 + 1 > \frac{5}{4}(3\beta_2 + 1)$$

as the last inequality is the same as $(\beta_2 - 1)(32\beta_2 + 1) > 0$. Then we easily observe that the right-hand side of (3) is larger than the left-hand side. Hence,

we need $\beta_2 = 1$, in which case $\frac{p_2^{\beta_2}}{3\beta_2 + 1} = \frac{5}{4}$. Then we cannot have additional prime factors. This gives one solution $n = 2^4 \cdot 5^3 = 2000$.

To conclude, the solutions are $n = 2, 128, 2000$. $\square$

Remarks. The binomial expansion is useful in estimating exponents. Induction can also be used to establish the same bounds, but usually the proof is more lengthy (unless one merely says the inequality clearly holds by induction!).

Problem 1.5. (CWMO 2004 Problem 4) Let $n \in \mathbb{N}$. Denote by $d(n)$ the number of positive divisors of n , and $\varphi(n)$ the number of integers among $1, 2, \dots, n$ which are relatively prime to n . Find all nonnegative integers c such that there exists a positive integer n satisfying

$$d(n) + \varphi(n) = n + c.$$

For each such c , find all positive integers n satisfying the above equation.

Analysis. We first use the definitions of $d(n)$ and $\varphi(n)$. Since $d(n)$ counts the number of positive divisors of n and $\varphi(n)$ counts the number of integers among $1, 2, \dots, n$ which are relatively prime to n , they cannot count the same integer except 1. Therefore, c must be small. To find all n satisfying the equation, we note that n cannot have too many prime factors, or otherwise some integers are not counted in both $d(n)$ and $\varphi(n)$.

Solution. Let A be the set of all positive divisors of n , and let B be the set of all positive integers from 1 to n relatively prime to n . Then A and B are subsets of $\{1, 2, \dots, n\}$ and $A \cap B = \{1\}$. This is because every positive divisor of n greater than 1 cannot be coprime to n . Therefore,

$$n + c = d(n) + \varphi(n) = |A| + |B| = |A \cup B| + |A \cap B| \leq n + 1. \quad (1)$$

This gives $c \leq 1$. As c is nonnegative, we have $c = 0$ or 1 . In view of the inequality in (1), at most one of $1, 2, \dots, n$ cannot belong to $A \cup B$.

- If $n = 1$, we have $d(1) + \varphi(1) = 1 + 1$. This is one of the solutions.
- If n is a prime, then $d(n) + \varphi(n) = 2 + (n - 1) = n + 1$. Hence, every prime n is a solution.
- If n is a composite number, let p be a prime factor of n . Note that $(n - p, n)$ and $(n - 2p, n)$ are both greater than 1 since they are divisible by p . If $n - 2p > \frac{n}{2}$, then $n - p$ and $n - 2p$ cannot be divisors of n . In that case, we would have $|A \cup B| \leq n - 2$ since at least 2 integers are not included in the union. Therefore, we need $n - 2p \leq \frac{n}{2}$, i.e. $n \leq 4p$. As $p \mid n$ and n is composite, this gives $n = 2p, 3p, 4p$. Using [proposition 1.8\(a\)](#) and [proposition 1.9](#), we check that

$$\begin{aligned}
d(2p) + \varphi(2p) &= 4 + (p - 1) = p + 3 \quad \text{for } p \neq 2, \\
d(4) + \varphi(4) &= 3 + 2 = 4 + 1, \\
d(3p) + \varphi(3p) &= 4 + 2(p - 1) = 2p + 2 \quad \text{for } p \neq 3, \\
d(9) + \varphi(9) &= 3 + 6 = 9, \\
d(4p) + \varphi(4p) &= 6 + 2(p - 1) = 2p + 4 \quad \text{for } p \neq 2, \\
d(8) + \varphi(8) &= 4 + 4 = 8.
\end{aligned}$$

From these, $2p$ is a solution iff $p = 2, 3$, $3p$ is a solution iff $p = 2, 3$, and $4p$ is a solution iff $p = 2$.

Therefore, the solutions are as follows. For $c = 1$, n can be a prime or 1, 4. For $c = 0$, n can be one of 6, 8, 9. □

1.3 Congruences

In many cases, we need to consider the division of an expression by some positive integers. In elementary number theory, there is a convenient notation to handle this.

Definition 1.7. Let n be a positive integer. If a and b leave the same remainder when divided by n , then we say that a is *congruent*(同餘) to b modulo n , or symbolically,

$$a \equiv b \pmod{n}.$$

Note that $a \equiv b \pmod{n}$ iff $n \mid a - b$. Therefore, this can be seen as another way to express a divisibility. However, when we use the congruence notation, we are emphasizing more on the modulus n . If a is not congruent to b modulo n , then we write $a \not\equiv b \pmod{n}$.

Here gives the properties of congruences.

Proposition 1.10. Let a, b, c, d, m, n be integers where m, n are positive. Then the following hold.

- (a) $a \equiv a \pmod{n}$
- (b) if $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$
- (c) if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$
- (d) if $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a \pm c \equiv b \pm d \pmod{n} \quad \text{and} \quad ac \equiv bd \pmod{n}$$

- (e) if $a \equiv b \pmod{n}$ and $c \mid a, c \mid b$, then $\frac{a}{c} \equiv \frac{b}{c} \pmod{\frac{n}{(n, c)}}$
- (f) if $a \equiv b \pmod{m}$ and $a \equiv b \pmod{n}$, then $a \equiv b \pmod{[m, n]}$

The first 3 parts are clear from the definition. We shall prove the remaining.

Proof.

- (d) We have $n \mid a - b$ and $n \mid c - d$. Then $n \mid (a - b) \pm (c - d)$. Therefore, we have $n \mid (a \pm c) - (b \pm d)$. Hence, $a \pm c \equiv b \pm d \pmod{n}$. Also, $n \mid a - b$ and $n \mid c - d$ give $n \mid c(a - b) + b(c - d) = ac - bd$. Thus, $ac \equiv bd \pmod{n}$.

- (e) We have $n \mid a - b$. By [proposition 1.5\(d\)](#), this yields $\frac{n}{(n, c)} \mid \frac{a}{c} - \frac{b}{c}$, and hence the result follows.

- (f) We have $m \mid a - b$ and $n \mid a - b$. By [proposition 1.7\(b\)](#), this yields $[m, n] \mid a - b$, and hence the result follows.

□

These properties suggest the congruence relation behaves almost the same as the usual equality, except that we need to be careful when doing division, as can be seen from part (e). Here, we list out some special cases of the above facts. As a corollary of (d), if $a \equiv b \pmod{n}$ and $k \in \mathbb{Z}$, then

$$\begin{aligned} a \pm k &\equiv b \pm k \pmod{n}, \\ ka &\equiv kb \pmod{n}, \\ a^k &\equiv b^k \pmod{n} \quad \text{for } k \geq 0. \end{aligned}$$

Thus, if $P(m)$ is a polynomial with integer coefficients, then $P(a) \equiv P(b) \pmod{n}$. This implies $a - b \mid P(a) - P(b)$.

For (e), if $a \equiv b \pmod{n}$ and $(n, c) = 1$, then $\frac{a}{c} \equiv \frac{b}{c} \pmod{n}$.

For (f), if $a \equiv b \pmod{m}$ and $a \equiv b \pmod{n}$, and $(m, n) = 1$, then $a \equiv b \pmod{mn}$.

The following shows when a linear congruence equation is solvable.

Proposition 1.11. Let a, b be integers and let n be a positive integer. Then the congruence equation $ax \equiv b \pmod{n}$ in x is solvable iff $(a, n) \mid b$.

Proof. Let $(a, n) = d$. If there exists $m \in \mathbb{Z}$ such that $am \equiv b \pmod{n}$, then $n \mid am - b$. Since $d \mid n$, we have $d \mid am - b$. As $d \mid a$, this shows $d \mid am - (am - b) = b$.

Conversely, suppose $d \mid b$. Since $d = (a, n)$, by Bézout's identity, there exist $k, \ell \in \mathbb{Z}$ such that $ak + n\ell = d$. Let $b = db'$ where $b' \in \mathbb{Z}$. Then

$$b = db' = (ak + n\ell)b' = a(kb') + n(\ell b').$$

Considering modulo n , we get $b \equiv a(kb') \pmod{n}$. Hence, $x = kb'$ is a solution to $ax \equiv b \pmod{n}$. $\square$

Note that in the case $(a, n) \mid b$, the solution to $ax \equiv b \pmod{n}$ is not unique. Indeed, it may not even be unique modulo n . For the special case when $n = p$ is a prime number, then $ax \equiv b \pmod{p}$ has a unique solution modulo p for any a which is not divisible by p . Indeed, if both x, y are solutions, then

$$ax \equiv b \equiv ay \pmod{p},$$

which implies $x \equiv y \pmod{p}$ by [proposition 1.10\(e\)](#) as $(a, p) = 1$.

In particular, there exists a unique solution to $ax \equiv 1 \pmod{p}$. Sometimes we denote this unique solution by a^{-1} (that means $aa^{-1} \equiv 1 \pmod{p}$). It is a convenient notation since we may then work on negative powers, as it behaves as the usual powers of integers. For example, we have

$$a^k \times a^\ell \equiv a^{k+\ell} \pmod{p} \quad \text{and} \quad (a^k)^\ell \equiv a^{k\ell} \pmod{p}$$

for any integers k, ℓ (positive or not, where we define $a^{-k} = (a^{-1})^k$ for $k \in \mathbb{N}$), as long as $(p, a) = 1$.

Next, we introduce a few useful theorems about congruences.

Theorem 1.2. ([Euler-Fermat theorem](#)(歐拉費馬定理)/[Euler's theorem](#)(歐拉定理))

Let n be a positive integer, and let a be an integer with $(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Here, $\varphi(n)$ is Euler's totient function.

Proof. Let $\{a_1, a_2, \dots, a_{\varphi(n)}\}$ be a reduced set of residues modulo n , i.e. a set of integers which are relatively prime to n such that there is exactly one integer in each residue class relatively prime to n . For example, we may take the integers from 1 to n which are relatively prime to n . There are exactly $\varphi(n)$ such integers.

Consider the set $\{aa_1, aa_2, \dots, aa_{\varphi(n)}\}$. Since $(a, n) = (a_i, n) = 1$, each aa_i is relatively prime to n . Therefore, for any fixed $1 \leq i \leq \varphi(n)$, there exists a unique index j such that $aa_i \equiv a_j \pmod{n}$. In addition, for any distinct i, j , we have $aa_i \not\equiv aa_j \pmod{n}$. Otherwise, $aa_i \equiv aa_j \pmod{n}$ implies $a_i \equiv a_j \pmod{n}$ by [proposition 1.10\(e\)](#) as $(a, n) = 1$. Therefore, each aa_i corresponds to a distinct a_j and hence the two sets $\{a_1, a_2, \dots, a_{\varphi(n)}\}$ and $\{aa_1, aa_2, \dots, aa_{\varphi(n)}\}$ are in one-to-one correspondence modulo n . In other words, one of these sets is just a permutation of the other by considering modulo n . In particular, the product of all elements inside the two sets should be equal, i.e.

$$a_1 a_2 \cdots a_{\varphi(n)} \equiv (aa_1)(aa_2) \cdots (aa_{\varphi(n)}) \pmod{n}.$$

Note that $(a_i, n) = 1$ for each i . Therefore, we can cancel out the factor $a_1 a_2 \cdots a_{\varphi(n)}$ from both sides by [proposition 1.10\(e\)](#). This yields $a^{\varphi(n)} \equiv 1 \pmod{n}$ as desired. $\square$

In particular, when n is a prime number, we have the following theorem.

Theorem 1.3. ([Fermat little theorem](#)(費馬小定理)) Let p be a prime number, and let a be an integer with $(a, p) = 1$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

We shall give an alternative proof of the Fermat little theorem.

Proof. Note that it suffices to consider the cases $a = 1, 2, \dots, p-1$. We shall prove that $a^p \equiv a \pmod{p}$ by induction on a , which clearly implies $a^{p-1} \equiv 1 \pmod{p}$ since $(a, p) = 1$.

When $a = 1$, we have $1^p \equiv 1 \pmod{p}$. Suppose $a^p \equiv a \pmod{p}$ holds for some a . By the binomial theorem, we have

$$(a+1)^p = a^p + \binom{p}{1}a^{p-1} + \binom{p}{2}a^{p-2} + \cdots + \binom{p}{p-1}a + 1.$$

Note that for $k = 1, 2, \dots, p-1$, $\binom{p}{k} = \frac{p(p-1) \cdots (p-k+1)}{k!}$ is an integer divisible by p since the numerator has a factor p while the denominator does not. Therefore, $(a+1)^p \equiv a^p + 1 \pmod{p}$, which, by the inductive hypothesis, is congruent to $a + 1$ modulo p . This completes the inductive step, and hence the result is true. $\square$

This alternative proof again shows the usefulness of the binomial theorem. The Euler-Fermat theorem and the Fermat little theorem are useful in dealing with high powers of integers. Sometimes it is more convenient to use the forms

$$a^p \equiv a \pmod{p} \quad \text{and} \quad a^{n+(p-1)d} \equiv a^n \pmod{p}$$

for any nonnegative integers n and d . Note that these also hold if $p \mid a$ unless $n = 0$. If $(p, a) = 1$, then the second congruence also holds for negative integers n, d using the convention discussed above.

Proposition 1.12. Let p and q be odd primes such that $(q, p-1) = 1$, and let a be an integer. Then the congruence $x^q \equiv a \pmod{p}$ in x has a unique solution modulo p .

Proof. Since $(q, p-1) = 1$, there exist $b, c \in \mathbb{Z}$ such that $qb + (p-1)c = 1$. Using the convention for negative powers, we have

$$a = a^{qb+(p-1)c} \equiv a^{qb} \equiv (a^b)^q \pmod{p}.$$

The first congruence holds by the Fermat little theorem. From this, we obtain a solution $x = a^b$ to $x^q \equiv a \pmod{p}$.

Now, suppose there is more than 1 solution modulo p for some $a \in \mathbb{Z}$. Then $0^q, 1^q, \dots, (p-1)^q$ must belong to at most $p-1$ different residue classes modulo p as two of them are the solutions of the same congruence equation. Then the equation $x^q \equiv b \pmod{p}$ has no solution for some $b \in \mathbb{Z}$ as there are p choices of b . This is a contradiction. Therefore, each congruence equation must have a unique solution. (In short, we can use the following fact: every surjective function $f : S \rightarrow S$ defined on a finite set S must be injective. We just need to define $f(x) = x^q$ and $S = \{0, 1, \dots, p-1\}$, where everything is taken modulo p .) $\square$

There is an uncommon name for this proposition, called the q th power lemma. One should memorize the proof since simply quoting this fact is usually not acceptable. Note that the condition that q is a prime is not necessary. However, this is usually the case in practice. The case $q \mid p-1$ will be discussed in [proposition 6.2](#).

The following theorem is about solving a system of linear congruence equations.

Theorem 1.4. ([Chinese remainder theorem](#)(中國剩餘定理)) Let k be a positive integer. Consider a system of congruence equations

$$\begin{cases} x \equiv a_1 \pmod{n_1}, \\ x \equiv a_2 \pmod{n_2}, \\ \dots \\ x \equiv a_k \pmod{n_k} \end{cases}$$

in x where $n_1, n_2, \dots, n_k$ are pairwise coprime positive integers, and $a_1, a_2, \dots, a_k$ are integers. Then there exists a unique solution modulo $n_1 n_2 \cdots n_k$.

Proof. We first prove the assertion when $k = 2$. To prove that there exists a solution, recall that there exist $u, v \in \mathbb{Z}$ such that

$$un_1 + vn_2 = 1$$

by Bézout's identity as $(n_1, n_2) = 1$. Then $vn_2 \equiv 1 \pmod{n_1}$ and $un_1 \equiv 1 \pmod{n_2}$. Consider the integer

$$m = a_2un_1 + a_1vn_2.$$

Then $m \equiv a_1vn_2 \equiv a_1 \pmod{n_1}$ and $m \equiv a_2un_1 \equiv a_2 \pmod{n_2}$. In other words, we get a solution $x = m$ to the system.

Suppose there are two solutions $x, y \pmod{n_1n_2}$. Then $x \equiv a_1 \equiv y \pmod{n_1}$ and $x \equiv a_2 \equiv y \pmod{n_2}$. These yield $x \equiv y \pmod{n_1n_2}$ since $(n_1, n_2) = 1$. Thus, they represent the same solution modulo n_1n_2 .

In the general case, our claim shows that there is a unique solution $b_1 \pmod{n_1n_2}$ to the first two congruence equations. Next, as $(n_1n_2, n_3) = 1$, the system

$$\begin{cases} x \equiv b_1 \pmod{n_1n_2}, \\ x \equiv a_3 \pmod{n_3} \end{cases}$$

has a unique solution modulo $n_1n_2n_3$. By induction, it is obvious that there is a unique solution modulo $n_1n_2 \cdots n_k$ to the original system of k congruence equations. $\square$

In view of [proposition 1.11](#), we can consider a more general system

$$\begin{cases} b_1x \equiv a_1 \pmod{n_1}, \\ b_2x \equiv a_2 \pmod{n_2}, \\ \dots \\ b_kx \equiv a_k \pmod{n_k}. \end{cases}$$

If $(b_j, n_j) = (n_i, n_j) = 1$ for any $1 \leq i < j \leq k$, then the system has a unique solution in x modulo $n_1n_2 \cdots n_k$. Indeed, we can solve each equation modulo n_j , and prove this using the Chinese remainder theorem.

In practice, the moduli n_j 's are usually powers of distinct primes, since in that case the conditions $(n_i, n_j) = 1$ are satisfied. This also suggests why we usually consider congruences modulo prime powers.

Problem 1.6. Find all square numbers all of whose digits are the same.

Analysis. It is easy to guess that the only solutions are 0, 1, 4, 9. We can consider different congruences to reject the remaining cases.

Solution. Let n be such an integer. The only one-digit square numbers are 0, 1, 4, 9, which are all possible values of n . We now consider those n with at least 2 digits. Note that the unit digit of n must be 0, 1, 4, 5, 6, 9 (we are actually considering modulo 10).

- $n \neq 00\cdots 0$ since the first digit cannot be 0.
- If $n = 55\cdots 5$, then $5 \mid n$. Since n is a square number, we have $5^2 \mid n$. However, this is impossible since the last 2 digits of multiples of 25 can only be 00, 25, 50, 75.
- If $n = 66\cdots 6$, then $n \equiv 2 \pmod{4}$. This is impossible as a square number is congruent to $0, 1 \pmod{4}$.
- If $n = 11\cdots 1$, then $n \equiv 3 \pmod{4}$ as n has at least 2 digits. Again, this is impossible.
- If $n = 44\cdots 4$ or $99\cdots 9$, then $\frac{n}{4}$ or $\frac{n}{9}$ is equal to $11\cdots 1$, which should be a square number. From above, this cannot happen.

Therefore, $n = 0, 1, 4, 9$. □

Remarks. To show that $m^2 \equiv 0, 1 \pmod{4}$, etc., it suffices to consider the squares of $0, 1, 2, 3$ since these are the only residue classes modulo 4.

Problem 1.7. (IMO HK TST 2009 Test 1 Problem 6) Show that the equation

$$y^{37} \equiv x^3 + 11 \pmod{p}$$

is solvable for every prime p , where $p \leq 100$.

Analysis. Since this is about the solvability of a congruence equation involving high powers, one may use the result of [proposition 1.12](#).

Solution. Since $p \leq 100$ and p is a prime, we have $37 \nmid p - 1$. By [proposition 1.12](#), there exists $y \in \mathbb{Z}$ such that $y^{37} \equiv 11 \pmod{p}$. Then we can take such a y together with $x = 0$ as a solution. □

Problem 1.8. (CGMO 2012 Problem 3) Find all the pairs (a, b) of integers satisfying the following condition: there exists an integer $d \geq 2$ such that $a^n + b^n + 1$ is divisible by d for all positive integers n .

Analysis. It is not hard to find some pairs of (a, b) satisfying the condition. For example, when a, b have different parities, then $a^n + b^n + 1$ must be even and we can take $d = 2$. There is a less trivial type of solution where we can take $d = 3$. Intuitively, it is expected that d should not be too large.

Note that it suffices to consider the case when d is a prime. We need to consider $a^n + b^n + 1 \pmod{d}$. As this involves powers of integers, it is natural to consider the Fermat little theorem. Indeed, if $(a, d) = (b, d) = 1$, then we always have

$$a^{d-1} + b^{d-1} + 1 \equiv 1 + 1 + 1 = 3 \pmod{d}.$$

This shows $d \nmid a^{d-1} + b^{d-1} + 1$ unless $d = 3$ or one of a, b is divisible by d . We can refine this argument to work out the following solution.

Solution. If a and b have different parities, then $2 \mid a^n + b^n + 1$ for any $n \in \mathbb{N}$. Hence, we can take $d = 2$.

If $a \equiv b \equiv 1 \pmod{3}$, then $a^n + b^n + 1 \equiv 1 + 1 + 1 \equiv 0 \pmod{3}$ for any $n \in \mathbb{N}$. Hence, we can take $d = 3$.

We consider the remaining cases. Since a, b now have the same parity, we have $2 \nmid a + b + 1$. Also, we have $3 \nmid a + b + 1$ unless $a \equiv 0 \pmod{3}$ and $b \equiv 2 \pmod{3}$ (or $b \equiv 0 \pmod{3}$ and $a \equiv 2 \pmod{3}$). In that case, we note that

$$a^2 + b^2 + 1 \equiv 0 + 1 + 1 \not\equiv 0 \pmod{3}.$$

Therefore, d cannot have the factors 2 and 3 in any case.

Consider any prime $p > 3$. If both a, b are not divisible by p , then

$$a^{p-1} + b^{p-1} + 1 \equiv 1 + 1 + 1 = 3 \not\equiv 0 \pmod{p}$$

by the Fermat little theorem. If exactly one of a, b is divisible by p , say $p \mid a$, then

$$a^{p-1} + b^{p-1} + 1 \equiv 0 + 1 + 1 = 2 \not\equiv 0 \pmod{p}.$$

If both a, b are divisible by p , then

$$a + b + 1 \equiv 0 + 0 + 1 = 1 \not\equiv 0 \pmod{p}.$$

Hence, d cannot have the prime factor p . These conclude such an integer d does not exist.

Therefore, the only pairs of (a, b) are those described at the beginning. $\square$

Problem 1.9. (CWMO 2008 Problem 3) Let $m \geq 2$ be an integer, and let $a_1, a_2, \dots, a_m$ be positive integers. Prove that there are infinitely many positive integers n such that $a_1 \cdot 1^n + a_2 \cdot 2^n + \dots + a_m \cdot m^n$ is a composite number.

Analysis. By the Fermat little theorem, we know that $k^n \equiv k^{n+d(p-1)} \pmod{p}$ for any $k \in \mathbb{Z}$ and $n, d \in \mathbb{N}$. Therefore, if $p \mid a_1 \cdot 1^n + a_2 \cdot 2^n + \dots + a_m \cdot m^n$, then we also have $p \mid a_1 \cdot 1^{n+d(p-1)} + a_2 \cdot 2^{n+d(p-1)} + \dots + a_m \cdot m^{n+d(p-1)}$. This is how we construct infinitely many n 's in the following solution.

Solution. Let $s_n = a_1 \cdot 1^n + a_2 \cdot 2^n + \dots + a_m \cdot m^n$. Note that $s_1 > 1$. We take any prime factor p of s_1 . By the Fermat little theorem, we have

$$s_{1+d(p-1)} \equiv s_1 \pmod{p}$$

for any $d \in \mathbb{N}$. Also, it is obvious that the sequence $\{s_n\}$ is strictly increasing. Hence, $s_{1+d(p-1)} > s_1 \geq p$. Then $s_{1+d(p-1)}$ is a multiple of p which is greater than p . Therefore, every such term is a composite number. $\square$

Problem 1.10. Let p be a prime number. Prove that there exists a positive integer n such that $n^n + 2$ is a multiple of p .

Analysis. Consider $n^n + 2 \pmod{p}$. We can first use the Fermat little theorem to make the exponent become a constant. For example, if we focus on those integers n with $n \equiv 1 \pmod{p-1}$, then we must have $n^n + 2 \equiv n + 2 \pmod{p}$. We can use the Chinese remainder theorem to find an n satisfying both congruences.

Solution. Since p and $p-1$ are relatively prime, by the Chinese remainder theorem, there exists a positive integer n such that

$$\begin{cases} n \equiv 1 \pmod{p-1}, \\ n \equiv -2 \pmod{p}. \end{cases}$$

For this n , we find that

$$n^n + 2 \equiv n + 2 \equiv 0 \pmod{p}$$

by the Fermat little theorem. This completes the proof. $\square$

Remarks. The use of the Chinese remainder theorem is not necessary for this problem, as we may simply take $n = 3p - 2$. However, the advantage of using the theorem is that we do not need to solve the system of congruences explicitly, which is in general hard to do when there are more congruence equations.

Problem 1.11. Let n be a positive integer. Define $a_k = 2^{2^k} + 1$ for $k = 0, 1, \dots, n$. Prove that there exist $n+1$ consecutive integers $b, b+1, \dots, b+n$ such that a_k divides $b+k$ for each $k = 0, 1, \dots, n$.

Analysis. Note that the conditions essentially give a system of congruence equations in b . To apply the Chinese remainder theorem, it suffices to show that the a_k 's are pairwise relatively prime.

Solution. Let $0 \leq j < k \leq n$. Consider modulo $a_j = 2^{2^j} + 1$. We find that

$$a_k = 2^{2^k} + 1 = (2^{2^j})^{2^{k-j}} + 1 \equiv (-1)^{2^{k-j}} + 1 \equiv 1 + 1 = 2 \pmod{a_j}.$$

Note that a_j is odd. Therefore, $(a_j, a_k) = (a_j, 2) = 1$. This shows $a_0, a_1, \dots, a_n$ are pairwise relatively prime. By the Chinese remainder theorem, there exists $b \in \mathbb{Z}$ such that $b \equiv -k \pmod{a_k}$ for $k = 0, 1, \dots, n$. For any such b , the integer a_k divides $b+k$ for each k as desired. $\square$

Here is a summary of the important concepts and theorems covered in this section.

- divisibilities (problem 1.2)
 - division algorithm (problem 1.2)
 - primes (problem 1.4)

- Bézout's identity ([problem 1.3](#))
 - greatest common divisors ([problems 1.1, 1.11](#))
- arithmetic functions
 - divisor functions ([problems 1.4, 1.5](#))
 - Euler's totient function ([problem 1.5](#))
- congruences ([problem 1.6](#))
 - Euler-Fermat theorem
 - Fermat little theorem ([problems 1.8, 1.9](#))
 - q th power lemma ([problem 1.7](#))
 - Chinese remainder theorem ([problems 1.10, 1.11](#))

2 Diophantine Equations

An equation with integer variables is called a *Diophantine equation* (不定方程). It is one kind of problems in number theory. Usually, the solutions of a Diophantine equation are obvious. The most difficult part is to prove that all other cases are not possible.

In this first section in Diophantine equations, we go through some well-known results in this topic. We shall also use some of the techniques introduced in section 1 to solve some simpler problems. More will be discussed in section 7.

To begin with, we consider linear Diophantine equations.

Proposition 2.1. Let a, b, c be integers. Then $am + bn = c$ has integer solutions in m, n iff $d \mid c$ where $d = (a, b)$. In that case, if $(m, n) = (m_0, n_0)$ is a solution, then all solutions are given by

$$m = m_0 + \frac{b}{d} \cdot k \quad \text{and} \quad n = n_0 - \frac{a}{d} \cdot k \quad (1)$$

where k is any integer.

Proof. If $d \mid c$, let $c = dc_1$ where $c_1 \in \mathbb{Z}$. By Bézout's identity, there are integers t and u such that $d = at + bu$. Then $(m, n) = (tc_1, uc_1)$ is a solution to $am + bn = c$ as

$$a(tc_1) + b(uc_1) = (at + bu)c_1 = dc_1 = c.$$

Conversely, if $am + bn = c$ is solvable, as $d \mid a$ and $d \mid b$, we have $d \mid am + bn = c$.

Now, suppose $(m, n) = (m_0, n_0)$ is a solution to $am + bn = c$. It is easy to check that all pairs of (m, n) given by (1) are solutions to the linear equation. Consider any solution (m, n) . We have

$$\begin{aligned} am_0 + bn_0 &= c = am + bn \\ \Rightarrow a(m_0 - m) &= b(n - n_0) \\ \Rightarrow \frac{a}{d}(m_0 - m) &= \frac{b}{d}(n - n_0). \end{aligned}$$

Since $\left(\frac{a}{d}, \frac{b}{d}\right) = 1$, we must have $\frac{b}{d} \mid m_0 - m$. We write $m = m_0 + \frac{b}{d} \cdot k$ for some $k \in \mathbb{Z}$.

Then $n = n_0 - \frac{a}{d} \cdot k$. In other words, every solution must be of the form given by (1).

This completes the proof of the proposition. $\square$

The linear case is not an interesting result. But it is a fundamental fact of Diophantine equations so that one should know this result. Also, the use of Bézout's identity is a useful technique.

Proposition 2.2. Let $a_1, a_2, \dots, a_n$ be nonzero integers with sum 0 and let p be a prime number. Let p^{α_j} be the largest power of p dividing a_j for each $j = 1, 2, \dots, n$. Then the smallest two integers among $\alpha_1, \alpha_2, \dots, \alpha_n$ are equal.

Proof. WLOG assume $0 \leq \alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_n$. Suppose on the contrary that $\alpha_1 < \alpha_2$. We write $a_j = b_j \cdot p^{\alpha_j}$ for each $j = 1, 2, \dots, n$. Then $p \nmid b_j$ by the definition of α_j . Note that

$$\begin{aligned} & b_1 \cdot p^{\alpha_1} + b_2 \cdot p^{\alpha_2} + \dots + b_n \cdot p^{\alpha_n} = 0 \\ \Rightarrow & b_1 + b_2 \cdot p^{\alpha_2 - \alpha_1} + b_3 \cdot p^{\alpha_3 - \alpha_1} + \dots + b_n \cdot p^{\alpha_n - \alpha_1} = 0. \end{aligned}$$

Every term except the first one is a multiple of p since $\alpha_j > \alpha_1$ for any $j > 1$. However, the first term b_1 is not divisible by p . This is clearly impossible as the left-hand side cannot be divisible by p . Thus, the assumption $\alpha_1 < \alpha_2$ is not true. Hence, we must have $\alpha_1 = \alpha_2$ as desired. $\square$

Almost all Diophantine equations can be rewritten in the form $a_1 + a_2 + \dots + a_n = 0$. This proposition simply says we can compare the exponents of a prime dividing each term to get some information.

Proposition 2.3. Let a, b, c, n be positive integers such that $n \geq 2$ and $ab = c^n$. If $(a, b) = 1$, then $a = u^n$ and $b = v^n$ for some positive integers u, v .

Proof. If at least one of a, b is 1, the result is obvious.

Assume $a, b > 1$. Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ and $b = q_1^{\beta_1} q_2^{\beta_2} \dots q_t^{\beta_t}$ be their prime factorizations. Note that p_i, q_j 's are all distinct as $(a, b) = 1$. The prime factors of c are exactly the p_i 's and the q_j 's from the equation. Hence, $c = p_1^{\gamma_1} p_2^{\gamma_2} \dots p_s^{\gamma_s} q_1^{\delta_1} q_2^{\delta_2} \dots q_t^{\delta_t}$. By comparing the two sides of $ab = c^n$, we get $\alpha_i = n\gamma_i$ and $\beta_j = n\delta_j$ for each $1 \leq i \leq s$ and $1 \leq j \leq t$. Therefore, we can take $u = p_1^{\gamma_1} p_2^{\gamma_2} \dots p_s^{\gamma_s}$ and $v = q_1^{\delta_1} q_2^{\delta_2} \dots q_t^{\delta_t}$ so that $a = u^n$ and $b = v^n$. $\square$

This is a useful fact since an equation of the form $ab = c^n$ can be decomposed into two equations if the conditions are satisfied. Note that there are many variations of this result. For example, if $(a, b) = 2$ instead of a, b being coprime, then we have $a = 2u^n$ and $b = 2^{n-1}v^n$ (or $a = 2^{n-1}u^n$ and $b = 2v^n$) for some $u, v \in \mathbb{N}$. The only trick is to compare the prime factorizations of both sides.

Before we proceed, recall from [problem 1.6](#) that there are some facts about square numbers and congruences. We note that square numbers can only belong to a few residue classes modulo small integers. In particular, we have $m^2 \equiv 0, 1 \pmod{3}$, $m^2 \equiv 0, 1 \pmod{4}$ and $m^2 \equiv 0, 1, 4 \pmod{8}$ for $m \in \mathbb{Z}$. This kind of fact is useful in solving some Diophantine equations. Indeed, by considering these remainders, we may restrict the variables to certain residue classes for further consideration. This gives us useful information to start.

We now go through some quadratic equations.

Theorem 2.1. (Pythagorean triples(畢氏三元數)) The positive integer solutions to

$$a^2 + b^2 = c^2 \quad (1)$$

are given by $(a, b, c) = (d(m^2 - n^2), 2dmn, d(m^2 + n^2))$ where d, m, n are positive integers, and $m > n$ are relatively prime integers with different parities. (Note that a and b can be interchanged.)

Proof. Let d be the greatest common divisor of a, b, c . By dividing both sides of (1) by d^2 , we get essentially the same equation. Therefore, we may assume $d = 1$. Then we can multiply d to any solution we find to get the general solution.

If any two of a, b, c have a common prime factor p , then p also divides the remaining variable by considering modulo p in (1). This contradicts $d = 1$. Hence, a, b, c are pairwise relatively prime. In particular, at least one of a, b is odd. If both a, b are odd, then c is even. However, in that case we have

$$1 + 1 \equiv a^2 + b^2 = c^2 \equiv 0 \pmod{4},$$

which is a contradiction. Therefore, we may assume b is even while a, c are odd.

Rewrite the equation as $b^2 = c^2 - a^2 = (c - a)(c + a)$. Note that

$$(c - a, c + a) = (c - a + c + a, c + a) = (2c, c + a) = 2.$$

The last equality holds since $(c, c + a) = (c, a) = 1$ and $c + a$ is even. Let $c - a = 2v$ and $c + a = 2u$. Then $(u, v) = 1$ and $b^2 = 4uv$. The last equation is the same as $\left(\frac{b}{2}\right)^2 = uv$. By proposition 2.3, there exist $m, n \in \mathbb{N}$ such that $u = m^2$ and $v = n^2$. This gives

$$(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2).$$

Since we have assumed a, b, c are pairwise relatively prime, we need $(m, n) = 1$ and m, n have different parities.

Conversely, suppose $(a, b, c) = (d(m^2 - n^2), 2dmn, d(m^2 + n^2))$. Then

$$\begin{aligned} a^2 + b^2 &= (d(m^2 - n^2))^2 + (2dmn)^2 = d^2(m^4 - 2m^2n^2 + n^4 + 4m^2n^2) \\ &= d^2(m^4 + 2m^2n^2 + n^4) = (d(m^2 + n^2))^2 = c^2. \end{aligned}$$

□

The result of this theorem is important, but it is rarely used since it is too well-known. However, the tricks used in the proof are the main things that we would like to introduce. Firstly, in dealing with a homogeneous equation, i.e. an equation where every term has the same degree, we can take out the common factor and assume the

variables are coprime. This simplifies the work in many cases. Secondly, we try to factorize the expression whenever possible. Then we can consider the divisibility and prime factorization, etc. Checking modulo 4, for example, is also useful as discussed above.

Theorem 2.2. (Pell's equation(佩爾方程)) Let $\{a_n\}$ and $\{b_n\}$ be integer sequences defined by $a_n + b_n\sqrt{2} = (3 + 2\sqrt{2})^n$. Then the positive integer solutions to $a^2 - 2b^2 = 1$ are given by $(a, b) = (a_n, b_n)$ where n can be any positive integer.

Proof. Firstly, we show that every (a_n, b_n) is a solution. From the given relation $a_n + b_n\sqrt{2} = (3 + 2\sqrt{2})^n$, we find that $a_n - b_n\sqrt{2} = (3 - 2\sqrt{2})^n$. Therefore, we have

$$\begin{aligned} a_n^2 - 2b_n^2 &= (a_n + b_n\sqrt{2})(a_n - b_n\sqrt{2}) \\ &= (3 + 2\sqrt{2})^n(3 - 2\sqrt{2})^n \\ &= ((3 + 2\sqrt{2})(3 - 2\sqrt{2}))^n \\ &= 1^n = 1. \end{aligned}$$

This shows (a_n, b_n) is a solution. Indeed, this also suggests why we define such a sequence.

Now, suppose we have a positive integer solution $(a, b) = (c, d)$ not of the form as suggested. We choose the pair with the smallest c . We seek for contradiction by finding a yet smaller solution. The answer suggests we should consider

$$\frac{c + d\sqrt{2}}{3 + 2\sqrt{2}} = (c + d\sqrt{2})(3 - 2\sqrt{2}) = (3c - 4d) + (3d - 2c)\sqrt{2}.$$

That means we shall show that $(3c - 4d, 3d - 2c)$ is another solution. Indeed, this arises from

$$(3c - 4d)^2 - 2(3d - 2c)^2 = c^2 - 2d^2 = 1$$

as (c, d) is a solution. Note that $c^2 - 2d^2 = 1$ implies $c > \sqrt{2}d$, and hence $3c - 4d > 0$. Also, we have $c < 2d$, and hence $3c - 4d < c$. Furthermore, we show that $3d - 2c \neq 0$. If $3d = 2c$, then since $c^2 - 2d^2 = 1$ suggests c, d are relatively prime, we must have $c = 3$ and $d = 2$. This is a contradiction as $(a_1, b_1) = (3, 2)$ is a solution found. Therefore, changing the sign of $3d - 2c$ if necessary (note that it does not change $(3d - 2c)^2$), we find a solution with a smaller positive a . This contradicts the minimality of c . Thus, there is no solution other than those we have found. $\square$

More generally, we can consider Pell's equation of the form $a^2 - db^2 = 1$ where d is a [squarefree](#)(無平方因子) integer, i.e. an integer without square divisors except 1. Let (a_1, b_1) be the positive integer solution with the smallest a_1 (it is true that such a solution must exist, but this is not trivial). Then all positive integer solutions are given by (a_n, b_n) defined by $a_n + b_n\sqrt{d} = (a_1 + b_1\sqrt{d})^n$ where $n \in \mathbb{N}$. Therefore, there are infinitely many solutions to Pell's equation.

The trick that we use in the proof to show that all solutions are of the desired form is useful. If we want to show that there are no other solutions, then we can assume there exists such a solution and choose the smallest one. This yields a contradiction if we can find a smaller solution than that. We would talk more about this trick in section 7.

We now proceed to work on some problems on Diophantine equations. We first start with some fundamental problems.

Problem 2.1. Find all integers a and b satisfying

$$2ab + 3a - b = 15. \quad (1)$$

Analysis. The first technique in solving Diophantine equations is factorization. If we can rewrite the expression into the form $f_1 f_2 \cdots f_k = c$ for some constant c and some integer expressions f_j , then we can decompose this equation into k equations since a nonzero integer c can be expressed as a product of k integers in finitely many ways. If $c = 0$, then we also have the information that one of the f_j 's is zero. In any case, this method can simplify the given equation.

Solution. We can try to factorize to get

$$2ab + 3a - b = a(2b + 3) - b = a(2b + 3) - \frac{1}{2}(2b + 3) + \frac{3}{2} = \left(a - \frac{1}{2}\right)(2b + 3) + \frac{3}{2}.$$

Since we only want to work on integers, we multiply both sides of (1) by 2. Then (1) becomes

$$(2a - 1)(2b + 3) = 27.$$

Then we have $(2a - 1, 2b + 3) = (\pm 1, \pm 27), (\pm 3, \pm 9), (\pm 9, \pm 3), (\pm 27, \pm 1)$. This gives the solutions

$$(a, b) = (-13, -2), (-4, -3), (-1, -6), (0, -15), (1, 12), (2, 3), (5, 0), (14, -1).$$

□

Remarks. Even if we want to find positive integer solutions only, we should consider the negative divisors of an integer, unless we can show that the factors must be positive.

Problem 2.2. Find all positive integers a, b, c satisfying

$$abc - a - b - c + 1 = 0. \quad (1)$$

Analysis. This time we cannot factorize the expression. We note that in general the product abc should be much larger than $a + b + c$. Therefore, a, b, c cannot be too large in order that the equation is satisfied. Using algebraic tools to bound the variables is another common trick in solving Diophantine equations, as there are only finitely many positive integers with a certain upper bound.

Solution. WLOG assume $a \geq b \geq c$ (we always do this for a symmetric expression to reduce the amount of work). Then

$$abc = a + b + c - 1 < a + a + a = 3a.$$

Hence, $bc < 3$. This gives $(b, c) = (1, 1), (2, 1)$. When $(b, c) = (1, 1)$, equation (1) is not true. When $(b, c) = (2, 1)$, equation (1) implies $2a - a - 2 - 1 + 1 = 0$, i.e. $a = 2$. Therefore, the only solution up to permutation is $(a, b, c) = (2, 2, 1)$. $\square$

Problem 2.3. Find all integers a and b satisfying

$$a^2 + 20ab + b^2 = 2345. \quad (1)$$

Analysis. Sometimes if the equation has no integer solution, we can simply consider a particular congruence to prove this.

Solution. By considering modulo 3, equation (1) implies

$$a^2 + 2ab + b^2 \equiv 2 \pmod{3}.$$

This is the same as $(a + b)^2 \equiv 2 \pmod{3}$. However, we have mentioned that a square number must be congruent to 0 or 1 modulo 3. Therefore, the above congruence has no solution. This clearly implies equation (1) has no solution. $\square$

Remarks. Note that we do not have the converse. In other words, if an equation is solvable in every congruence, then it may not be solvable in integers.

Problem 2.4. Find all integers a and b satisfying

$$4a^2 - 3ab - b^2 + 1 = 0. \quad (1)$$

Analysis. This time the equation has some solutions. For example, $(a, b) = (0, \pm 1)$ is a simple one. Therefore, there is no hope to obtain any contradiction by considering modulo a certain integer. For quadratic equation, one way is to solve it using the quadratic formula for real solutions.

Solution. Viewing (1) as a quadratic equation in a , we have

$$a = \frac{3b \pm \sqrt{(-3b)^2 - 4(4)(-b^2 + 1)}}{2(4)} = \frac{3b \pm \sqrt{25b^2 - 16}}{8}.$$

Since a is an integer, $25b^2 - 16$ must be a square number. Let $25b^2 - 16 = k^2$ for some $k \in \mathbb{N}_0$. Then

$$16 = 25b^2 - k^2 = (5b - k)(5b + k).$$

Hence, $(5b - k, 5b + k) = (\pm 1, \pm 16), (\pm 2, \pm 8), (\pm 4, \pm 4), (\pm 8, \pm 2), (\pm 16, \pm 1)$. The only integer solutions with $k \geq 0$ are $(b, k) = (\pm 1, 3)$. This gives $a = 0$. Therefore, the solutions are $(a, b) = (0, \pm 1)$. $\square$

Problem 2.5. Find all integers a and b satisfying

$$a^2 + 4ab + b^2 = 1.$$

Analysis. We can do the same thing for this problem. However, we cannot do factorization as the previous problem. Indeed, we get an equation which is Pell's equation. But then we know how to solve it.

Solution. We have

$$a = \frac{-4b \pm \sqrt{(4b)^2 - 4(1)(b^2 - 1)}}{2} = -2b \pm \sqrt{3b^2 + 1}.$$

Since $a \in \mathbb{Z}$, we have $3b^2 + 1 = k^2$ for some $k \in \mathbb{N}_0$. This can be rewritten as

$$k^2 - 3b^2 = 1. \quad (1)$$

Observe that $(k, b) = (2, 1)$ is the smallest positive integer solution to (1). Therefore, the general positive integer solutions to (1) are given by $(k, b) = (k_n, b_n)$ where $k_n + b_n\sqrt{3} = (2 + \sqrt{3})^n$. Here, we introduce one method to find an explicit formula for k_n, b_n .

Note that $k_{n+1} + b_{n+1}\sqrt{3} = (k_n + b_n\sqrt{3})(2 + \sqrt{3})$. Therefore, we have the recurrence formulae

$$\begin{cases} k_{n+1} = 2k_n + 3b_n, \\ b_{n+1} = k_n + 2b_n. \end{cases}$$

These combine to give

$$b_{n+2} - 2b_{n+1} = k_{n+1} = 2k_n + 3b_n = 2(b_{n+1} - 2b_n) + 3b_n.$$

Hence, we get

$$b_{n+2} = 4b_{n+1} - b_n.$$

The roots to the characteristic equation $\lambda^2 - 4\lambda + 1 = 0$ are $\lambda = 2 \pm \sqrt{3}$. Therefore, $b_n = A(2 + \sqrt{3})^n + B(2 - \sqrt{3})^n$ for some constants A, B . Using $b_1 = 1$ and $b_2 = 4$, we easily find $A = \frac{1}{2\sqrt{3}}$ and $B = -\frac{1}{2\sqrt{3}}$. Therefore, $b_n = \frac{1}{2\sqrt{3}}((2 + \sqrt{3})^n - (2 - \sqrt{3})^n)$.

Then

$$\begin{aligned} k_n &= b_{n+1} - 2b_n \\ &= \frac{1}{2\sqrt{3}}((2 + \sqrt{3})^{n+1} - (2 - \sqrt{3})^{n+1} - 2((2 + \sqrt{3})^n - (2 - \sqrt{3})^n)) \\ &= \frac{1}{2\sqrt{3}}(\sqrt{3}(2 + \sqrt{3})^n + \sqrt{3}(2 - \sqrt{3})^n) \\ &= \frac{1}{2}((2 + \sqrt{3})^n + (2 - \sqrt{3})^n). \end{aligned}$$

Now, for positive b , we have a list of solutions $(a, b) = (a_n^\pm, b_n)$ where

$$a_n^\pm = -2b_n \pm k_n = -\frac{1}{\sqrt{3}}((2 + \sqrt{3})^n - (2 - \sqrt{3})^n) \pm \frac{1}{2}((2 + \sqrt{3})^n + (2 - \sqrt{3})^n).$$

For negative b , we have another list of solutions $(a, b) = (c_n^\pm, -b_n)$ where

$$c_n^\pm = 2b_n \pm k_n = \frac{1}{\sqrt{3}}((2 + \sqrt{3})^n - (2 - \sqrt{3})^n) \pm \frac{1}{2}((2 + \sqrt{3})^n + (2 - \sqrt{3})^n).$$

Lastly, we also have the solution $(a, b) = (\pm 1, 0)$ corresponding to $b = 0$ and $k = 1$. $\square$

Remarks. Alternatively, we can rewrite the given equation as $(a + 2b)^2 - 3b^2 = 1$ and solve this Pell's equation directly.

Usually, we do not need to use Pell's equation. The reason is that it has infinitely many solutions and it is tedious to write down the explicit form of all solutions. If one obtains Pell's equation as an intermediate step, then one should better find another method to solve the problem unless the solution is obvious once Pell's equation is solved.

Problem 2.6. Find all integers a, b, c satisfying

$$a^2 + b^2 = 2c^2. \quad (1)$$

Analysis. The equation looks like the one for Pythagorean triples. However, the same trick cannot be applied since we cannot factorize the expression easily. The closest way is to rewrite (1) as $(a - c)(a + c) = (c - b)(c + b)$, but this does not lead to a solution in an obvious way. For the following solution, we introduce another useful technique, namely, substitution. Sometimes changing an equation to another form may simplify the problem.

Solution. Since $a^2 + b^2$ is even, a and b have the same parity. Let $u = \frac{a+b}{2}$ and $v = \frac{a-b}{2}$ so that $u, v \in \mathbb{Z}$. This implies $a = u + v$ and $b = u - v$. Hence, equation (1) becomes

$$(u + v)^2 + (u - v)^2 = 2c^2.$$

This is the same as $u^2 + v^2 = c^2$. Then we can use the result of Pythagorean triples to get

$$(u, v, c) = (d(m^2 - n^2), 2dmn, d(m^2 + n^2)) \text{ or } (2dmn, d(m^2 - n^2), d(m^2 + n^2))$$

where $d, m, n \in \mathbb{Z}$. (Note that we allow d, m, n to be negative so that we can obtain all integer solutions. It is not necessary to specify that m, n are relatively prime and have different parities. This means some solutions can be expressed in this form in more than one way, but the important thing is to make sure that we have got all the solutions.) Thus,

$$(a, b, c) = (d(m^2 + 2mn - n^2), \pm d(m^2 - 2mn - n^2), d(m^2 + n^2)).$$

$\square$

Problem 2.7. Suppose an acute triangle has integer side lengths and integer area. Find the smallest possible perimeter of this triangle.

Analysis. To relate the side lengths to the area, we can use Heron's formula. This gives us a Diophantine equation. This time we are not going to solve the equation (note that it has infinitely many solutions), but to find the smallest solution in some sense. Therefore, we can just test from the smallest case. The acute condition can be checked easily after solutions are found.

Solution. Let a, b, c be the side lengths and let S be the area. By Heron's formula, we have

$$(a + b + c)(a + b - c)(a - b + c)(-a + b + c) = 16S^2. \quad (1)$$

Since the only explicit integer appearing in (1) is 16, which is a power of 2, we first consider the parity on both sides. Note that each factor on the left-hand side of (1) has the same parity as $a + b + c$. As their product is even, $a + b + c$ must be even.

Case 1. exactly one of a, b, c is even

WLOG assume $2 \mid c$ and $2 \nmid a, b$. Let $a = 2\ell - 1$, $b = 2m - 1$ and $c = 2n$ where $\ell, m, n \in \mathbb{N}$. Then equation (1) becomes

$$(\ell + m + n - 1)(\ell + m - n - 1)(\ell - m + n)(-\ell + m + n) = S^2. \quad (2)$$

We write $w = \ell + m + n - 1$, $x = \ell + m - n - 1$, $y = \ell - m + n$ and $z = -\ell + m + n$. Then $wxyz$ is a square number and $w > x, y, z > 0$. This shows $w > 1$. Our goal is to minimize w .

- If w is a prime, then $w \mid S^2$ implies $w^2 \mid S^2$ since every prime factor of a square number must occur an even number of times. Then w has to divide one of x, y, z , which cannot be the case as $w > x, y, z > 0$.
- If $w = 4$, as $w \equiv x \pmod{2}$ and $w > x > 0$, we must have $x = 2$. Also, y, z are odd as $w \not\equiv y, z \pmod{2}$. Then $wxyz$ is divisible by 8 but not 16. This is impossible since it is a square number.
- If $w = 6$, then similarly x is even and y, z are odd. Since $3 \mid wxyz$, we need $9 \mid wxyz$. Note that $3 \nmid x$ since $x = 2$ or 4 . Then exactly one of y, z is divisible by 3, and hence one of them is equal to 3. This easily yields the only possible solutions $(w, x, y, z) = (6, 2, 3, 1)$ or $(6, 2, 1, 3)$. We use these to solve (2) and hence (1). We find that $(a, b, c) = (5, 3, 4)$ up to permutation. This is a right-angled triangle and hence is not acute.
- If $w = 8$, we can find one solution $(w, x, y, z) = (8, 2, 3, 3)$ by using similar consideration (there are other solutions, but this does not matter). This gives $(a, b, c) = (5, 5, 6)$ up to permutation. It is an isosceles triangle. It suffices to check that the angle opposite to the base is acute. Indeed, this follows from the cosine formula as $\frac{5^2 + 5^2 - 6^2}{2 \cdot 5 \cdot 5} > 0$. In this case, we find a triangle with perimeter 16.

Case 2. all of a, b, c are even

Let $a = 2\ell$, $b = 2m$ and $c = 2n$ where $\ell, m, n \in \mathbb{N}$. Then equation (1) becomes

$$(\ell + m + n)(\ell + m - n)(\ell - m + n)(-\ell + m + n) = S^2. \quad (3)$$

We write $w = \ell + m + n$, $x = \ell + m - n$, $y = \ell - m + n$ and $z = -\ell + m + n$. Then $wxyz$ is a square number and $w > x, y, z > 0$. This shows $w > 1$. Again, our goal is to minimize w .

- Similar to **case 1**, w cannot be a prime.
- If w is even, then so are x, y, z . This shows $16 \mid S^2$. Then (3) can be rewritten as

$$(\ell + m + n)(\ell + m - n)(\ell - m + n)(-\ell + m + n) = 16 \left(\frac{S}{4} \right)^2.$$

This is essentially the same as the original equation (1) as $\frac{S}{4}$ is a positive integer. Then any solution to this cannot be the smallest one.

- For the remaining cases, we must have $w \geq 9$. Then $a + b + c = 2w \geq 18$. So this case cannot give the smallest solution.

Therefore, the smallest possible perimeter is 16. □

Remarks. The main fact that we have used in this solution is that the exponents of primes in the prime factorization of a square number are even. In particular, whenever a prime p divides S^2 , then $p \mid S$ and $p^2 \mid S^2$. This is a useful fact related to square numbers. Note that there is an analogous result for higher powers of integers.

Problem 2.8. (Baltic Way 2016 Problem 1) Find all pairs of primes (p, q) such that

$$p^3 - q^5 = (p + q)^2. \quad (1)$$

Analysis. One simple way to solve Diophantine equations in prime variables is to show that one of the primes is divisible by some integer greater than 1 by considering congruences. Since this immediately tells us the only possible choice for that prime. Usually, the solutions, if exist, are small primes like 2, 3, 5. Therefore, it is natural to consider modulo these primes.

Solution. We consider modulo 3. By the Fermat little theorem, we have

$$p^3 - q^5 \equiv p - q \pmod{3}.$$

Also, $(p + q)^2 \equiv 0, 1 \pmod{3}$.

- If $(p+q)^2 \equiv 0 \pmod{3}$, then $p-q \equiv 0 \pmod{3}$ from (1). Hence, $p \equiv q \pmod{3}$ and

$$0 \equiv (p+q)^2 \equiv (2p)^2 \equiv p^2 \pmod{3}.$$

This implies $p \equiv 0 \pmod{3}$ and hence $p = q = 3$. This is not a solution to (1).

- If $(p+q)^2 \equiv 1 \pmod{3}$, then $p-q \equiv 1 \pmod{3}$ from (1). If none of p, q is divisible by 3, then we must have $p \equiv 2 \pmod{3}$ and $q \equiv 1 \pmod{3}$. However, this contradicts $(p+q)^2 \equiv 1 \pmod{3}$. Therefore, one of p, q is divisible by 3, and hence is equal to 3.

– If $p = 3$, then equation (1) becomes $27 - q^5 = (3+q)^2$. Note that the right-hand side is positive while the left-hand side is negative as $q \geq 2$. Thus, there is no solution.

– If $q = 3$, then equation (1) becomes $p^3 - 243 = (p+3)^2$. This is a polynomial equation in p . It can be rewritten as $p^3 - p^2 - 6p - 252 = 0$. It is not hard to factorize and get

$$(p-7)(p^2 + 6p + 36) = 0.$$

The quadratic equation $p^2 + 6p + 36 = 0$ has no real solution. The only solution is $p = 7$, i.e. $(p, q) = (7, 3)$.

□

Problem 2.9. Find all positive integers m and n satisfying

$$n^3 = n^2 + m^3n + m^4. \quad (1)$$

Analysis. If m, n are relatively prime, then the problem is rather simple. For example, from $n \mid m^4$, we may conclude $n = 1$ in that case. For the general case, we usually take out the greatest common divisor of m, n so that we have a pair of relatively prime integers. Then we can use a similar consideration, perhaps with the help of [proposition 2.2](#) to get more information about the prime components of the integers.

Solution. Let $d = (m, n)$. Let $m = da$ and $n = db$ so that $(a, b) = 1$. Dividing both sides of (1) by d^2 , the equation becomes

$$db^3 = b^2 + d^2a^3b + d^2a^4. \quad (2)$$

From (2), we note that $b \mid d^2a^4$. Since $(a, b) = 1$, this implies $b \mid d^2$. Also, (2) implies $d \mid b^2$. Therefore, the prime divisors of b and d are exactly the same.

Consider any prime p dividing both b and d . Then $p \nmid a$ as $(a, b) = 1$. Let $v_p(\ell)$ be the greatest nonnegative integer for which $p^{v_p(\ell)}$ divides a nonzero integer ℓ . We note

that

$$\begin{aligned}v_p(db^3) &= v_p(d) + 3v_p(b), \\v_p(b^2) &= 2v_p(b), \\v_p(d^2a^3b) &= 2v_p(d) + v_p(b), \\v_p(d^2a^4) &= 2v_p(d).\end{aligned}$$

By [proposition 2.2](#), the smallest two of these numbers are equal. From the relations $v_p(d) + 3v_p(b) > 2v_p(b)$ and $2v_p(d) + v_p(b) > 2v_p(d)$, the smallest two must be $2v_p(b)$ and $2v_p(d)$. Therefore, $v_p(b) = v_p(d)$. Since this holds for every prime divisor p , we have $b = d$. Then equation [\(2\)](#) is simplified to

$$b^2 = 1 + a^3b + a^4.$$

Rewrite this as a quadratic equation in b , i.e. $b^2 - a^3b - (a^4 + 1) = 0$. This gives $b = \frac{a^3 \pm \sqrt{a^6 + 4a^4 + 4}}{2}$. As b is an integer, we have

$$a^6 + 4a^4 + 4 = k^2 \tag{3}$$

for some $k \in \mathbb{N}_0$. The following gives a standard way to find all solutions to [\(3\)](#). Firstly, we have one solution $(a, k) = (1, 3)$. Suppose $a \geq 2$. Then

$$\begin{aligned}(a^3 + 2a - 1)^2 &= a^6 + 4a^4 - 2a^3 + 4a^2 - 4a + 1 \\&< a^6 + 4a^4 + 4 \\&< a^6 + 4a^4 + 4a^2 = (a^3 + 2a)^2\end{aligned}$$

since the first inequality is the same as $2a^3 - 4a^2 + 4a + 3 > 0$, which is true from $2a^3 \geq 4a^2$. This shows k^2 is a square number strictly between two consecutive square numbers. This is impossible. Therefore, $a \geq 2$ yields no solution.

This concludes the only solution is $(a, k) = (1, 3)$. One can easily check that the corresponding pair $(m, n) = (2, 4)$ is a solution. $\square$

Problem 2.10. (IMO Shortlist 2006 N1/Problem 4) Determine all pairs (x, y) of integers such that

$$1 + 2^x + 2^{2x+1} = y^2. \tag{1}$$

Analysis. Since there are some powers of 2, we should probably consider the exponents of 2 of each term. It is better to put the 1 on the right-hand side first and do factorization. Otherwise, we only know that y is odd in general. Note that we need to consider the case when x is negative.

Solution. If $x < -1$, then the left-hand side of [\(1\)](#) has denominator $2^{2(-x)-1}$ when written in the simplest form, which means it is not an integer. If $x = -1$, we get $1 + 2^{-1} + 2^{-1} = 2$, which is not a square. If $x = 0$, we get the solutions $(x, y) = (0, \pm 2)$.

It remains to consider $x > 0$, in which case we may assume $y \geq 0$ by changing its sign if necessary.

Note that y is odd. Equation (1) can be rewritten as

$$2^x + 2^{2x+1} = (y-1)(y+1). \quad (2)$$

Consider the exponent of 2 of each term in (2). By [proposition 2.2](#), $(y-1)(y+1)$ must be equal to 2^x times an odd number. As $(y-1, y+1) = (y-1, 2) = 2$, one of $y-1$ and $y+1$ is 2 times an odd number, while the other is 2^{x-1} times an odd number.

Case 1. $y-1 = 2^{x-1}a$ for some odd $a \in \mathbb{N}$

Then (2) becomes $1 + 2^{x+1} = a(2^{x-2}a + 1)$. Note that a cannot be too large. Indeed, if $a \geq 3$, then we get the contradiction

$$a(2^{x-2}a + 1) \geq 2^{x-2}a^2 + 3 > 2^{x+1} + 3 > 2^{x+1} + 1.$$

As a is odd, we have $a = 1$, which gives $2^{x+1} = 2^{x-2}$. There is no solution.

Case 2. $y+1 = 2^{x-1}b$ for some odd $b \in \mathbb{N}$

Then (2) becomes $1 + 2^{x+1} = b(2^{x-2}b - 1)$. This can be rewritten as

$$2^{x-2}(b^2 - 8) = b + 1. \quad (3)$$

The left-hand side must be positive, so we have $b \geq 3$.

If $x \geq 3$, then

$$b + 1 = 2^{x-2}(b^2 - 8) \geq 2(b^2 - 8) \geq 6b - 16$$

so that $b \leq 3$. Therefore, we have $b = 3$. In that case (3) becomes $2^{x-2} = 4$. This gives $x = 4$ and hence $y = 23$.

If $x = 2$, then (3) becomes $b^2 = b + 9$, which has no solution. Also x cannot be 1 since $y + 1 = 2^{x-1}b$ should be even.

Therefore, the solutions are $(x, y) = (0, \pm 2), (4, \pm 23)$. □

Problem 2.11. Suppose m and n are positive integers with $m \geq n$. Let a, b, c, d be the permutation of the sum $m + n$, the difference $m - n$, the product mn and the quotient $\frac{m}{n}$ such that $a \leq b \leq c \leq d$. Find all pairs of m, n for which

$$b - a = \frac{d}{c}.$$

Analysis. The first step is to determine what a, b, c, d are. In general, it should be

quotient < difference < sum < product.

But we need to take care of the exceptional cases.

Afterwards, we shall obtain a Diophantine equation in m, n . It is a quadratic equation in m and one may solve it using the same tricks as in [problems 2.4](#) and [2.5](#). We will slightly mention the ideas along this way. However, this time we shall apply the trick used in [problem 2.9](#) and examine the prime divisors of m, n . Both methods are useful to solve problems of this kind.

Solution. If $n = 1$, then $(a, b, c, d) = (m - 1, m, m, m + 1)$. The equation is equivalent to $1 = \frac{m+1}{m}$, which has no solution.

If $m = n \geq 2$, then $(a, b, c, d) = (0, 1, 2m, m^2)$. The equation becomes $1 = \frac{m}{2}$. The only solution is $m = 2$. It is obvious that $(m, n) = (2, 2)$ is a solution.

If $m = n + 1$ with $n \geq 2$, then $(a, b, c, d) = \left(1, \frac{n+1}{n}, 2n+1, n(n+1)\right)$. The equation becomes $\frac{1}{n} = \frac{n(n+1)}{2n+1}$, which is the same as $n^3 + n^2 - 2n - 1 = 0$. However, we have

$$n^3 + n^2 - 2n - 1 = n(n^2 - 2) + (n^2 - 1) \geq 2(2) + 3 > 0$$

as $n \geq 2$. There is no solution.

In general, if $m \geq n + 2$ with $n \geq 2$, then $(a, b, c, d) = \left(\frac{m}{n}, m - n, m + n, mn\right)$ as

$$\begin{aligned} \frac{m}{n} &= \frac{m-n}{n} + 1 \leq \frac{m-n}{2} + 1 \leq m-n, \\ m-n &< m+n, \\ m+n &= mn - (m-1)(n-1) + 1 \leq mn. \end{aligned}$$

Clearing denominators, the equation $m - n - \frac{m}{n} = \frac{mn}{m+n}$ can be rewritten as

$$m^2n = m^2 + mn^2 + mn + n^3. \quad (1)$$

If one views it as a quadratic equation in m , then the discriminant $n^2(5n^2 - 2n + 1)$ should be a perfect square. This gives $(2n)^2 + (n-1)^2 = k^2$ for some $k \in \mathbb{N}_0$, and then we can use the result of Pythagorean triples to work out a proof.

Alternatively, observe that (1) implies $m \mid n^3$ and $n \mid m^2$. So m, n have the same prime divisors. For each prime p dividing m, n , let $v_p(\ell)$ be the greatest nonnegative integer for which $p^{v_p(\ell)}$ divides a nonzero integer ℓ . Then we have

$$\begin{aligned} v_p(m^2n) &= 2v_p(m) + v_p(n), \\ v_p(m^2) &= 2v_p(m), \\ v_p(mn^2) &= v_p(m) + 2v_p(n), \\ v_p(mn) &= v_p(m) + v_p(n), \\ v_p(n^3) &= 3v_p(n). \end{aligned}$$

By [proposition 2.2](#), the smallest two among these numbers are equal. Note that

$$v_p(m) + v_p(n) < 2v_p(m) + v_p(n), v_p(m) + 2v_p(n)$$

and

$$v_p(m) + v_p(n) < \frac{1}{2}(2v_p(m) + 3v_p(n)).$$

Therefore, $v_p(m) + v_p(n)$ must be one of the smallest two numbers. The other one can only be $2v_p(m)$ or $3v_p(n)$. Thus, we have $v_p(m) = v_p(n)$ or $v_p(m) = 2v_p(n)$.

From this, we can partition the prime divisors of m, n into 2 groups. Then there exist coprime positive integers P, Q such that $n = PQ$ and $m = PQ^2$. After simplification, equation (1) becomes

$$PQ^2 = Q + PQ + 1 + P. \quad (2)$$

Note that $(P, Q) = 1$ and $n = PQ > 1$ implies $P \neq Q$.

Suppose $P > Q$. From (2), we have $P \mid Q + 1$. This yields $P \leq Q + 1$. Therefore, we have $P = Q + 1$. Then (2) gives $Q^3 - 3Q - 2 = 0$, i.e. $(Q + 1)^2(Q - 2) = 0$. The only positive solution is $Q = 2, P = 3$. This gives $(m, n) = (12, 6)$.

Suppose $P < Q$. From (2), we have $Q \mid P + 1$. This yields $Q \leq P + 1$. Therefore, we have $Q = P + 1$. Then (2) gives $P^3 + P^2 - 2P - 2 = 0$, i.e. $(P + 1)(P^2 - 2) = 0$. There is no positive integer solution.

By checking, the only solutions are $(m, n) = (2, 2), (12, 6)$. □

Here sums up various basic techniques in solving Diophantine equations.

- factorization ([problems 2.1, 2.4, 2.10](#))
- deriving bounds ([problems 2.2, 2.9, 2.10](#))
- considering congruences ([problems 2.3, 2.8](#))
- considering prime components ([problems 2.7, 2.9, 2.10, 2.11](#))
- Pell's equation ([problem 2.5](#))
- Pythagorean triples ([problem 2.6](#))

3 More on Divisibilities

In this section, we go through more difficult divisibility problems. Before that, we introduce two more useful results involving powers of integers. As in [problems 2.9](#) and [2.11](#), we shall use the following common notations.

Definition 3.1. For any prime p and nonzero integer n , the *p -adic valuation* (p 進賦値) of n is the greatest nonnegative integer $v_p(n)$ for which $p^{v_p(n)}$ divides n . We also write $p^\alpha \parallel n$ if $\alpha = v_p(n)$.

Theorem 3.1. ([Lifting the exponent lemma](#)) Let p be an odd prime, let n be a positive integer, and let a, b be integers not divisible by p .

- (a) If $p \mid a - b$ and $a \neq b$, then $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$.
- (b) If n is odd, $p \mid a + b$ and $a \neq -b$, then $v_p(a^n + b^n) = v_p(a + b) + v_p(n)$.

Proof. We only prove part (a) here since the proofs of the two parts are similar. Let $p^\alpha \parallel n$ and $p^\beta \parallel a - b$. We need to prove $p^{\alpha+\beta} \parallel a^n - b^n$. We shall show this by induction on α .

When $\alpha = 0$, we have

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1}).$$

Note that

$$a^{n-1} + a^{n-2}b + \cdots + b^{n-1} \equiv a^{n-1} + a^{n-1} + \cdots + a^{n-1} = na^{n-1} \not\equiv 0 \pmod{p}$$

as $p \nmid n, a$. Therefore, $v_p(a^n - b^n) = v_p(a - b) = \beta$. This establishes the base case.

Assume the case $\alpha = k$ holds. When $\alpha = k + 1$, let $\ell = \frac{n}{p}$ and let $s = a^\ell$, $t = b^\ell$. Then we have

$$a^n - b^n = s^p - t^p = (s - t)(s^{p-1} + s^{p-2}t + \cdots + t^{p-1}). \quad (1)$$

Note that $p^k \parallel \ell$. By the inductive hypothesis, $p^{\alpha+\beta} \parallel a^\ell - b^\ell = s - t$. Let $s - t = p^{\alpha+\beta}c$ where $p \nmid c$. For the second factor of (1), using the binomial theorem, we find that

$$\begin{aligned} & s^{p-1} + s^{p-2}t + \cdots + t^{p-1} \\ &= (t + p^{\alpha+\beta}c)^{p-1} + (t + p^{\alpha+\beta}c)^{p-2}t + \cdots + t^{p-1} \\ &\equiv (t^{p-1} + (p-1)t^{p-2}p^{\alpha+\beta}c) + (t^{p-1} + (p-2)t^{p-2}p^{\alpha+\beta}c) + \cdots + t^{p-1} \\ &= pt^{p-1} + ((p-1) + (p-2) + \cdots + 1)t^{p-2}p^{\alpha+\beta}c \\ &= pt^{p-1} + p \cdot \frac{p-1}{2}t^{p-2}p^{\alpha+\beta}c \\ &\equiv p \pmod{p^2}. \end{aligned}$$

Here, we have used the conditions that $a \equiv b \pmod{p}$ (hence $\beta \geq 1$), p is odd (hence $\frac{p-1}{2}$ is an integer) and $p \nmid b$ (hence we can apply the Fermat little theorem to the term t^{p-1}). From this, the second factor in (1) is divisible by p but not p^2 . Therefore, $p^{(\alpha+1)+\beta} \parallel a^n - b^n$. By induction, the result is proved. $\square$

The following gives the analogous result for the prime 2.

Theorem 3.2. ([Lifting the exponent lemma](#)) Let n be a positive integer, and let a, b be distinct odd integers.

- (a) If n is odd, then $v_2(a^n - b^n) = v_2(a - b)$.
- (b) If n is even, then $v_2(a^n - b^n) = v_2(a^2 - b^2) + v_2(n) - 1$.

Proof.

- (a) Suppose n is odd. We have

$$a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1}).$$

Since a, b, n are odd, we have

$$a^{n-1} + a^{n-2}b + \dots + b^{n-1} \equiv n \equiv 1 \pmod{2}.$$

This shows $v_2(a^n - b^n) = v_2(a - b)$.

- (b) Suppose n is even. Let $n = 2^\alpha m$ where m is odd, so that $\alpha \geq 1$. We prove the assertion by induction on α as in the proof of [theorem 3.1](#). When $\alpha = 1$, we have

$$a^{2m} - b^{2m} = (a^2 - b^2)(a^{2(m-1)} + a^{2(m-2)}b^2 + \dots + b^{2(m-1)}).$$

Again, the second factor is odd. Therefore, $v_2(a^{2m} - b^{2m}) = v_2(a^2 - b^2)$. This establishes the base case.

Assume the case $\alpha = k$ holds. When $\alpha = k + 1$, let $\ell = \frac{n}{2}$. Then we have

$$a^n - b^n = (a^\ell)^2 - (b^\ell)^2 = (a^\ell - b^\ell)(a^\ell + b^\ell).$$

Note that $2^k \parallel \ell$. By the inductive hypothesis, $v_2(a^\ell - b^\ell) = v_2(a^2 - b^2) + k - 1$. Since a, b are odd, we have $a^2 \equiv b^2 \equiv 1 \pmod{8}$. Therefore, $v_2(a^\ell - b^\ell) \geq v_2(a^2 - b^2) \geq 3$. If $4 \mid a^\ell + b^\ell$, then $4 \mid (a^\ell + b^\ell) + (a^\ell - b^\ell) = 2a^\ell$. This is impossible since a is odd. This shows $2 \parallel a^\ell + b^\ell$ as a, b are odd. Then

$$v_2(a^n - b^n) = v_2(a^\ell - b^\ell) + v_2(a^\ell + b^\ell) = v_2(a^2 - b^2) + k$$

as desired. By induction, we have completed the proof. $\square$

Theorem 3.3. (Zsigmondy's theorem(席格蒙迪定理)) Let $a > b \geq 1$ be relatively prime integers and let n be a positive integer. Then there exists a prime p such that $p \mid a^n - b^n$ and $p \nmid a^k - b^k$ for each $k = 1, 2, \dots, n-1$ except the following cases:

- (i) $a - b = 1$ and $n = 1$;
- (ii) $a + b$ is a power of 2 and $n = 2$;
- (iii) $a = 2, b = 1$ and $n = 6$.

Proof. When $n = 1$, $a - b$ has a prime divisor iff $a - b \neq 1$. Hence, only (i) is the exceptional case.

When $n = 2$, we have $a^2 - b^2 = (a - b)(a + b)$. Note that

$$(a - b, a + b) = (a - b, 2a) = (a - b, 2).$$

If a, b have different parities, then $(a - b, a + b) = 1$. As $a + b > 1$, it must contain a prime divisor not dividing $a - b$. If a, b have the same parity, then they must both be odd. Then $(a - b, a + b) = 2$. In this case, every prime factor of $a^2 - b^2$ divides $a - b$ iff $a + b$ is a power of 2, which is the case given in (ii).

In the following, we shall denote $f(m) = a^m - b^m$ for any $m \in \mathbb{N}$ for convenience.

Suppose $n \geq 3$ and every prime divisor of $f(n)$ divides some $f(k)$ where $1 \leq k < n$. For any $1 \leq k < n$, by using the same trick as in [problem 1.2](#), we get

$$(a^k - b^k, a^n - b^n) = a^{(k,n)} - b^{(k,n)}.$$

Therefore, those primes which divide both $f(k)$ and $f(n)$ must divide $f(d)$ where d is some proper divisor of n . In particular, if we let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ be the prime factorization, then the prime divisors of $f(n)$ must divide $f(d)$ for some d of the form $\frac{n}{p_j}$.

We first consider the special case when $n = 2^{\alpha_1}$ where $\alpha_1 \geq 2$. Then we have

$$a^n - b^n = (a^{\frac{n}{2}} - b^{\frac{n}{2}})(a^{\frac{n}{2}} + b^{\frac{n}{2}}).$$

As in the case $n = 2$, the greatest common divisor of the 2 factors is 1 or 2. In order to find the exceptional cases, we only need to consider the case when $a^{\frac{n}{2}} + b^{\frac{n}{2}}$ is a power of 2. Note that a, b are odd. Also, $\frac{n}{2}$ is even since $\alpha_1 \geq 2$. Then

$$a^{\frac{n}{2}} + b^{\frac{n}{2}} \equiv 1 + 1 = 2 \pmod{8},$$

which leads to $a^{\frac{n}{2}} + b^{\frac{n}{2}} = 2$, contradiction. Hence, there is no exceptional case when n is a power of 2 greater than 2.

For the remaining cases, consider any odd prime divisor p of $f(n)$. Suppose it is also a divisor of $f\left(\frac{n}{p_j}\right)$. Note that $p \nmid a, b$ since otherwise p has to divide both a, b ,

which contradicts the relatively prime condition. Therefore, we have $p \mid a^{\frac{n}{p_j}} - b^{\frac{n}{p_j}}$, and $a^{\frac{n}{p_j}}, b^{\frac{n}{p_j}}$ are not divisible by p . Then we can use the lifting the exponent lemma (where we replace a and b in the result by $a^{\frac{n}{p_j}}$ and $b^{\frac{n}{p_j}}$ respectively) to get

$$v_p(f(n)) = v_p\left(f\left(\frac{n}{p_j}\right)\right) + v_p(p_j).$$

This is equal to $v_p\left(f\left(\frac{n}{p_j}\right)\right)$ if $p \neq p_j$, and is equal to $v_p\left(f\left(\frac{n}{p_j}\right)\right) + 1$ if $p = p_j$.

This shows the exponent of each prime divisor of $f(n)$ cannot be too large, and so we can use this to obtain an upper bound for $f(n)$. Indeed, if $f(n)$ is odd, then we get

$$f(n) \leq p_1 p_2 \cdots p_s \left[f\left(\frac{n}{p_1}\right), f\left(\frac{n}{p_2}\right), \dots, f\left(\frac{n}{p_s}\right) \right]. \quad (1)$$

Here, we use $[m_1, m_2, \dots, m_s]$ to denote the least common multiple of $m_1, m_2, \dots, m_s$. If $f(n)$ is odd and n is even, say $p_1 = 2$, then the prime 2 cannot appear. Thus, we have a slightly stronger inequality

$$f(n) \leq p_2 \cdots p_s \left[f\left(\frac{n}{p_1}\right), f\left(\frac{n}{p_2}\right), \dots, f\left(\frac{n}{p_s}\right) \right]. \quad (2)$$

Similar results hold if $f(n)$ is even. If $f(n)$ is even and n is odd, then a, b must both be odd. By [theorem 3.2\(a\)](#), we have $v_2(f(n)) = v_2\left(f\left(\frac{n}{p_s}\right)\right)$. This means the contribution from the prime 2 is included in the least common multiple. Therefore, formula (1) still holds.

If $f(n)$ and n are both even, we can apply [theorem 3.2\(b\)](#). Note that p_s is odd since $s \geq 2$. Thus, $v_2(f(n)) = v_2\left(f\left(\frac{n}{p_s}\right)\right)$. Again, this shows the contribution from the prime 2 is counted in the least common multiple. The prime $p_1 = 2$ needs not be included. This leads to the formula (2).

To continue, we need to find a formula to express the least common multiple. Indeed, we claim the following formula

$$[m_1, m_2, \dots, m_s] = \left(\prod m_{i_1}\right) \left(\prod (m_{i_1}, m_{i_2})^{-1}\right) \left(\prod (m_{i_1}, m_{i_2}, m_{i_3})\right) \cdots \left((m_1, m_2, \dots, m_s)^{(-1)^{s+1}}\right). \quad (3)$$

Here, the k th term on the right is the product of terms $(m_{i_1}, m_{i_2}, \dots, m_{i_k})^{(-1)^{k+1}}$ running through all indices $1 \leq i_1 < i_2 < \dots < i_k \leq s$. This is simply a generalization of [proposition 1.7\(a\)](#) where $s = 2$. To prove this formula, we consider each prime component p on both sides. Let $p^{\beta_j} \parallel m_j$ for $1 \leq j \leq s$. WLOG assume

$$\beta_1 \geq \beta_2 \geq \dots \geq \beta_s.$$

Then the exponent of p on the left-hand side is β_1 . The term p^{β_1} appears only once on the right-hand side from the term m_1 . For the other term p^{β_j} where $2 \leq j \leq s$, the number of times it appears on the right-hand side is

$$1 - \binom{j-1}{1} + \binom{j-1}{2} - \dots + (-1)^{j-1} \binom{j-1}{j-1}.$$

This is equal to $(1-1)^{j-1} = 0$ by the binomial theorem. Thus, the exponent of p on the right-hand side of (3) is also β_1 . This proves (3).

We return to our main proof. Since $\left(f\left(\frac{n}{p_{i_1}}\right), \dots, f\left(\frac{n}{p_{i_k}}\right)\right) = f\left(\frac{n}{p_{i_1} \cdots p_{i_k}}\right)$, we apply (3) to (1) and (2) to obtain

$$\begin{aligned} f(n) &\leq (\prod p_j) \left(\prod f\left(\frac{n}{p_{i_1}}\right) \right) \left(\prod f\left(\frac{n}{p_{i_1} p_{i_2}}\right)^{-1} \right) \left(\prod f\left(\frac{n}{p_{i_1} p_{i_2} p_{i_3}}\right) \right) \\ &\quad \cdots \left(f\left(\frac{n}{p_1 \cdots p_s}\right)^{(-1)^{s+1}} \right). \end{aligned} \quad (4)$$

Note that $\prod p_j$ runs through $1 \leq j \leq s$ when n is odd and runs through $2 \leq j \leq s$ when n is even, where $p_1 = 2$. We shall show that (4) is not true in the general case. Indeed, we can use the loose bounds

$$a^k > f(k) \geq a^k - (a-1)^k > a^{k-1}.$$

We apply this upper bound to each $f\left(\frac{n}{p_{i_1} \cdots p_{i_k}}\right)$ with odd k , and the lower bound to those with even k and to the term $f(n)$. Note that there are 2^{s-1} terms which require lower bounds. Hence, by putting everything except $\prod p_j$ to the left-hand side, the exponent of a is equal to

$$n - \sum \frac{n}{p_{i_1}} + \sum \frac{n}{p_{i_1} p_{i_2}} - \dots + (-1)^s \frac{n}{p_1 \cdots p_s} - 2^{s-1} = \varphi(n) - 2^{s-1}.$$

Thus, we deduce

$$a^{\varphi(n) - 2^{s-1}} \leq \prod p_j. \quad (5)$$

We consider a few cases.

Case 1. $p_1 > 2$ and $s \geq 2$

Then we have

$$p_1 p_2 \cdots p_s \geq a^{\varphi(n) - 2^{s-1}} \geq a^{(p_1-1) \cdots (p_{s-1}-1) - 2^{s-1}} \geq a^{(p_1-1) \cdots (p_{s-1}-1)(p_s-2)}$$

since each $p_j - 1$ is at least 2. Note that $p_j - 1 \geq 2$ and $p_s - 2 \geq 2$. Therefore, the exponent $(p_1-1) \cdots (p_{s-1}-1)(p_s-2)$ is at least $(p_1-1) + \dots + (p_{s-1}-1) + (p_s-2)$. This leads to

$$p_1 p_2 \cdots p_s \geq a^{p_1-1} \cdots a^{p_{s-1}-1} a^{p_s-2}. \quad (6)$$

However, we can prove by induction that $m < a^{m-1}$ for $m \geq 3$ since $a \geq 2$, and $m < a^{m-2}$ for $m \geq 5$. This shows (6) cannot hold. Therefore, $f(n)$ must contain a prime divisor other than those of $f(k)$ where $1 \leq k < n$.

Case 2. $p_1 > 2$ and $s = 1$

In this case, (5) becomes $a^{p_1^{\alpha_1-1}(p_1-1)-1} \leq p_1$. This implies $a^{p_1-2} \leq p_1$. As above, we can use induction to show $a^{m-2} > m$ for $m \geq 5$. Therefore, we can only have $p_1 = 3$ and $a \leq 3$. From (5), we can only have $n = 3$. We examine these possibilities.

- For $2^3 - 1^3$, the prime factor 7 does not divide $2 - 1$ and $2^2 - 1^2$.
- For $3^3 - 1^3$, the prime factor 13 does not divide $3 - 1$ and $3^2 - 1^2$.
- For $3^3 - 2^3$, the prime factor 19 does not divide $3 - 2$ and $3^2 - 2^2$.

Hence, there is no exceptional case in this case.

Case 3. $p_1 = 2$ and $s \geq 3$

As in case 1, we have

$$\begin{aligned} p_2 p_3 \cdots p_s &\geq a^{\varphi(n)-2^{s-1}} \geq a^{(p_2-1)\cdots(p_s-1)-2^{s-1}} \geq a^{(p_2-1)\cdots(p_{s-1}-1)(p_s-3)} \\ &\geq a^{p_2-1} \cdots a^{p_{s-1}-1} a^{p_s-3}. \end{aligned} \quad (7)$$

Note that $m < a^{m-1}$ for $m \geq 3$ and $m < a^{m-3}$ for $m \geq 7$. In order that (7) is satisfied, we can only have $s = 3$ and $p_3 = 5$ (then $p_2 = 3$). Since $5 < a^{5-3}$ for $a \geq 3$, we can also restrict to $a = 2$. In that case, (7) is still not satisfied since $3 \times 5 = 15 < 16 = 2^{3-1}2^{5-3}$. There is no exceptional case in this case.

Case 4. $p_1 = 2$ and $s = 2$

Note that this is the last case since n is not a power of 2. In this case, (5) becomes $a^{2^{\alpha_1-1}p_2^{\alpha_2-1}(p_2-1)-2} \leq p_2$. This implies $a^{p_2-3} \leq p_2$. Note that $a^{m-3} > m$ for $m \geq 7$. Therefore, we can only have $p_2 = 3, 5$.

- If $p_2 = 5$, then $a = 2$. From (5), we have $n = 10$. We check that $2^{10} - 1^{10} = 1023$ has a prime factor 11, which divides none of $2^k - 1^k$ for $k = 1, 2, \dots, 9$.
- If $p_2 = 3$, then (5) shows $n = 6$. We cannot get further information from (5). Therefore, we return to the stronger inequality (4), which gives

$$a^6 - b^6 \leq \frac{3(a^3 - b^3)(a^2 - b^2)}{a - b}$$

in this case. By removing equal factors, this is the same as

$$a^2 - ab + b^2 \leq 3.$$

As $a > b$, $a^2 - ab + b^2 \geq b^2 + b + 1$. Thus, we must have $b = 1$ and $a = 2$. We check that the only prime factors of $2^6 - 1^6$ are 3, 7. As $3 \mid 2^2 - 1^2$ and $7 \mid 2^3 - 1^3$, this serves as the last exceptional case (iii).

The proof is complete. □

Although the proof is lengthy, the result of Zsigmondy's theorem is rather nice. Also, there are a lot of useful tricks and facts introduced in the proof. In addition, one may view it as an application of the lifting the exponent lemma.

There is an analogous result for the sequence $a^n + b^n$ where a, b are distinct coprime positive integers. The only exceptional case for which it does not have any prime divisor relatively prime to $a^k + b^k$ (for $k = 1, 2, \dots, n-1$) is $a = 2$, $b = 1$ and $n = 3$. One can deduce this easily from the original form of Zsigmondy's theorem. Indeed, there is no exceptional case when $n = 1$, while for every $a^n + b^n$ with $n > 1$ except $(a, b, n) = (2, 1, 3)$, there exists a prime p dividing $a^{2n} - b^{2n}$ such that $p \nmid a^k - b^k$ for $1 \leq k < 2n$. In particular, $p \nmid a^n - b^n$. This implies $p \mid a^n + b^n$. For this prime p , we have $p \nmid a^k + b^k$ for $1 \leq k < n$ since $p \nmid a^{2k} - b^{2k}$.

In the remainder of this section, we work on a number of divisibility problems.

Problem 3.1. (IMO Shortlist 2008 N2) Let $a_1, a_2, \dots, a_n$ be distinct positive integers, $n \geq 3$. Prove that there exist distinct indices i and j such that $a_i + a_j$ does not divide any of the numbers $3a_1, 3a_2, \dots, 3a_n$.

Analysis. In order that $a_i + a_j$ does not divide $3a_k$, a simple sufficient condition is that $a_i + a_j > 3a_k$. Therefore, we should first consider large pairs of a_i, a_j . Note that we may assume the greatest common divisor of the variables is 1. This is sometimes a useful assumption for this kind of homogeneous problem.

Solution. Suppose on the contrary that no such pair of a_i and a_j exists. Suppose $a_1 < a_2 < \dots < a_n$. Consider $a_i + a_n$ for some $1 \leq i \leq n-1$. Note that $a_i + a_n \mid 3a_k$ for some $1 \leq k \leq n$. If $a_i + a_n$ is not divisible by 3, then this implies $a_i + a_n \mid a_k$. However, this is impossible since

$$a_i + a_n > a_n \geq a_k.$$

Therefore, we have $3 \mid a_i + a_n$ for all $1 \leq i \leq n-1$. Then for any $1 \leq i < j \leq n-1$, we have $a_i \equiv -a_n \equiv a_j \pmod{3}$. WLOG assume $\gcd(a_1, a_2, \dots, a_n) = 1$, since otherwise we can divide each term by the common divisor to get an equivalent problem with greatest common divisor 1. In particular, at least one of the terms is not divisible by 3. From the above congruences, we must have $3 \nmid a_n$ and hence $3 \nmid a_i$ for all $1 \leq i \leq n$.

Then we consider the next largest pair $a_{n-2} + a_{n-1}$ (note that $n \geq 3$). Firstly, since $a_{n-2} + a_{n-1} \equiv -2a_n \not\equiv 0 \pmod{3}$, we know that $a_{n-2} + a_{n-1}$ does not divide any of $3a_1, 3a_2, \dots, 3a_{n-1}$ for the same reason as before. By our assumption, we must have $a_{n-2} + a_{n-1} \mid a_n$. From this, we observe that a_n is relatively large, and hence $a_{n-1} + a_n$ probably does not divide any $3a_k$.

- Indeed, if $a_n \geq 2(a_{n-2} + a_{n-1})$, then $a_{n-1} + a_n \nmid 3a_k$ for $1 \leq k \leq n-1$ since

$$a_{n-1} + a_n > 3a_{n-1} \geq 3a_k.$$

Thus, we must have $a_{n-1} + a_n \mid 3a_n$. This implies

$$a_{n-1} + a_n \mid 3(a_{n-1} + a_n) - 3a_n = 3a_{n-1},$$

which is again impossible. This yields a contradiction.

- It remains to consider the case $a_n = a_{n-2} + a_{n-1}$. Again, $a_{n-1} + a_n > 3a_k$ for $1 \leq k \leq n-2$ so that $a_{n-1} + a_n \nmid 3a_k$. Also, both $a_{n-1} + a_n \mid 3a_{n-1}$ and $a_{n-1} + a_n \mid 3a_n$ are the same as $a_{n-2} + 2a_{n-1} \mid 3a_{n-1}$. However, since

$$a_{n-2} + 2a_{n-1} < 3a_{n-1} < 2(a_{n-2} + 2a_{n-1}),$$

the divisibility cannot hold. This is again a contradiction.

Therefore, the assumption is not true. In other words, we can always find i, j such that $a_i + a_j \nmid 3a_k$ for all $1 \leq k \leq n$. $\square$

Remarks. This is a standard example of applying algebra to tackle number theory problems.

Problem 3.2. Find all primes p and positive integers n such that $(p^2 + 1)(n^3 + 1)$ divides $pn(pn + 1)^2$.

Analysis. For divisibility problems involving polynomials, one usual method is to reduce the degree of the dividend. Then the dividend is smaller than the divisor for large cases, which cannot be true. This usually leaves us a finite number of cases.

Solution. Observe that

$$\begin{aligned} & (p^2 + 1)(n^3 + 1) \mid pn(pn + 1)^2 \\ \Rightarrow & p^2n^3 + p^2 + n^3 + 1 \mid p^3n^3 + 2p^2n^2 + pn \\ \Rightarrow & p^2n^3 + p^2 + n^3 + 1 \mid p(p^2n^3 + p^2 + n^3 + 1) + 2p^2n^2 + pn - p^3 - pn^3 - p \\ \Rightarrow & p^2n^3 + p^2 + n^3 + 1 \mid 2p^2n^2 + pn - p^3 - pn^3 - p. \end{aligned} \tag{1}$$

There are 3 possible cases.

Case 1. $2p^2n^2 + pn - p^3 - pn^3 - p > 0$

Then we have

$$p^2n^3 + p^2 + n^3 + 1 \leq 2p^2n^2 + pn - p^3 - pn^3 - p$$

from (1). Note that this shows $2p^2n^2 + pn$ dominates all other terms, which should be impossible since p^2n^3 is the term with the largest degree. Indeed, since $n \geq 1$, we have the inequality

$$p^2(n-1)(n^2-n-1) \geq 0,$$

which gives $2p^2n^2 \leq p^2n^3 + p^2$. Also, we have $pn \leq pn^3$. These yield the contradiction

$$n^3 + 1 + p^3 + p \leq (2p^2n^2 - p^2n^3 - p^2) + (pn - pn^3) \leq 0.$$

Case 2. $2p^2n^2 + pn - p^3 - pn^3 - p < 0$

Then we have $p^2n^3 + p^2 + n^3 + 1 \leq -(2p^2n^2 + pn - p^3 - pn^3 - p)$ and hence

$$p^2n^3 + p^2 + n^3 + 1 + 2p^2n^2 + pn \leq p^3 + pn^3 + p. \quad (2)$$

This time we cannot get the same contradiction as in **case 1** since the term p^3 could possibly be the dominating term. However, in that case p should be quite large compared to n . Then p might be relatively prime to the left-hand side of (1), so that we can remove the factor p on the right-hand side. This is the idea of the following steps.

Firstly, from (2) and $p \leq pn$, we get

$$p^2n^3 + p^2 + n^3 + 1 + 2p^2n^2 \leq p^3 + pn^3. \quad (3)$$

Since $pn^3 < p^2n^3$, this implies

$$2p^2n^2 < p^2 + n^3 + 1 + 2p^2n^2 < p^3,$$

and hence $2n^2 < p$. From this, we get $pn^3 < 4pn^4 < 2p^2n^2$. Together with (3), we have $p^2n^3 + p^2 + n^3 + 1 \leq p^3$. This implies $p^2n^3 + p^2 < p^3$, and thus $n^3 + 1 < p$.

Now, return to the divisibility (1). Note that $p \nmid p^2 + 1$ and $p \nmid n^3 + 1$. As p is a prime, these show p is relatively prime to the left-hand side of the divisibility. Therefore, we can remove the prime p on the right-hand side to get

$$p^2n^3 + p^2 + n^3 + 1 \mid 2pn^2 + n - p^2 - n^3 - 1.$$

Since the right-hand side is negative in this case, we deduce

$$p^2n^3 + p^2 + n^3 + 1 \leq -(2pn^2 + n - p^2 - n^3 - 1),$$

which is clearly impossible.

Case 3. $2p^2n^2 + pn - p^3 - pn^3 - p = 0$

This is the same as $p^2 - 2n^2p + n^3 - n + 1 = 0$. Viewing it as a quadratic equation in p , the discriminant must be a square. Let

$$\Delta = 4n^4 - 4n^3 + 4n - 4 = m^2$$

for some $m \in \mathbb{N}_0$. If $n \geq 3$, then

$$(2n^2 - n)^2 = 4n^4 - 4n^3 + n^2 > \Delta > 4n^4 - 4n^3 - 3n^2 + 2n + 1 = (2n^2 - n - 1)^2.$$

This is impossible since m^2 lies strictly between two consecutive square numbers. If $n = 2$, we get $p = 7$ as the only prime solution. If $n = 1$, p cannot be a prime.

We check that $(p, n) = (7, 2)$ is the only solution. □

Problem 3.3. (IMO Shortlist 2007 N1) Find all pairs (k, n) of positive integers for which $7^k - 3^n$ divides $k^4 + n^2$.

Analysis. In general, exponential terms like 7^k and 3^n are much larger than monomials like k^4 and n^2 . However, we cannot apply the bound directly since $7^k - 3^n$ can well be positive or negative. We can first find more information on k and n through simple congruences.

Solution. Note that $7^k - 3^n$ is even. Therefore, $k^4 + n^2$ is even. This implies k, n have the same parity. If k, n are odd, then $7^k - 3^n \equiv 7 - 3 \equiv 0 \pmod{4}$. Then $4 \mid k^4 + n^2$. However, we also have $k^4 + n^2 \equiv 1 + 1 \equiv 2 \pmod{4}$. This is a contradiction.

Therefore, k, n are even. Let $k = 2a$ and $n = 2b$. Then we have

$$7^{2a} - 3^{2b} \mid 16a^4 + 4b^2. \quad (1)$$

Note that

$$7^{2a} - 3^{2b} = (7^a - 3^b)(7^a + 3^b)$$

is divisible by $2(7^a + 3^b)$ since $7^a - 3^b$ is even. Hence, we get $7^a + 3^b \mid 8a^4 + 2b^2$. This time we are sure that both sides are positive. This yields

$$7^a + 3^b \leq 8a^4 + 2b^2. \quad (2)$$

Intuitively, (2) cannot hold for large a, b . Firstly, for $a \geq 5$, we have

$$\begin{aligned} 7^a &= (6+1)^a > 6^4 \binom{a}{4} = \frac{6^4 a^4}{24} \cdot \frac{a-1}{a} \cdot \frac{a-2}{a} \cdot \frac{a-3}{a} \\ &\geq \frac{6^4 a^4}{24} \cdot \frac{4}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} = \frac{1296}{125} a^4 > 8a^4. \end{aligned}$$

We check directly that $7^4 = 2401 > 2048 = 8 \cdot 4^4$. Similarly, for $b \geq 3$, we have

$$\begin{aligned} 3^b &= (2+1)^b > 2^2 \binom{b}{2} + 2^3 \binom{b}{3} = \frac{2^2 b^2}{2} \cdot \frac{b-1}{b} + \frac{2^3 b^3}{6} \cdot \frac{b-1}{b} \cdot \frac{b-2}{b} \\ &\geq \frac{2^2 b^2}{2} \cdot \frac{2}{3} + \frac{2^3 \cdot 3b^2}{6} \cdot \frac{2}{3} \cdot \frac{1}{3} = \frac{4}{3} b^2 + \frac{8}{9} b^2 > 2b^2. \end{aligned}$$

We check directly that $3^2 > 2 \cdot 2^2$ and $3^1 > 2 \cdot 1^2$. Therefore, (2) cannot hold if $a \geq 4$. Hence, we only need to work on the small cases.

- If $a = 3$, then (2) implies $3^b \leq 2b^2 + 305$. We can show that this only holds when $b \leq 5$. Indeed, for $b = 6$, we have $3^6 = 729 > 377 = 2 \cdot 6^2 + 305$. Suppose $3^b > 2b^2 + 305$ holds for some b . Then we have

$$3^{b+1} = 3 \cdot 3^b > 3(2b^2 + 305) = 6b^2 + 915 > 2(b+1)^2 + 305,$$

since the last inequality is the same as $b(b-1) + 152 > 0$. Thus, by induction, we deduce $3^b > 2b^2 + 305$ for $b \geq 6$.

Now, for $1 \leq b \leq 5$, we focus on (1). We find that

$$7^6 - 3^{2b} \geq 7^6 - 3^{10} = 117649 - 59049 > 50000$$

and

$$1296 + 4b^2 \leq 1296 + 100 < 10000.$$

Hence, the divisibility cannot hold.

- If $a = 2$, then (2) implies $3^b \leq 2b^2 + 79$. Again, we can prove by induction that this only holds when $b \leq 4$. We find that none of these satisfies (1).
- If $a = 1$, then (2) implies $3^b \leq 2b^2 + 1$. By induction, this only holds when $b \leq 2$. We check that only $b = 2$ satisfies (1).

Therefore, the only solution is $(k, n) = (2, 4)$. □

Problem 3.4. (Baltic Way 2015 Problem 17) Find all positive integers n for which $n^{n-1} - 1$ is divisible by 2^{2015} , but not by 2^{2016} .

Analysis. Since it is about the 2-adic valuation of some integer and it involves exponents, we should use the lifting the exponent lemma.

Solution. Note that n must be odd. By the lifting the exponent lemma (theorem 3.2(b)), we have

$$2015 = v_2(n^{n-1} - 1) = v_2(n^2 - 1) + v_2(n - 1) - 1 = v_2(n + 1) + 2v_2(n - 1) - 1 \quad (1)$$

since $n^2 - 1 = (n - 1)(n + 1)$. Note that exactly one of $n + 1$ and $n - 1$ is divisible by 2 but not 4.

- If $v_2(n - 1) = 1$, then (1) gives $v_2(n + 1) = 2014$. This means $n = 2^{2014}m - 1$ for some odd m .
- If $v_2(n + 1) = 1$, then (1) gives $2v_2(n - 1) = 2015$. This is impossible.

Therefore, n can be any positive integer of the form $2^{2014}m - 1$ for some odd m . □

Remarks. The lifting the exponent lemma is very useful, apart from this simple application. We shall see more of its applications in later sections.

Problem 3.5. (IMO Shortlist 2000 N4) Find all triplets of positive integers (a, m, n) such that $a^m + 1 \mid (a + 1)^n$.

Analysis. This is a trivial consequence of Zsigmondy's theorem. One just needs to be careful about the exceptional cases.

Solution. When $a = 1$ or $m = 1$, the divisibility always holds. When $a = 2$ and $m = 3$, we need $9 \mid 3^n$. Therefore, n can be any integer greater than 1.

For the remaining cases, we can apply Zsigmondy's theorem to find a prime p such that $p \mid a^m + 1$ but $p \nmid a + 1$ (see the discussion after the proof of that theorem). This implies $a^m + 1 \nmid (a + 1)^n$ for any $n \in \mathbb{N}$.

Therefore, the only triplets are $(1, m, n)$ for $m, n \in \mathbb{N}$, $(a, 1, n)$ for $a, n \in \mathbb{N}$ and $(2, 3, n)$ for $n \geq 2$. $\square$

Problem 3.6. (IMO Shortlist 2002 N3) Let $p_1, p_2, \dots, p_n$ be distinct primes greater than 3. Show that $2^{p_1 p_2 \cdots p_n} + 1$ has at least 4^n divisors.

Analysis. In other words, we need to find at least $2n$ distinct prime divisors of $2^{p_1 p_2 \cdots p_n} + 1$. But we can indeed do a lot more by Zsigmondy's theorem.

Solution. Let $d_1 < d_2 < \cdots < d_m$ be all the positive divisors of $p_1 p_2 \cdots p_n$. Then $m = d(p_1 p_2 \cdots p_n) = 2^n$. Since the primes are greater than 3, each d_j is not equal to 3. By Zsigmondy's theorem, for $2 \leq j \leq m$, we can find a prime q_j such that $q_j \mid 2^{d_j} + 1$ and $q_j \nmid 2^k + 1$ for $1 \leq k < d_j$. Also, we can take $q_1 = 3$ so that $q_1 \mid 2^1 + 1$. Then all the primes $q_1, q_2, \dots, q_m$ are distinct. As $2^{d_j} + 1 \mid 2^{p_1 p_2 \cdots p_n} + 1$, we have found 2^n distinct prime divisors of $2^{p_1 p_2 \cdots p_n} + 1$. Hence, this integer has at least 2^{2^n} divisors, which is at least $2^{2^n} = 4^n$ as $2^n \geq 2n$ for $n \in \mathbb{N}$ by induction. $\square$

Remarks. Although Zsigmondy's theorem looks powerful, it is seldom used in recent MO problems because problems which can be solved by a direct consequence of the theorem are avoided. (Not quite, because it can be used in IMO 2022!)

Problem 3.7. (CWMI 2013 Problem 8) Find all positive integers a such that for any positive integer $n \geq 5$ we have $(2^n - n^2) \mid (a^n - n^a)$.

Analysis. In general, $a^n - n^a$ is larger than $2^n - n^2$. Therefore, there is no hope to derive a contradiction using the bounds for large a . This time we try to use a more number theoretic method to tackle this divisibility problem. As in [problem 3.3](#), we try to find a divisor of $2^n - n^2$ so that it must also divide $a^n - n^a$. It is not easy to do so for a fixed n . But we have more flexibility since the divisibility holds for every large n . This means we can fix a prime number p , and find $n \geq 5$ such that $p \mid 2^n - n^2$. This can be done as in [problem 1.10](#).

Solution. Let p be any odd prime number. We take $n = p(p-1) + 2 > 5$. Then

$$2^n - n^2 \equiv 2^2 - n^2 \equiv 0 \pmod{p}$$

by the Fermat little theorem. (We can also apply the Chinese remainder theorem. Indeed, it suffices to find n for which $n \equiv m \pmod{p-1}$ and $n^2 \equiv 2^m \pmod{p}$ for some m . This can be done as long as the congruence modulo p is solvable, say when $m = 2$.)

Thus, we have $p \mid a^n - n^a$. Note that

$$a^n - n^a \equiv a^2 - n^a \equiv a^2 - 2^a \pmod{p}$$

again by the Fermat little theorem. Therefore, we find that $p \mid a^2 - 2^a$. Since this holds for infinitely many primes p , we must have $a^2 - 2^a = 0$.

Clearly, $a = 2, 4$ are solutions to $a^2 - 2^a = 0$, while $a = 1, 3$ are not. For $a = 5$, we have $2^5 > 5^2$. Suppose $2^a > a^2$ for some $a \geq 5$. Then

$$2^{a+1} = 2 \cdot 2^a > 2a^2 \geq a^2 + 5a > a^2 + 2a + 1 = (a+1)^2.$$

By induction, $2^a > a^2$ for every $a \geq 5$. Therefore, the only possible solutions are $a = 2, 4$.

For $a = 2$, $2^n - n^2 \mid 2^n - n^2$ holds trivially. For $a = 4$, we have

$$4^n - n^4 = 2^{2n} - n^4 = (2^n)^2 - (n^2)^2 = (2^n - n^2)(2^n + n^2).$$

This is always divisible by $2^n - n^2$. Hence, both $a = 2, 4$ are solutions. $\square$

Problem 3.8. (CMO 2010 Problem 6) Let $a_1, a_2, a_3, b_1, b_2, b_3$ be distinct positive integers, satisfying

$$(n+1)a_1^n + na_2^n + (n-1)a_3^n \mid (n+1)b_1^n + nb_2^n + (n-1)b_3^n$$

for all positive integers n . Prove that there exists a positive integer k such that $b_i = ka_i$ for each $i = 1, 2, 3$.

Analysis. The trick that we use is similar to the one in [problem 3.7](#). We first fix a prime p and find an n satisfying $p \mid (n+1)a_1^n + na_2^n + (n-1)a_3^n$. This can be done by fixing the exponent to a certain residue class modulo $p-1$ so that it becomes a linear congruence equation. At this point, we do not know what the exponent should be. But we may add extra conditions later.

Solution. Fix an $M \in \mathbb{N}$ and a prime p . We consider $n \in \mathbb{N}$ with $n \equiv M \pmod{p-1}$. By the Fermat little theorem, we have

$$(n+1)a_1^n + na_2^n + (n-1)a_3^n \equiv (n+1)a_1^M + na_2^M + (n-1)a_3^M \pmod{p}.$$

Since we would like this to be congruent to 0 modulo p , we would like n to satisfy

$$(a_1^M + a_2^M + a_3^M)n + (a_1^M - a_3^M) \equiv 0 \pmod{p}.$$

If $p \nmid a_1^M + a_2^M + a_3^M$, then this congruence is solvable in n . One simple way is to assume p is sufficiently large so that $p > a_1^M + a_2^M + a_3^M$. This can be done since M and p are arbitrary at this stage. Now, we would like to have n satisfying

$$\begin{cases} n \equiv M \pmod{p-1}, \\ (a_1^M + a_2^M + a_3^M)n + (a_1^M - a_3^M) \equiv 0 \pmod{p}. \end{cases}$$

This is solvable by the Chinese remainder theorem as $(p-1, p) = 1$. We choose any such $n \in \mathbb{N}$. Then $p \mid (n+1)a_1^n + na_2^n + (n-1)a_3^n$ and hence $p \mid (n+1)b_1^n + nb_2^n + (n-1)b_3^n$. Applying the Fermat little theorem again, we can change the exponents to M . We rewrite these as

$$\begin{cases} (a_1^M + a_2^M + a_3^M)n \equiv a_3^M - a_1^M \pmod{p}, \\ (b_1^M + b_2^M + b_3^M)n \equiv b_3^M - b_1^M \pmod{p}. \end{cases} \quad (1)$$

Since n is an unknown and we aim at some relations on a_i, b_j 's, we should eliminate n using the two congruences. Indeed, we can assume $p > |a_3^M - a_1^M|$ so that $p \nmid n$. By cross multiplying the 2 congruences in (1) and dividing by n , we get

$$(a_1^M + a_2^M + a_3^M)(b_3^M - b_1^M) \equiv (b_1^M + b_2^M + b_3^M)(a_3^M - a_1^M) \pmod{p}. \quad (2)$$

It seems that we do not get any information related to our conclusion. However, one should note that we still have not chosen our M and p yet. Indeed, one will soon realize that (2) does give us surprisingly useful information.

Firstly, for fixed M , we have infinitely many choices for p . Since (2) holds for infinitely many primes p , we must have

$$(a_1^M + a_2^M + a_3^M)(b_3^M - b_1^M) = (b_1^M + b_2^M + b_3^M)(a_3^M - a_1^M).$$

Expanding and simplifying, we get

$$(a_1b_2)^M + 2(a_1b_3)^M + (a_2b_3)^M = (a_2b_1)^M + 2(a_3b_1)^M + (a_3b_2)^M. \quad (3)$$

Note that this holds for every $M \in \mathbb{N}$. Therefore, it is natural to guess that the terms on the two sides are equal in pairs. Indeed, we shall prove a more general result as follows.

Claim. Let $t_1 \geq t_2 \geq \dots \geq t_s$ and $u_1 \geq u_2 \geq \dots \geq u_s$ be positive integers. If

$$t_1^M + t_2^M + \dots + t_s^M = u_1^M + u_2^M + \dots + u_s^M \quad (4)$$

for all $M \in \mathbb{N}$, then $t_j = u_j$ for $j = 1, 2, \dots, s$.

Proof. Suppose on the contrary that $t_1 \neq u_1$. WLOG assume $t_1 > u_1$. Then t_1^M is the dominating term of (4) for large M , and hence it cannot be always true. Indeed, (4) implies

$$t_1^M \leq t_1^M + t_2^M + \dots + t_s^M = u_1^M + u_2^M + \dots + u_s^M \leq su_1^M$$

so that $M \log t_1 \leq \log s + M \log u_1$. This yields $M \leq \frac{\log s}{\log t_1 - \log u_1}$, which does not hold for large M . Therefore, we must have $t_1 = u_1$. By eliminating the equal terms t_1^M and u_1^M , we can prove the **claim** by induction. $\square$

Applying the **claim** to (3), the terms on both sides of the equation are equal in pairs. As $a_1, a_2, a_3, b_1, b_2, b_3$ are distinct, we have $a_2b_3 \neq a_2b_1$ and $a_1b_2 \neq a_3b_2$.

If $a_1b_3 \neq a_3b_1$, then $a_1b_2 = a_3b_1 = a_2b_3$. The remaining pairs of equal terms are $a_2b_1 = a_1b_3 = a_3b_2$. Using the two relations repeatedly, we get

$$\frac{a_1}{a_2} = \frac{b_1}{b_3} = \frac{a_2}{a_3} = \frac{b_2}{b_1} = \frac{a_3}{a_1}.$$

From $\frac{a_1}{a_2} = \frac{a_2}{a_3} = \frac{a_3}{a_1}$, we easily deduce the contradiction $a_1 = a_2 = a_3$.

If $a_1b_3 = a_3b_1$, then the remaining pairs are $a_1b_2 = a_2b_1$ and $a_2b_3 = a_3b_2$. Therefore, we have $\frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3} = k$ for some real number $k > 0$. We need to show that $k \in \mathbb{N}$. Indeed, by replacing b_j by ka_j in the given divisibility, we have

$$(n+1)a_1^n + na_2^n + (n-1)a_3^n \mid k((n+1)a_1^n + na_2^n + (n-1)a_3^n),$$

and hence $1 \mid k$. This means $k \in \mathbb{N}$ as $k > 0$. Thus we are done. $\square$

Remarks. The main question that one may ask is how we can think of this solution (probably the same for **problem 3.7**). The initial steps seem to be very far from the conclusion that we want. One probably does not realize we are almost finished after getting (3).

In fact, this is only one of the many different possible directions to tackle this problem. It just turns out that this method does provide a relatively neat solution to us. Therefore, we should always think of different ways to work on a problem, no matter they seem to relate to the problem or not. The situation is similar to a lucky draw. It is difficult to win the draw, but there is always someone who can win. With luck, everyone can figure out a correct way to work out the problem.

The same trick used in this solution can be applied to the following problem.

(IMO Shortlist 2005 N6) Let a, b be positive integers such that $b^n + n$ is a multiple of $a^n + n$ for all positive integers n . Prove that $a = b$.

The solution is much simpler. One may try to solve this N6 problem in one minute!

The tricks used to solve divisibility problems are mostly those introduced in previous sections. As we have seen, we only need to combine these tricks together to solve some contest problems. Nevertheless, we recall these tricks in the summary below.

- factorization (problem 3.3)
- deriving bounds (problems 3.1, 3.2, 3.3, 3.7, 3.8)
- considering congruences modulo small integers or primes (problems 3.1, 3.3, 3.7, 3.8)
- lifting the exponent lemma (problem 3.4)
- Zsigmondy's theorem (problems 3.5, 3.6)
- Chinese remainder theorem (problems 3.7, 3.8)
- Fermat little theorem (problems 3.7, 3.8)

4 More on Congruences

In this section, we revisit the topic of congruences. In section 1, we mainly talked about linear congruences, the Euler-Fermat theorem and the Chinese remainder theorem. We have seen the usefulness of these things to solve number theory problems. This time we shall go through more advanced results. They are less useful in MO because advanced knowledge is not required in solving MO problems. However, it makes no harm to learn more results as they may give us some insight in solving various problems.

Firstly, we consider congruence equations with polynomials. In view of the Chinese remainder theorem, it suffices to consider congruences modulo a power of prime. Let p be a prime number and let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

be a polynomial with integer coefficients. There is no general method to determine whether $f(m) \equiv 0 \pmod{p}$ is solvable or not. The most straightforward method is to test all possibilities of $f(m) \pmod{p}$. Indeed, since $a \equiv b \pmod{p}$ implies $f(a) \equiv f(b) \pmod{p}$, it suffices to test $f(0), f(1), \dots, f(p-1)$. If none of these satisfies the congruence, then it is not solvable. Certainly, this is not a good method as we do not know how large p is in general. The only useful fact we know is the following.

Proposition 4.1. Let p be a prime number. Define

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

where $a_n, a_{n-1}, \dots, a_0$ are integers. If $p \nmid a_n$, then $f(x) \equiv 0 \pmod{p}$ has at most n distinct solutions modulo p .

Proof. We prove this by induction on the degree n . When $n = 0$, there is no solution since $p \nmid a_0$. When $n = 1$, the linear congruence equation $a_1 x + a_0 \equiv 0 \pmod{p}$ has a unique solution since $p \nmid a_1$.

Suppose a degree k polynomial congruence equation has at most k solutions modulo p . Consider the case $n = k + 1$. If it has no solution, then we are done. Suppose it has a solution α . Then we can write

$$f(x) = (x - \alpha)q(x) + r$$

where $q(x)$ is a degree k polynomial with integer coefficients and leading coefficient not divisible by p , and where $r \in \mathbb{Z}$. Putting $x = \alpha$, we get $p \mid r$ since $p \mid f(\alpha)$. Thus, $f(x) \equiv 0 \pmod{p}$ is equivalent to $(x - \alpha)q(x) \equiv 0 \pmod{p}$. As p is a prime, this implies $x \equiv \alpha \pmod{p}$ or $q(x) \equiv 0 \pmod{p}$. Note that $q(x) \equiv 0 \pmod{p}$ has at most k solutions by the inductive hypothesis. In total, we have at most $k + 1$ solutions.

By induction, the proposition is proved. □

There is one useful corollary of this proposition. Suppose

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0.$$

If $f(m) \equiv 0 \pmod{p}$ has more than n solutions, then f has to be the zero polynomial modulo p , i.e. $p \mid a_j$ for $0 \leq j \leq n$.

For higher powers of p , we have the following more concrete result.

Theorem 4.1. (Hensel's lifting lemma(亨澤爾引理)) Let p be a prime number. Define

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

where $a_n, a_{n-1}, \dots, a_0$ are integers. If $x \equiv m \pmod{p}$ is a solution to $f(x) \equiv 0 \pmod{p}$ such that $f'(m) \not\equiv 0 \pmod{p}$, then for every $k \geq 1$, there exists a unique solution to $f(x) \equiv 0 \pmod{p^k}$ satisfying $x \equiv m \pmod{p}$.

Here, $f'(x)$ denotes the derivative of $f(x)$.

Proof. We shall prove by induction on k . The base case $k = 1$ is trivial. Suppose there exists a unique $\ell \pmod{p^k}$ such that $\ell \equiv m \pmod{p}$ and $f(\ell) \equiv 0 \pmod{p^k}$. To solve for $f(x) \equiv 0 \pmod{p^{k+1}}$ with $x \equiv m \pmod{p}$, it suffices to consider those integers x with $x \equiv \ell \pmod{p^k}$. Let $x = tp^k + \ell$. Then

$$\begin{aligned} f(x) &= a_n (tp^k + \ell)^n + a_{n-1} (tp^k + \ell)^{n-1} + \cdots + a_0 \\ &\equiv a_n (ntp^k \ell^{n-1} + \ell^n) + a_{n-1} ((n-1)tp^k \ell^{n-2} + \ell^{n-1}) + \cdots + a_0 \\ &= (na_n \ell^{n-1} + (n-1)a_{n-1} \ell^{n-2} + \cdots + a_1) tp^k + (a_n \ell^n + a_{n-1} \ell^{n-1} + \cdots + a_0) \\ &= f'(\ell) tp^k + f(\ell) \pmod{p^{k+1}}. \end{aligned}$$

Since $p^k \mid f(\ell)$, we have $p^{k+1} \mid f(x)$ iff

$$f'(\ell)t + \frac{f(\ell)}{p^k} \equiv 0 \pmod{p}.$$

As $f'(\ell) \equiv f'(m) \not\equiv 0 \pmod{p}$, there exists a unique $t \pmod{p}$ satisfying this congruence. Hence, there exists a unique solution $x \pmod{p^{k+1}}$ with $x \equiv m \pmod{p}$, as desired. $\square$

In general, the condition $f'(m) \not\equiv 0 \pmod{p}$ is usually satisfied. Therefore, if there exists $m \in \mathbb{Z}$ with $p \mid f(m)$, then $f(x) \equiv 0 \pmod{p^k}$ is usually solvable. This suggests why we usually focus on congruences modulo a prime only.

Theorem 4.2. (Wilson's theorem(威爾遜定理)) Let p be a prime number. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. The cases $p = 2$ and $p = 3$ can be checked directly.

For $p \geq 5$, consider any $2 \leq k \leq p-2$. As $(k, p) = 1$, the linear congruence equation $km_k \equiv 1 \pmod{p}$ has a unique solution $m_k \pmod{p}$. Note that $(m_k, p) = 1$. Also, we have $k \not\equiv m_k \pmod{p}$ since the only solutions to $k^2 \equiv 1 \pmod{p}$ are $k \equiv \pm 1 \pmod{p}$.

Next, we show that for $k \not\equiv \ell \pmod{p}$, we have $m_k \not\equiv m_\ell \pmod{p}$. Indeed, suppose $m_k \equiv m_\ell \pmod{p}$. Then $km_k \equiv 1 \equiv \ell m_\ell \pmod{p}$ yields $k \equiv \ell \pmod{p}$. Therefore, the $p-3$ integers $2, 3, \dots, p-2$ can be paired up into $\frac{p-3}{2}$ groups so that the product of integers in each pair is congruent to 1 modulo p . Thus,

$$(p-1)! = 1 \times (2 \times 3 \times \dots \times (p-2)) \times (p-1) \equiv 1 \times 1^{\frac{p-3}{2}} \times (-1) \equiv -1 \pmod{p}.$$

□

The idea of the proof is far more important than the result of the theorem.

Theorem 4.3. ([Legendre's formula](#)(勒讓德定理)) Let p be a prime number and let n be a positive integer. Then

$$v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n-s}{p-1} \quad (1)$$

where s is the sum of digits of the base p representation of n .

Note that the sum is indeed a finite sum since $\frac{n}{p^i} < 1$ for sufficiently large i .

Proof. Among the integers $1, 2, \dots, n$, there are $\left\lfloor \frac{n}{p} \right\rfloor$ of them which are divisible by p , there are $\left\lfloor \frac{n}{p^2} \right\rfloor$ of them which are divisible by p^2 , etc. Each integer of the form $p^\alpha m$ where $p \nmid m$ is counted α times since it is divisible by $p, p^2, \dots, p^\alpha$ but not higher powers of p . Therefore, the sum in (1) counts the sum of the p -adic valuations of $1, 2, \dots, n$. This is exactly

$$v_p(1) + v_p(2) + \dots + v_p(n) = v_p(n!).$$

For the second equality, let $\overline{a_t a_{t-1} \dots a_0}_{(p)}$ be the base p representation of n . This means

$$n = a_t p^t + a_{t-1} p^{t-1} + \dots + a_1 p + a_0.$$

Therefore, we have $\left\lfloor \frac{n}{p^i} \right\rfloor = a_t p^{t-i} + a_{t-1} p^{t-1-i} + \dots + a_i$ for each $1 \leq i \leq t$. Thus,

$$\sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor = \sum_{i=1}^t \sum_{j=i}^t a_j p^{j-i} = \sum_{j=1}^t \sum_{i=1}^j a_j p^{j-i} = \sum_{j=1}^t a_j \frac{p^j - 1}{p - 1} = \frac{1}{p - 1} \left(\sum_{j=0}^t a_j p^j - \sum_{j=0}^t a_j \right) = \frac{n-s}{p-1}.$$

□

For example, consider $p = 5$ and $n = 67 = 232_{(5)}$. Then $v_5(67!) = \left\lfloor \frac{67}{5} \right\rfloor + \left\lfloor \frac{67}{5^2} \right\rfloor = 15$, which is also equal to $\frac{67 - (2 + 3 + 2)}{5 - 1}$.

Theorem 4.4. (Kummer's theorem (庫默爾定理)) Let p be a prime number and let $m \geq n$ be positive integers. Then $v_p\binom{m}{n}$ equals the number of carries when n is added to $m - n$ in base p .

Here, we write $v_p\binom{m}{n}$ to mean $v_p\left(\binom{m}{n}\right)$ (where $\binom{m}{n}$ is the binomial coefficient), just to avoid the clumsy notations.

Proof. Let a, b, c be the sums of digits of the base p representations of $m - n, n, m$ respectively. By Legendre's formula, since $\binom{m}{n} = \frac{m!}{n!(m-n)!}$, we have

$$v_p\binom{m}{n} = v_p(m!) - v_p(n!) - v_p((m-n)!) = \frac{m-c}{p-1} - \frac{n-b}{p-1} - \frac{(m-n)-a}{p-1} = \frac{a+b-c}{p-1}.$$

When we sum up n and $m-n$ in base p , the total sum of digits is unchanged if there is no carry. Whenever there is one carry, this means the sum s of that digit is changed to $s-p$, and we add 1 to the next digit. This reduces the sum of digits by $p-1$. Therefore, the resultant sum of digits is reduced by $p-1$ times the number of carries in this addition. This means exactly that $\frac{a+b-c}{p-1}$ is the number of carries. $\square$

As an example, to find the 7-adic valuation of $\binom{348}{128}$, we first express the numbers 128 and $348 - 128 = 220$ in base 7 representation. We get $242_{(7)}$ and $433_{(7)}$. Their sum is $1005_{(7)}$, where the number of carries is 2. Thus, $v_7\binom{348}{128} = 2$, meaning that $\binom{348}{128}$ is divisible by 7^2 but not 7^3 .

We consider a particular case when $m = p^k$. Then the number of carries when n is added to $m - n$ is $k - v_p(n)$. This gives $v_p\binom{p^k}{n} = k - v_p(n)$.

Theorem 4.5. (Lucas' theorem (盧卡斯定理)) Let p be a prime number and let m, n be positive integers. Let $\overline{a_s a_{s-1} \cdots a_0}_{(p)}$ and $\overline{b_s b_{s-1} \cdots b_0}_{(p)}$ be the base p representations of m and n respectively. Then

$$\binom{m}{n} \equiv \binom{a_s}{b_s} \binom{a_{s-1}}{b_{s-1}} \cdots \binom{a_0}{b_0} \pmod{p}.$$

Proof. Note that $\binom{m}{n}$ is the coefficient of x^n in the expansion of $(x+1)^m$. As

$$m = a_s p^s + a_{s-1} p^{s-1} + \cdots + a_0,$$

we have

$$(x+1)^m = (x+1)^{a_s p^s} (x+1)^{a_{s-1} p^{s-1}} \cdots (x+1)^{a_0}.$$

By Kummer's theorem, we have $\binom{p^j}{k} \equiv 0 \pmod{p}$ for $1 \leq k \leq p^j - 1$. This gives

$$(x+1)^{p^j} = \binom{p^j}{p^j} x^{p^j} + \binom{p^j}{p^j-1} x^{p^j-1} + \cdots + \binom{p^j}{0} \equiv x^{p^j} + 1 \pmod{p}.$$

This holds for $0 \leq j \leq s$. It follows that

$$(x+1)^m \equiv (x^{p^s} + 1)^{a_s} (x^{p^{s-1}} + 1)^{a_{s-1}} \cdots (x+1)^{a_0} \pmod{p}.$$

In view of the right-hand side, to form the monomial x^n , as

$$n = b_s p^s + b_{s-1} p^{s-1} + \cdots + b_0$$

corresponds to the unique base p representation, we must choose b_j factors x^{p^j} for each $0 \leq j \leq s$. The number of ways to do so is exactly

$$\binom{a_s}{b_s} \binom{a_{s-1}}{b_{s-1}} \cdots \binom{a_0}{b_0}.$$

This establishes the result. □

As an example, we have

$$\binom{1000}{361} = \binom{2626_{(7)}}{1024_{(7)}} \equiv \binom{2}{1} \binom{6}{0} \binom{2}{2} \binom{6}{4} = 2 \cdot 1 \cdot 1 \cdot 15 \equiv 2 \pmod{7}.$$

Note that if $b_j > a_j$ for some j , then we must have $\binom{m}{n} \equiv 0 \pmod{p}$.

Theorem 4.6. ([Cauchy-Davenport theorem](#)(柯西-達文波特定理)) Let p be a prime number and let A, B be subsets of $\{0, 1, \dots, p-1\}$. Define

$$A + B = \{a + b \pmod{p} : a \in A, b \in B\}.$$

Then

$$|A + B| \geq \min\{p, |A| + |B| - 1\}.$$

Proof. Let $\alpha = |A|$, $\beta = |B|$ and $\gamma = |A + B|$. Define

$$f(x, y) = \prod_{c \in A+B} (x + y - c) = \sum_{i+j \leq \gamma} f_{ij} x^i y^j \quad (1)$$

where the product runs through all elements c in $A+B$. This suggests why the degree of f is γ . Suppose on the contrary that $\gamma < \min\{p, \alpha + \beta - 1\}$.

Analogous to the Lagrange interpolation formula for polynomials, for each $i \in \mathbb{N}_0$, we can find a polynomial $p_i(x)$ of degree at most $\alpha - 1$ with integer coefficients such that $p_i(a) \equiv a^i \pmod{p}$ for all $a \in A$. Similarly, for each $j \in \mathbb{N}_0$, there exists a polynomial $q_j(y)$ of degree at most $\beta - 1$ such that $q_j(b) \equiv b^j \pmod{p}$ for all $b \in B$.

Now, define

$$g(x, y) = \sum_{\substack{i < \alpha \\ j < \beta}} f_{ij} x^i y^j + \sum_{\substack{\alpha \leq i \leq \gamma \\ j \leq \gamma - i}} f_{ij} p_i(x) y^j + \sum_{\substack{\beta \leq j \leq \gamma \\ i \leq \gamma - j}} f_{ij} x^i q_j(y). \quad (2)$$

For any $a \in A$ and $b \in B$, we have

$$g(a, b) \equiv \sum_{\substack{i < \alpha \\ j < \beta}} f_{ij} a^i b^j + \sum_{\substack{\alpha \leq i \leq \gamma \\ j \leq \gamma - i}} f_{ij} a^i b^j + \sum_{\substack{\beta \leq j \leq \gamma \\ i \leq \gamma - j}} f_{ij} a^i b^j = f(a, b) \pmod{p}$$

since every pair of (i, j) with $i + j \leq \gamma$ belongs to exactly one of the types $i < \alpha, j < \beta$; $\alpha \leq i \leq \gamma, j \leq \gamma - i$; $\beta \leq j \leq \gamma, i \leq \gamma - j$ (note that we cannot have $i \geq \alpha$ and $j \geq \beta$ at the same time since $i + j \leq \gamma < \alpha + \beta - 1$).

In the definition of f in (1), since we have the term $a + b - c$ in the product where $c \equiv a + b \pmod{p}$, we find that $f(a, b) \equiv 0 \pmod{p}$. This leads to $g(a, b) \equiv 0 \pmod{p}$.

If we fix an $a \in A$, then we can rewrite $g(a, y)$ as

$$g_a(y) = g_{a, \beta-1} y^{\beta-1} + g_{a, \beta-2} y^{\beta-2} + \cdots + g_{a, 0},$$

which is a polynomial in y of degree at most $\beta - 1$ (in the first sum of (2), the largest term has degree $\beta - 1$, while in the second sum, the largest term has degree $\gamma - \alpha < \beta - 1$). Since $g_a(y) \equiv 0 \pmod{p}$ has β distinct solutions in y (the elements in B), [proposition 4.1](#) shows that $g_a(y) \equiv 0 \pmod{p}$. Thus, $g_{a, j} \equiv 0 \pmod{p}$ for every j .

Similarly, we note that $g_{a, j}$ is a polynomial in a of degree at most $\alpha - 1$. Since this is 0 for α distinct values of a , it is identically zero modulo p . Therefore, we get $g(x, y) \equiv 0 \pmod{p}$.

Consider the coefficient of $x^{\alpha-1} y^{\gamma-\alpha+1}$ of $g(x, y)$. Note that $\gamma - \alpha + 1 \leq \beta - 1$. From equation (2), this coefficient is $f_{(\alpha-1)(\gamma-\alpha+1)}$. By definition, this is equal to

$$\binom{\gamma}{\alpha-1} = \frac{\gamma(\gamma-1) \cdots (\gamma-\alpha+2)}{(\alpha-1)!}.$$

Since $\gamma < p$, the numerator is not divisible by p . Therefore, this integer is not divisible by p . This contradicts $g(x, y) \equiv 0 \pmod{p}$.

Thus, we must have $\gamma \geq \min\{p, \alpha + \beta - 1\}$. □

The Cauchy-Davenport theorem shows that the sumset $A + B$ modulo p cannot have too few elements. The result does not hold if p is replaced by a power of prime. For example, by considering modulo 9 and the set $A = \{0, 3, 6\}$, we have $A + A = \{0, 3, 6\}$ so that $|A + A| = 3$ but $\min\{9, |A| + |A| - 1\} = 5$.

Problem 4.1. (IMO Prelim 2008 Problem 20) When $(1 + x)^{38}$ is expanded in ascending powers of x , N_1 of the coefficients leave a remainder of 1 when divided by 3, while N_2 of the coefficients leave a remainder of 2 when divided by 3. Find $N_1 - N_2$.

Analysis. Lucas' theorem is useful to solve this simple problem because the coefficients are just some binomial coefficients, and we are finding their remainders modulo 3. The following solution illustrates the use of this theorem.

Solution. The coefficients are $\binom{38}{k}$ where $k = 0, 1, \dots, 38$. We write $38 = 1102_{(3)}$ and $k = \overline{abcd}_{(3)}$ in base 3 representation. By Lucas' theorem, we have

$$\binom{38}{k} \equiv \binom{1}{a} \binom{1}{b} \binom{0}{c} \binom{2}{d} \pmod{3}.$$

To have a nonzero remainder, we must have $a \leq 1$, $b \leq 1$ and $c = 0$. For each of $(a, b) = (0, 0), (0, 1), (1, 0), (1, 1)$, the product $\binom{1}{a} \binom{1}{b} \binom{0}{0}$ equals 1. Note that 2 of the choices $d = 0, 2$ give a remainder 1 and the remaining choice $d = 1$ gives a remainder 2. Thus, we get

$$N_1 - N_2 = 8 - 4 = 4.$$

□

Problem 4.2. Let p be a prime number. Prove that for every integer n , there exist positive integers a, b, c (not necessarily distinct) such that p divides $a!b!c! - n$.

Analysis. Since the expression involves factorials and we are considering modulo a prime, we should think of Wilson's theorem. From $(p - 1)! \equiv -1 \pmod{p}$, we can compute other similar congruences like $(p - 2)! \equiv 1 \pmod{p}$ since $p - 1 \equiv -1 \pmod{p}$. So we can try to see how to combine different $(p - m)!$ together so that their product covers all residue classes modulo p .

Solution. Firstly, we find that for any $1 \leq m \leq p - 1$,

$$\begin{aligned} (p - m)! &= (p - m)(p - m - 1) \cdots (1) \equiv (-m)(-m - 1) \cdots (-p + 1) \\ &= (-1)^{p-m} (m)(m + 1) \cdots (p - 1) \pmod{p}. \end{aligned}$$

Thus, we have

$$m!(p - m)! \equiv (-1)^{p-m} m(p - 1)! \equiv (-1)^{p-m+1} m \pmod{p} \quad (1)$$

by Wilson's theorem. We can now work out a proof based on (1).

If $p \mid n$, then we can simply take $a = b = c = p$. If $p \nmid n$, we take $1 \leq m \leq p-1$ such that $n \equiv m \pmod{p}$. If $p-m+1$ is even, then we take $a = m$, $b = p-m$ and $c = 1$ so that

$$a!b!c! = m!(p-m)! \equiv m \pmod{p}$$

by (1). If $p-m+1$ is odd, then we take $a = m$, $b = p-m$ and $c = p-1$ so that

$$a!b!c! = m!(p-m)!(p-1)! \equiv m \pmod{p}$$

by (1) and Wilson's theorem. This completes the proof. $\square$

Problem 4.3. Let p be a prime number. Prove that for every integer n , there exist positive integers a, b (not necessarily distinct) such that p divides $a^2 + b^2 - n$.

Analysis. Equivalently, we need to find a, b such that $a^2 + b^2 \equiv n \pmod{p}$. As n is arbitrary, this means the set $C = \{a^2 + b^2 \pmod{p}\}$ should contain all residue classes modulo p . Noting that C is just the sumset $A + A$ where $A = \{c^2 \pmod{p}\}$, it remains to find $|A|$ and apply the Cauchy-Davenport theorem.

Solution. Let $A = \{c^2 \pmod{p} : c \in \mathbb{Z}\}$. Note that $c^2 \equiv d^2 \pmod{p}$ iff $p \mid c^2 - d^2$ iff $p \mid c-d$ or $p \mid c+d$. This holds iff $c \equiv \pm d \pmod{p}$. Therefore, $0^2, 1^2, \dots, \left(\frac{p-1}{2}\right)^2$ are pairwise incongruent modulo p , while $\left(\frac{p+1}{2}\right)^2, \dots, (p-1)^2$ are congruent to these. This shows

$$A = \left\{0^2, 1^2, \dots, \left(\frac{p-1}{2}\right)^2 \pmod{p}\right\},$$

so that $|A| = \frac{p+1}{2}$. By the Cauchy-Davenport theorem, we have

$$|A + A| \geq \min \left\{ p, \frac{p+1}{2} + \frac{p+1}{2} - 1 \right\} = p.$$

This shows $A + A$ is a complete set of residues modulo p . In other words, $a^2 + b^2$ runs through every residue class modulo p . Thus, we are done. $\square$

Problem 4.4. (IMO HK TST 2009 Test 2 Problem 3) Prove that there are infinitely many primes p such that $N_p = p^2$, where N_p is the total number of solutions mod p to the equation

$$3x^3 + 4y^3 + 5z^3 - y^4z \equiv 0. \quad (1)$$

Analysis. Since there is only one term involving x , we should first fix y and z and solve the congruence. Note that there are p^2 choices for y, z . Therefore, it suffices to find p such that the equation is uniquely solvable in x . One simple way is to apply [proposition 1.12](#).

Solution. Suppose p is a prime such that $p \equiv 2 \pmod{3}$. Then there exists an inverse of 3 $\pmod{p}$, i.e. there exists r such that $3r \equiv 1 \pmod{p}$. For each of the p^2 choices of y and z , the congruence equation (1) is the same as

$$x^3 \equiv r(-4y^3 - 5z^3 + y^4z) \pmod{p}.$$

By [proposition 1.12](#), as $(3, p-1) = 1$, this congruence has a unique solution modulo p . Therefore, $N_p = p^2$.

It remains to show there are infinitely many primes p such that $p \equiv 2 \pmod{3}$. One may apply Dirichlet's theorem to be introduced in section 8 to get this result directly. However, there is a more elementary proof analogous to [proposition 1.2](#).

Suppose on the contrary that there are only finitely many primes congruent to 2 modulo 3. We order all these primes as $p_1 < p_2 < \dots < p_m$. Consider $n = (p_1 p_2 \dots p_m)^2 - 1$. Clearly, $n > 1$ and $n \equiv 2 \pmod{3}$. Note that n must have a prime divisor which is congruent to 2 modulo 3, since the product of any primes of the form $3k+1$ is still of the form $3k+1$. However, n is not divisible by any p_j since the remainder when n is divided by p_j is -1 . This is a contradiction. Thus, there are infinitely many primes of the form $3k+2$ as desired. $\square$

Problem 4.5. (CMO 2008 Problem 6) Find all triples (p, q, n) satisfying

$$q^{n+2} \equiv 3^{n+2} \pmod{p^n}, \quad p^{n+2} \equiv 3^{n+2} \pmod{q^n},$$

where p, q are odd prime numbers and n is an integer greater than 1.

Analysis. One cannot apply the lifting the exponent lemma to find the p -adic valuation of $q^{n+2} - 3^{n+2}$. The reason is that we do not know whether $q \equiv 3 \pmod{p}$ holds or not. Indeed, this may not be the case in general.

We observe that the moduli p^n, q^n are quite large. Therefore, it is expected that we can use some bounds to obtain useful relations between p and q like what we have done in divisibility problems. Also, note that the special prime 3 should be handled separately, as this probably constitutes the only kind of solution.

Solution. If one of p, q is equal to 3, say $p = 3$, then $q^{n+2} \equiv 3^{n+2} \pmod{3^n}$, which implies $3 \mid q^{n+2}$. Therefore, q is divisible by 3 and it must be equal to 3. The case $p = q = 3$ is clearly a solution for any $n > 1$.

Now, suppose $p, q > 3$. The congruences can be rewritten as

$$p^n \mid q^{n+2} - 3^{n+2} \tag{1}$$

and

$$q^n \mid p^{n+2} - 3^{n+2}. \tag{2}$$

Taking the product of (1) and (2), we get

$$p^n q^n \mid (q^{n+2} - 3^{n+2})(p^{n+2} - 3^{n+2}) = p^{n+2} q^{n+2} - 3^{n+2}(p^{n+2} + q^{n+2} - 3^{n+2}).$$

This implies $p^n q^n \mid 3^{n+2}(p^{n+2} + q^{n+2} - 3^{n+2})$ and hence

$$p^n q^n \mid p^{n+2} + q^{n+2} - 3^{n+2} \quad (3)$$

as $(pq, 3) = 1$. Thus, we get $p^n q^n \leq p^{n+2} + q^{n+2} - 3^{n+2}$ as the right-hand side is positive. WLOG we may assume $p \geq q$. Then we observe that p has to be much larger than q . Indeed, we have

$$p^n q^n < p^{n+2} + q^{n+2} \leq 2p^{n+2} < p^{n+2} q$$

so that $p > q^{\frac{n-1}{2}}$. Combining with (1), we have

$$q^{\frac{n(n-1)}{2}} < p^n \leq q^{n+2} - 3^{n+2} < q^{n+2}.$$

Thus, $\frac{n(n-1)}{2} < n+2$. This is equivalent to $(n-4)(n+1) < 0$, and hence $n = 2, 3$ from $n > 1$.

Case 1. $n = 3$

We need to refine our argument. Firstly, equation (1) becomes

$$p^3 \mid q^5 - 3^5 = (q-3)(q^4 + 3q^3 + 9q^2 + 27q + 81).$$

If $p \mid q-3$, then we can apply the lifting the exponent lemma to get $p^2 \mid q-3$ (since if $p \parallel q-3$, then $p^2 \parallel q^5 - 3^5$ or $p \parallel q^5 - 3^5$ depending on $p = 5$ or not). This yields $q^2 \leq p^2 \leq q-3$ so that $q(q-1) + 3 \leq 0$, which is clearly impossible for $q > 3$. Therefore, we have $p \nmid q-3$ and hence

$$p^3 \mid q^4 + 3q^3 + 9q^2 + 27q + 81.$$

On the other hand, (3) becomes $p^3 q^3 \mid p^5 + q^5 - 3^5$, so that

$$p^3 q^3 \leq p^5 + q^5 - 3^5 < 2p^5.$$

This gives $p > \frac{1}{\sqrt{2}} q^{\frac{3}{2}}$. Then

$$q^4 + 3q^3 + 9q^2 + 27q + 81 \geq p^3 > \frac{1}{2\sqrt{2}} q^{\frac{9}{2}}.$$

The degree on the right-hand side is larger. This means q cannot be large. Indeed, for $q \geq 17$, we have

$$\frac{1}{2\sqrt{2}} q^{\frac{9}{2}} \geq \frac{\sqrt{17}}{2\sqrt{2}} q^4 > \sqrt{2} q^4 > 1.4q^4.$$

Also,

$$\begin{aligned} 1.4q^4 - (q^4 + 3q^3 + 9q^2 + 27q + 81) &\geq 6.8q^3 - (3q^3 + 9q^2 + 27q + 81) \\ &\geq 64.6q^2 - (9q^2 + 27q + 81) \\ &\geq 945.2q - (27q + 81) > 0. \end{aligned}$$

This contradicts the inequality. It suffices to check $q = 5, 7, 11, 13$.

- When $q = 5$, we have $5^5 - 3^5 = 3125 - 243 = 2882 = 2 \times 11 \times 131$.
- When $q = 7$, we have $7^5 - 3^5 = 16807 - 243 = 16564 = 2^2 \times 41 \times 101$.
- When $q = 11$, we have $11^5 - 3^5 = 161051 - 243 = 160808 = 2^3 \times 20101$.
- When $q = 13$, we have $13^5 - 3^5 = 371293 - 243 = 371050 = 2 \times 5^2 \times 41 \times 181$.

None of these is divisible by p^3 .

Case 2. $n = 2$

Equation (1) becomes $p^2 \mid (q-3)(q+3)(q^2+9)$. Note that $(q-3, q+3) = (q-3, 6)$, $(q-3, q^2+9) = (q-3, 18)$ and $(q+3, q^2+9) = (q+3, 18)$. Each of these cannot have prime divisors other than 2, 3. Therefore, at most one of $q-3$, $q+3$ and q^2+9 is divisible by p . If $p^2 \mid q \pm 3$, we have the contradiction $q^2 \leq p^2 \leq q+3$ since $q > 3$. Thus, we obtain $p^2 \mid q^2+9$.

If $p = q$, then $p^2 \mid 9$, which contradicts $p > 3$. If $p \geq q+2$, then

$$(q+2)^2 \leq p^2 \leq q^2+9.$$

This is the same as $4q \leq 5$, which is again impossible.

Therefore, the only solutions are $(p, q, n) = (3, 3, n)$ for any integer $n > 1$. $\square$

Problem 4.6. (IMO Shortlist 2008 N4) Let n be a positive integer. Show that the numbers

$$\binom{2^n-1}{0}, \binom{2^n-1}{1}, \binom{2^n-1}{2}, \dots, \binom{2^n-1}{2^{n-1}-1} \quad (1)$$

are congruent modulo 2^n to $1, 3, 5, \dots, 2^n-1$ in some order.

Analysis. By Lucas' theorem, we only know that all these numbers are odd. Therefore, it suffices to show that any 2 of them are incongruent modulo 2^n . However, we cannot use that theorem to find each $\binom{2^n-1}{k}$ directly since the theorem only applies to prime modulus but not power of a prime.

One way to tackle this problem is to rewrite the binomial coefficients in terms of some $\binom{2^n}{k}$. This makes things easier to handle since we are considering modulo 2^n .

Under this consideration, we find that it is easier to compare $\binom{2^n-1}{a}$ and $\binom{2^n-1}{b}$ if a, b have the same parity.

Solution. The binary representation of 2^n-1 is $\underbrace{11\dots 1}_{n \text{ times}}$. By Lucas' theorem, $\binom{2^n-1}{k}$

is congruent to a product of $\binom{1}{0}$ and $\binom{1}{1}$ modulo 2, which means it must be odd.

When k runs through 0 to $2^{n-1} - 1$, we have 2^{n-1} odd numbers $\binom{2^n - 1}{k}$. Therefore, if any 2 of them are not congruent modulo 2^n , then they must be congruent to $1, 3, 5, \dots, 2^n - 1$ modulo 2^n in some order. As $\binom{2^n - 1}{k} = \binom{2^n - 1}{2^n - 1 - k}$, the numbers in the list (1) are equal to

$$\binom{2^n - 1}{1}, \binom{2^n - 1}{3}, \binom{2^n - 1}{5}, \dots, \binom{2^n - 1}{2^n - 1}.$$

Now, suppose

$$\binom{2^n - 1}{a} \equiv \binom{2^n - 1}{b} \pmod{2^n}$$

for some odd numbers a, b where $1 \leq a < b \leq 2^n - 1$. We use the identity

$$\begin{aligned} & \binom{2^n - 1}{b} - \binom{2^n - 1}{a} \\ &= \left(\binom{2^n - 1}{b} - \binom{2^n - 1}{b-2} \right) + \left(\binom{2^n - 1}{b-2} - \binom{2^n - 1}{b-4} \right) + \dots + \left(\binom{2^n - 1}{a+2} - \binom{2^n - 1}{a} \right) \\ &= \left(\binom{2^n}{b} - \binom{2^n}{b-1} \right) + \left(\binom{2^n}{b-2} - \binom{2^n}{b-3} \right) + \dots + \left(\binom{2^n}{a+2} - \binom{2^n}{a+1} \right). \end{aligned}$$

This shows

$$\binom{2^n}{b} - \binom{2^n}{b-1} + \binom{2^n}{b-2} - \binom{2^n}{b-3} + \dots + \binom{2^n}{a+2} - \binom{2^n}{a+1} \equiv 0 \pmod{2^n}. \quad (2)$$

There are at least 2 terms here since $b \geq a+2$. Note that we can use Kummer's theorem to find the 2-adic valuation of each term. Indeed, we have $v_2\left(\binom{2^n}{k}\right) = n - v_2(k)$. We choose $a+1 \leq c \leq b$ such that $v_2(c)$ is the largest among the numbers in this range. Note that $v_2(c) \geq 1$ and c is uniquely determined, since if $v_2(c) = v_2(d) = \alpha$ for some $a+1 \leq c < d \leq b$, then $d \geq c + 2^\alpha$, and we can take $c' = c + 2^\alpha$ so that $a+1 \leq c' \leq b$ and

$$v_2(c') = v_2\left(2^\alpha \left(\frac{c}{2^\alpha} + 1\right)\right) \geq \alpha + 1$$

as $\frac{c}{2^\alpha}$ is odd. This contradicts the maximal choice of c .

Now, for every $a+1 \leq k \leq b$ where $k \neq c$, we have

$$v_2\left(\binom{2^n}{k}\right) = n - v_2(k) \geq n - (v_2(c) - 1)$$

by the choice of c . Also, we have $v_2\left(\binom{2^n}{c}\right) = n - v_2(c)$. From these, the left-hand side of (2) is divisible by $2^{n-v_2(c)}$ but not $2^{n-v_2(c)+1}$. Thus, it is not divisible by 2^n . This contradicts (2). Hence,

$$\binom{2^n - 1}{a} \not\equiv \binom{2^n - 1}{b} \pmod{2^n}$$

for any odd a, b with $1 \leq a < b \leq 2^n - 1$. This completes the proof from the above arguments. $\square$

Problem 4.7. (IMO Shortlist 2014 N5) Find all triples (p, x, y) consisting of a prime number p and two positive integers x and y such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p .

Analysis. The special case $p = 2$ can be easily done. It is also simple if one of x, y is a multiple of p . For the general case, we find that $x \equiv y \equiv -1 \pmod{p}$ by the Fermat little theorem. Here, we only use the fact that $x^{p-1} + y$ and $x + y^{p-1}$ are divisible by p . We can consider modulo higher powers of p by induction. This is indeed a lifting procedure like the proof of Hensel's lifting lemma.

Solution. If $p = 2$, then we need $x + y$ to be a power of 2. Clearly, we can simply find any $x, y \in \mathbb{N}$ such that $x + y = 2^n$ for some $n \in \mathbb{N}$. In the following, suppose p is an odd prime. Let $x^{p-1} + y = p^\alpha$ and $x + y^{p-1} = p^\beta$ where $\alpha, \beta \in \mathbb{N}$.

If one of x, y is divisible by p , say $p \mid x$, then $p \mid y$ from $x^{p-1} + y = p^\alpha$. Let $p^\gamma \parallel x$ and $p^\delta \parallel y$. Applying **proposition 2.2** to $x^{p-1} + y = p^\alpha$, the smallest 2 among $\gamma(p-1)$, δ and α are equal. Since x^{p-1} and y are both smaller than p^α , α must be greater than the other integers. Thus, we must have $\gamma(p-1) = \delta$. By symmetry, we have $\delta(p-1) = \gamma$. These yield $\delta = \gamma(p-1) = \delta(p-1)^2$. This cannot hold since $\delta > 0$ and $p > 2$.

If $p \nmid x, y$, we apply the Fermat little theorem to get $1 + y \equiv 0 \pmod{p}$ and similarly $1 + x \equiv 0 \pmod{p}$. We write $x = x_1p - 1$ and $y = y_1p - 1$ where $x_1, y_1 \in \mathbb{N}$. Then

$$p^\alpha = x^{p-1} + y \geq (p-1)^{p-1} + (p-1) > (p-1) + (p-1) > p$$

so that $\alpha \geq 2$ and similarly $\beta \geq 2$. Then $x^{p-1} + y = p^\alpha$ implies $x^{p-1} + y \equiv 0 \pmod{p^2}$. Note that we have

$$\begin{aligned} & x^{p-1} + y \equiv 0 \pmod{p^2} \\ \Rightarrow & (x_1p - 1)^{p-1} + y_1p - 1 \equiv 0 \pmod{p^2} \\ \Rightarrow & -x_1p(p-1) + 1 + y_1p - 1 \equiv 0 \pmod{p^2} \\ \Rightarrow & x_1 + y_1 \equiv 0 \pmod{p}. \end{aligned} \tag{1}$$

If one of α, β is 2, say $\alpha = 2$, then we must have $p = 3$ since otherwise

$$p^2 = x^{p-1} + y \geq (p-1)^{p-1} + (p-1) \geq (p-1)^4 + (p-1) = p(p-1)(p^2 - 3p + 3) > p^2.$$

When $p = 3$, from $x^2 + y = 9$, the only possible solutions are $(x, y) = (1, 8), (2, 5)$. One checks that only $(x, y) = (2, 5)$ is a solution as $1 + 8^2 = 65$ is not a power of 3 while $2 + 5^2 = 27$ is.

Now, suppose $\alpha, \beta \geq 3$. We further consider modulo p^3 . Note that

$$\begin{aligned} & x^{p-1} + y \equiv 0 \pmod{p^3} \\ \Rightarrow & (x_1p - 1)^{p-1} + y_1p - 1 \equiv 0 \pmod{p^3} \\ \Rightarrow & x_1^2p^2 \frac{(p-1)(p-2)}{2} - x_1p(p-1) + 1 + y_1p - 1 \equiv 0 \pmod{p^3} \\ \Rightarrow & x_1^2p - x_1p + x_1 + y_1 \equiv 0 \pmod{p^2}. \end{aligned}$$

By symmetry, we have $y_1^2 p - y_1 p + x_1 + y_1 \equiv 0 \pmod{p^2}$. Taking their difference, we get $x_1^2 p - x_1 p \equiv y_1^2 p - y_1 p \pmod{p^2}$ and hence

$$x_1 - y_1 \equiv x_1^2 - y_1^2 \equiv 0 \pmod{p}$$

as $p \mid x_1 + y_1$ from (1). From $x_1 \equiv y_1 \pmod{p}$ and $x_1 + y_1 \equiv 0 \pmod{p}$, we get $p \mid x_1, y_1$ since p is odd. Therefore, we can write $x = x_2 p^2 - 1$ and $y = y_2 p^2 - 1$ for some $x_2, y_2 \in \mathbb{N}$. This motivates the following **claim**.

Claim. For any $2 \leq k < \min\{\alpha, \beta\}$, we have $x \equiv y \equiv -1 \pmod{p^k}$.

Proof. We prove this by induction on k . The base case $k = 2$ is established above. Suppose $x = x_k p^k - 1$ and $y = y_k p^k - 1$ for some $x_k, y_k \in \mathbb{N}$ where $k \geq 2$. Consider the case $k + 1$ and suppose $k + 1 < \min\{\alpha, \beta\}$. As before, we can consider modulo p^{k+2} to get $p^{k+2} \mid x^{p-1} + y$. Note that

$$\begin{aligned} & x^{p-1} + y \equiv 0 \pmod{p^{k+2}} \\ \Rightarrow & (x_k p^k - 1)^{p-1} + y_k p^k - 1 \equiv 0 \pmod{p^{k+2}} \\ \Rightarrow & -x_k p^k (p-1) + 1 + y_k p^k - 1 \equiv 0 \pmod{p^{k+2}} \\ \Rightarrow & -x_k p + x_k + y_k \equiv 0 \pmod{p^2}. \end{aligned} \tag{2}$$

By symmetry, we have $-y_k p + x_k + y_k \equiv 0 \pmod{p^2}$. Together with the above congruence, we get $x_k p \equiv y_k p \pmod{p^2}$ so that $x_k \equiv y_k \pmod{p}$. At the same time, (2) implies $p \mid x_k + y_k$. Therefore, we must have $p \mid x_k, y_k$. From this, it follows that $x \equiv y \equiv -1 \pmod{p^{k+1}}$. The **claim** follows by induction. $\square$

WLOG assume $\alpha \leq \beta$. By the **claim**, we get $x \equiv y \equiv -1 \pmod{p^{\alpha-1}}$. Then

$$p^\alpha = x^{p-1} + y \geq (p^{\alpha-1} - 1)^{p-1} + (p^{\alpha-1} - 1) \geq (p^{\alpha-1} - 1)^2 + (p^{\alpha-1} - 1) = (p^{\alpha-1} - 1)p^{\alpha-1}.$$

It follows that $p \geq p^{\alpha-1} - 1 \geq p^2 - 1$. However, this cannot hold for $p \geq 3$ since $p^2 \geq 3p > p + 1$. There is no solution for $\alpha, \beta \geq 3$.

Thus, the solutions are $(p, x, y) = (2, k, 2^n - k)$ where $n \in \mathbb{N}$ and $1 \leq k \leq 2^n - 1$, and $(p, x, y) = (3, 2, 5), (3, 5, 2)$. $\square$

Here sums up several results and ideas used in the problems in this section. Again, the more useful techniques are those introduced in earlier sections.

- deriving bounds (**problems 4.5, 4.7**)
- Fermat little theorem (**problem 4.7**)
- lifting the exponent lemma (**problem 4.5**)
- q th power lemma (**problem 4.4**)
- Wilson's theorem (**problem 4.2**)

- Kummer's theorem ([problem 4.6](#))
- Lucas' theorem ([problems 4.1, 4.6](#))
- Cauchy-Davenport theorem ([problem 4.3](#))

5 Quadratic Residues

In this section, we focus on quadratic congruence equations. We shall mainly consider such an equation modulo a prime number. If we know how to solve this kind of equations, then we can probably use Hensel's lifting lemma to solve congruence equations modulo higher power of a prime. Afterwards, we can use the Chinese remainder theorem to solve the general case.

The case $p = 2$ is trivial. Suppose p is an odd prime and a, b, c are integers such that $p \nmid a$. Consider the equation

$$am^2 + bm + c \equiv 0 \pmod{p}.$$

By completing square, since $(p, 4a) = 1$, this is the same as

$$(2am + b)^2 \equiv b^2 - 4ac \pmod{p}. \quad (1)$$

If we know how to solve the simpler equation $n^2 \equiv d \pmod{p}$ with given $d \in \mathbb{Z}$, then (1) is reduced to a linear equation, which we know how to solve. Therefore, in the following, we shall focus on the equation

$$n^2 \equiv d \pmod{p}. \quad (2)$$

If $p \mid d$, then the only solution to (2) is $n \equiv 0 \pmod{p}$. If $p \nmid d$, then we say that d is a *quadratic residue* (二次剩餘) modulo p if (2) is solvable. Otherwise, d is called a *quadratic nonresidue* (二次非剩餘). From the solution of [problem 4.3](#), we know that

$$1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2 \pmod{p}$$

are pairwise incongruent. Also, we have $(p-k)^2 \equiv k^2 \pmod{p}$ for $k = 1, 2, \dots, \frac{p-1}{2}$.

Therefore, there are exactly $\frac{p-1}{2}$ quadratic residues modulo p . For each quadratic residue d , equation (2) has exactly 2 solutions modulo p , which are k and $p-k$ for some k . Note that the other $p-1 - \frac{p-1}{2} = \frac{p-1}{2}$ residue classes are quadratic nonresidues.

However, if we only want to know whether a particular d is a quadratic residue, then testing all of $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ is not a good method. There is a more convenient way to do so. We first introduce the following symbol.

Definition 5.1. Let p be an odd prime and let a be an integer. The *Legendre symbol* (勒讓德符號) is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

Although it looks like a fraction, they are totally different things. Therefore, one should specify when one is using the Legendre symbol.

We have a few elementary facts about this symbol, which help us find its value easily in many cases.

Proposition 5.1. Let p be an odd prime and let a, b be integers. Then the following hold.

- (a) if $p \nmid a$, then $\left(\frac{a^2}{p}\right) = 1$
- (b) if $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$
- (c) $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$
- (d) ([Euler's criterion](#)(歐拉判別法)) $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$
- (e) $\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv -1 \pmod{4} \end{cases}$
- (f) $\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{if } p \equiv \pm 3 \pmod{8} \end{cases}$

Proof.

(a) This is true by definition.

(b) If $p \mid a$, then $p \mid a, b$ so that $\left(\frac{a}{p}\right) = 0 = \left(\frac{b}{p}\right)$.

If a is a quadratic residue modulo p , then $n^2 \equiv a \pmod{p}$ for some $n \in \mathbb{Z}$. This implies $n^2 \equiv b \pmod{p}$. Therefore, both a, b are quadratic residues modulo p . Thus, $\left(\frac{a}{p}\right) = 1 = \left(\frac{b}{p}\right)$.

If a is a quadratic nonresidue modulo p , then b cannot be a quadratic residue from above, and vice versa. Hence, both a, b are quadratic nonresidues, and we get $\left(\frac{a}{p}\right) = -1 = \left(\frac{b}{p}\right)$.

(c) A simple proof is to assume Euler's criterion at the moment. In that case, we get

$$\left(\frac{ab}{p}\right) \equiv (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}.$$

Note that both sides can only be $-1, 0, 1$. As $p \geq 3$, the two sides must be equal.

(d) If $p \mid a$, then $\left(\frac{a}{p}\right) = 0 \equiv a^{\frac{p-1}{2}} \pmod{p}$.

If a is a quadratic residue modulo p , then $n^2 \equiv a \pmod{p}$ for some $n \in \mathbb{Z}$ with $p \nmid n$. We have

$$a^{\frac{p-1}{2}} \equiv (n^2)^{\frac{p-1}{2}} = n^{p-1} \equiv 1 = \left(\frac{a}{p}\right) \pmod{p}$$

by the Fermat little theorem.

If a is a quadratic nonresidue modulo p , we pick for each $1 \leq m \leq p-1$ the unique b such that $mb \equiv a \pmod{p}$. Note that $m \not\equiv b \pmod{p}$ since otherwise a is a quadratic residue. This shows the integers $1, 2, \dots, p-1$ can be paired up so that the product of each pair is congruent to a modulo p . Thus,

$$a^{\frac{p-1}{2}} \equiv (p-1)! \equiv -1 = \left(\frac{a}{p}\right) \pmod{p}$$

by Wilson's theorem.

(e) This easily follows from Euler's criterion.

(f) If $p \equiv 1 \pmod{4}$, then we multiply the congruences

$$p-1 \equiv -1 \pmod{p},$$

$$2 \equiv 2 \pmod{p},$$

$$p-3 \equiv -3 \pmod{p},$$

$$4 \equiv 4 \pmod{p},$$

...

$$p - \frac{p-3}{2} \equiv -\frac{p-3}{2} \pmod{p},$$

$$\frac{p-1}{2} \equiv \frac{p-1}{2} \pmod{p}.$$

This gives $2 \times 4 \times \dots \times (p-1) \equiv (-1)^{\frac{p-1}{4}} \left(\frac{p-1}{2}\right)! \pmod{p}$ so that

$$2^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{4}} \pmod{p}$$

since $2 \times 4 \times \dots \times (p-1) = (2 \cdot 1) \times (2 \cdot 2) \times \dots \times \left(2 \cdot \frac{p-1}{2}\right) = 2^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)!$. The result for this case follows from Euler's criterion.

Similarly, if $p \equiv 3 \pmod{4}$, then we multiply the congruences

$$\begin{aligned} p-1 &\equiv -1 \pmod{p}, \\ 2 &\equiv 2 \pmod{p}, \\ p-3 &\equiv -3 \pmod{p}, \\ 4 &\equiv 4 \pmod{p}, \\ &\dots \\ \frac{p-3}{2} &\equiv \frac{p-3}{2} \pmod{p}, \\ p - \frac{p-1}{2} &\equiv -\frac{p-1}{2} \pmod{p}. \end{aligned}$$

This gives $2 \times 4 \times \dots \times (p-1) \equiv (-1)^{\frac{p+1}{4}} \left(\frac{p-1}{2}\right)! \pmod{p}$ so that

$$2^{\frac{p-1}{2}} \equiv (-1)^{\frac{p+1}{4}} \pmod{p}.$$

Again, the result follows from Euler's criterion. □

Among these facts, we usually use (a), (b) and (c) to simplify the Legendre symbols. Facts (e) and (f) can be used to find the values of some simple Legendre symbols. Only Euler's criterion provides a reliable way to determine $\left(\frac{a}{p}\right)$ for the general case. However, this is still not simple in case a and p are large. Finding the remainder of $a^{\frac{p-1}{2}}$ modulo p is rather tedious.

Another useful fact is the following, which can be used to find the values of Legendre symbols easily in many cases.

Theorem 5.1. (Quadratic reciprocity law(二次互反律)) Let p, q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Proof. We first show the following.

Claim. For any $p \nmid a$, we have $\left(\frac{a}{p}\right) = (-1)^s$ where

$$s = \left\lfloor \frac{2a}{p} \right\rfloor + \left\lfloor \frac{4a}{p} \right\rfloor + \dots + \left\lfloor \frac{(p-1)a}{p} \right\rfloor.$$

Here is the proof. For each $t = 1, 2, \dots, \frac{p-1}{2}$, we write $2ta = b_t p + r_t$ where $0 \leq r_t < p$ using the division algorithm. Then

$$s = b_1 + b_2 + \dots + b_{\frac{p-1}{2}}.$$

As p is odd, b_t, r_t have the same parity. Thus,

$$(-1)^s = (-1)^{r_1+r_2+\dots+r_{\frac{p-1}{2}}}. \quad (1)$$

Note that if $r_t \equiv \pm r_u \pmod{p}$, then $2ta \equiv \pm 2ua \pmod{p}$ so that $t \equiv \pm u \pmod{p}$. From the range $1 \leq t, u \leq \frac{p-1}{2}$, this implies $t = u$. Therefore,

$$(-1)^{r_1}r_1, (-1)^{r_2}r_2, \dots, (-1)^{r_{\frac{p-1}{2}}}r_{\frac{p-1}{2}} \pmod{p} \quad (2)$$

are pairwise distinct. If r_j is even, then $(-1)^{r_j}r_j = r_j$ is even. If r_j is odd, then $(-1)^{r_j}r_j \equiv p - r_j$ where $p - r_j$ is even. Therefore, the numbers in (2) are congruent to $2, 4, \dots, p-1$ modulo p in some order. Hence, taking their product, we have

$$(-1)^{r_1+r_2+\dots+r_{\frac{p-1}{2}}}r_1r_2\dots r_{\frac{p-1}{2}} \equiv 2 \times 4 \times \dots \times (p-1) \pmod{p}.$$

Recall that $2ta = b_tp + r_t$. Therefore, $r_t \equiv 2ta \pmod{p}$ so that the above congruence yields

$$(-1)^{r_1+r_2+\dots+r_{\frac{p-1}{2}}} \times (2a) \times (4a) \times \dots \times ((p-1)a) \equiv 2 \times 4 \times \dots \times (p-1) \pmod{p}.$$

This leads to

$$a^{\frac{p-1}{2}} \equiv (-1)^{r_1+r_2+\dots+r_{\frac{p-1}{2}}} \pmod{p}. \quad (3)$$

Combining (1) and (3), and using Euler's criterion, we have proved the [claim](#).

We rewrite the formula given in the [claim](#). Suppose a is odd. For each $1 \leq k \leq \frac{p-1}{2}$, we have

$$\left\lfloor \frac{(p-k)a}{p} \right\rfloor = \left\lfloor a - \frac{ka}{p} \right\rfloor = a - \left\lfloor \frac{ka}{p} \right\rfloor - 1 \equiv \left\lfloor \frac{ka}{p} \right\rfloor \pmod{2}.$$

Therefore, we can replace $\left\lfloor \frac{(p-1)a}{p} \right\rfloor, \left\lfloor \frac{(p-3)a}{p} \right\rfloor, \dots$ by $\left\lfloor \frac{a}{p} \right\rfloor, \left\lfloor \frac{3a}{p} \right\rfloor, \dots$ in the exponent of the formula. This gives $\left(\frac{a}{p}\right) = (-1)^{s'}$ where

$$s' = \left\lfloor \frac{a}{p} \right\rfloor + \left\lfloor \frac{2a}{p} \right\rfloor + \dots + \left\lfloor \frac{\frac{p-1}{2} \cdot a}{p} \right\rfloor.$$

Therefore, to complete the proof of the theorem, it suffices to show

$$\left(\left\lfloor \frac{q}{p} \right\rfloor + \left\lfloor \frac{2q}{p} \right\rfloor + \dots + \left\lfloor \frac{\frac{p-1}{2} \cdot q}{p} \right\rfloor \right) + \left(\left\lfloor \frac{p}{q} \right\rfloor + \left\lfloor \frac{2p}{q} \right\rfloor + \dots + \left\lfloor \frac{\frac{q-1}{2} \cdot p}{q} \right\rfloor \right) = \frac{p-1}{2} \cdot \frac{q-1}{2}. \quad (4)$$

We shall show that each side of (4) counts the number N of lattice points (m, n) with $1 \leq m \leq \frac{p}{2}$ and $1 \leq n \leq \frac{q}{2}$. Firstly, since p, q are odd, it is clear that $N = \frac{p-1}{2} \cdot \frac{q-1}{2}$.

Next, we consider those points (m, n) with $1 \leq m \leq \frac{p}{2}$, $1 \leq n \leq \frac{q}{2}$ and $\frac{n}{m} < \frac{q}{p}$. For each $1 \leq m \leq \frac{p}{2}$, we need $n < \frac{mq}{p}$. Since $(mq, p) = 1$, this holds iff $n \leq \frac{mq}{p}$. If this is true, then we immediately get $n \leq \frac{mq}{p} \leq \frac{q}{2}$. Therefore, the number of such n is $\left\lfloor \frac{mq}{p} \right\rfloor$. Thus, there are

$$\left\lfloor \frac{q}{p} \right\rfloor + \left\lfloor \frac{2q}{p} \right\rfloor + \cdots + \left\lfloor \frac{\frac{p-1}{2} \cdot q}{p} \right\rfloor \quad (5)$$

lattice points in this case.

Similarly, we consider those points (m, n) with $1 \leq m \leq \frac{p}{2}$, $1 \leq n \leq \frac{q}{2}$ and $\frac{n}{m} > \frac{q}{p}$. By symmetry, there are

$$\left\lfloor \frac{p}{q} \right\rfloor + \left\lfloor \frac{2p}{q} \right\rfloor + \cdots + \left\lfloor \frac{\frac{q-1}{2} \cdot p}{q} \right\rfloor \quad (6)$$

lattice points in this case.

Note that there are no lattice points (m, n) with $1 \leq m \leq \frac{p}{2}$, $1 \leq n \leq \frac{q}{2}$ and $\frac{n}{m} = \frac{q}{p}$, since in that case we need $q \mid n$ and $p \mid m$. Hence, the sum of (5) and (6) equals N . This establishes (4) and hence the theorem. $\square$

The quadratic reciprocity law can be rewritten as $\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \left(\frac{q}{p}\right)$. It follows that $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ if one of p, q is of the form $4k+1$, and $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$ if both p, q are of the form $4k+3$.

This theorem relates quadratic residues modulo different prime moduli, which is an unexpected result from its definition. This helps simplify the Legendre symbol in many cases.

There are analogous results for higher powers. However, those extensions require investigations in complex numbers, which do not simplify the things in practice. Thus, we only talk about quadratic things here.

Problem 5.1. Determine whether a is a quadratic residue modulo p in each of the following cases.

- (a) $a = 193$ and $p = 37$
- (b) $a = 3210$ and $p = 113$

Analysis. The following solution provides some standard ways to find the values of Legendre symbols. Firstly, we can use [proposition 5.1\(a\)](#), [\(b\)](#) and [\(c\)](#) to reduce the size of the number. When it is small enough, we can use [proposition 5.1\(e\)](#), [\(f\)](#), Euler's criterion and simple facts to find its value. In addition, we can use the quadratic reciprocity law to change the modulus, so that the same process can be repeated until we get the answer.

Solution.

(a) Since $193 \equiv 8 \pmod{37}$, we have $\left(\frac{193}{37}\right) = \left(\frac{8}{37}\right)$ by [proposition 5.1\(b\)](#). Next, [proposition 5.1\(c\)](#) and [\(a\)](#) give $\left(\frac{8}{37}\right) = \left(\frac{2}{37}\right)\left(\frac{4}{37}\right) = \left(\frac{2}{37}\right)$. Hence, we can use [proposition 5.1\(f\)](#) to find $\left(\frac{193}{37}\right) = \left(\frac{2}{37}\right) = -1$ as $37 \equiv -3 \pmod{8}$. This means 193 is a quadratic nonresidue modulo 37.

(b) Similar to part [\(a\)](#), we find that

$$\left(\frac{3210}{113}\right) = \left(\frac{46}{113}\right) = \left(\frac{2}{113}\right)\left(\frac{23}{113}\right) = \left(\frac{23}{113}\right)$$

as $113 \equiv 1 \pmod{8}$. To continue, we use the quadratic reciprocity law. This gives

$$\left(\frac{23}{113}\right) = \left(\frac{113}{23}\right) = \left(\frac{-2}{23}\right) = \left(\frac{-1}{23}\right)\left(\frac{2}{23}\right) = -1.$$

Again, this shows 3210 is a quadratic nonresidue modulo 113.

□

Problem 5.2. Find all prime numbers p such that $p! + p$ is a perfect square.

Analysis. For any prime q smaller than p (if exists), we know that p is a quadratic residue modulo q from the condition. This shows $\left(\frac{p}{q}\right) = 1$ for any prime $q < p$.

It follows that $\left(\frac{q}{p}\right) = 1$ in general, since we can easily show that $p \equiv 1 \pmod{4}$. Therefore, every odd prime smaller than p is a quadratic residue modulo p , which should not be the case for large p , since otherwise we would have a lot of quadratic residues.

Solution. Clearly, $p = 2, 3$ are solutions since $2! + 2 = 4$ and $3! + 3 = 9$.

Suppose $p \geq 5$ and let $p! + p = k^2$ for some $k \in \mathbb{N}$. Note that p divides the left-hand side. Therefore, $p \mid k^2$ so that $p \mid k$. Let $k = pm$. Then $(p-1)! + 1 = pm^2$. As $p \geq 5$, we have $4 \mid (p-1)!$. This shows pm^2 is of the form $4n+1$. As $m^2 \equiv 0, 1 \pmod{4}$, we must have $p \equiv 1 \pmod{4}$.

Consider any prime q smaller than p . As $q \mid p!$, we have $p \equiv k^2 \pmod{q}$. Hence, $\left(\frac{p}{q}\right) = 1$. By the quadratic reciprocity law, as $p \equiv 1 \pmod{4}$, this implies $\left(\frac{q}{p}\right) = 1$.

In general, for any odd integer a such that $1 < a < p$, we can write $a = q_1^{\alpha_1} q_2^{\alpha_2} \cdots q_s^{\alpha_s}$ with distinct odd primes q_j smaller than p . Then

$$\left(\frac{a}{p}\right) = \left(\frac{q_1}{p}\right)^{\alpha_1} \left(\frac{q_2}{p}\right)^{\alpha_2} \cdots \left(\frac{q_s}{p}\right)^{\alpha_s} = 1$$

as each $\left(\frac{q_j}{p}\right)$ is 1. In other words, every odd integer from 3 to $p-2$ is a quadratic residue modulo p . Certainly, 1 and 4 are also quadratic residues. Thus, we have obtained at least $\frac{(p-2)-3}{2} + 1 + 2 = \frac{p+1}{2}$ quadratic residues modulo p . This is a contradiction since there are exactly $\frac{p-1}{2}$ quadratic residues.

Therefore, there are no solutions other than $p = 2, 3$. $\square$

Problem 5.3. (USAMO 2008 Problem 1) Prove that for each positive integer n , there are pairwise relatively prime integers $k_0, k_1, \dots, k_n$, all strictly greater than 1, such that $k_0 k_1 \cdots k_n - 1$ is the product of two consecutive integers.

Analysis. It suffices to find $n+1$ primes p such that $m(m+1)+1 \equiv 0 \pmod{p}$ is solvable. Once this is done, we can use the Chinese remainder theorem to prove the existence of m such that $m(m+1)+1$ has at least $n+1$ distinct prime divisors. This clearly implies what we want. Thus, the problem is reduced to a problem in quadratic congruence equation. The solvability can be checked using the Legendre symbol.

Solution. Consider $m^2 + m + 1 \equiv 0 \pmod{p}$ for some odd prime $p > 3$. This is equivalent to $(2m+1)^2 \equiv -3 \pmod{p}$. As p is odd, this is solvable iff -3 is a quadratic residue modulo p .

We find that

$$\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{3}{p}\right) = \left(\frac{p}{3}\right). \quad (1)$$

Here, we have applied the quadratic reciprocity law. Note that when we invert $\left(\frac{3}{p}\right)$ to $\left(\frac{p}{3}\right)$, a negative sign is added iff $p \equiv -1 \pmod{4}$, which holds iff $\left(\frac{-1}{p}\right) = -1$.

Therefore, the sign must cancel out with $\left(\frac{-1}{p}\right)$ in any case.

From (1), the equation $m^2 + m + 1 \equiv 0 \pmod{p}$ is solvable iff $p \equiv 1 \pmod{3}$. If we can show that there are infinitely many primes of the form $3t+1$, we can pick $n+1$ of them $p_0, p_1, \dots, p_n$. Then for each $0 \leq j \leq n$, there exists m_j such that

$$m_j^2 + m_j + 1 \equiv 0 \pmod{p_j}$$

from above. By the Chinese remainder theorem, we have $m \equiv m_j \pmod{p_j}$ for $0 \leq j \leq n$ for some $m \in \mathbb{N}$. Then $m^2 + m + 1$ is a multiple of $p_0 p_1 \cdots p_n$. Let the prime factorization be $m^2 + m + 1 = p_0^{\alpha_0} p_1^{\alpha_1} \cdots p_s^{\alpha_s}$ with $s \geq n$. Then we can simply take

$$k_0 = p_0^{\alpha_0}, \quad k_1 = p_1^{\alpha_1}, \quad \dots, \quad k_{n-1} = p_{n-1}^{\alpha_{n-1}}, \quad k_n = p_n^{\alpha_n} \cdots p_s^{\alpha_s}$$

so that $k_0, k_1, \dots, k_n$ are pairwise relatively prime and $k_0 k_1 \cdots k_n - 1 = m^2 + m = m(m+1)$ as desired.

Thus, it remains to show that there are infinitely many primes of the form $3t + 1$. This again follows from Dirichlet's theorem. But we shall give an alternative proof here. Note that the same argument used in [problem 4.4](#) does not apply in this case, since an integer of the form $3t + 1$ may not have prime divisors of the same form.

Suppose on the contrary that there are only finitely many primes congruent to 1 modulo 3. We order all these primes as $q_1 < q_2 < \cdots < q_\ell$. Consider $N = (q_1 q_2 \cdots q_\ell)^2 + 3$. Note that $3 \nmid N$ and $N > 1$. Also, we have $(N, q_j) = (3, q_j) = 1$ for any j . Therefore, any prime divisor of N is of the form $3t + 2$. We pick any such prime divisor p . Then $p \mid Q^2 + 3$ where $Q = q_1 q_2 \cdots q_\ell$ so that -3 is a quadratic residue modulo p . This contradicts to what we have found early in this solution.

Hence, there are infinitely many primes of the form $3t + 1$ and the proof is complete. $\square$

Problem 5.4. (Romania TST 2008 Day 3 Problem 3) Let $m, n \geq 3$ be positive odd integers. Prove that $2^m - 1$ doesn't divide $3^n - 1$.

Analysis. Consider any prime p dividing both $2^m - 1$ and $3^n - 1$. We note that $2 \equiv 2^{m+1} \pmod{p}$ and $3 \equiv 3^{n+1} \pmod{p}$ are even powers modulo p . Thus, 2 and 3 are quadratic residues modulo p . This is how we relate this problem to quadratic residues.

Solution. Suppose there exists a prime p such that $2^m \equiv 3^n \equiv 1 \pmod{p}$. Firstly, this implies $3 \equiv 3^{n+1} \pmod{p}$. As $n + 1$ is even, this shows 3 is a quadratic residue modulo p . Thus,

$$1 = \left(\frac{3}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{p}{3}\right).$$

If $\left(\frac{-1}{p}\right) = \left(\frac{p}{3}\right) = 1$, then $p \equiv 1 \pmod{4}$ and $p \equiv 1 \pmod{3}$. If $\left(\frac{-1}{p}\right) = \left(\frac{p}{3}\right) = -1$, then $p \equiv 3 \pmod{4}$ and $p \equiv 2 \pmod{3}$. Hence, we have $p \equiv \pm 1 \pmod{12}$.

Now, we note that $2^m - 1 \equiv -1 \pmod{4}$ and $2^m - 1 \equiv 1 \pmod{3}$ since $m \geq 3$ and m is odd. Thus, $2^m - 1 \equiv 7 \pmod{12}$. Therefore, not all prime divisors of $2^m - 1$ are of the form $12k \pm 1$. If we pick a prime divisor p of $2^m - 1$ with $p \not\equiv \pm 1 \pmod{12}$, then this prime p cannot divide $3^n - 1$ from above. Hence, $2^m - 1 \nmid 3^n - 1$. $\square$

Problem 5.5. (IMO Shortlist 1998 N5) Determine all positive integers n for which there exists an integer m such that $2^n - 1$ is a divisor of $m^2 + 9$.

Analysis. For any prime divisor p of $2^n - 1$, we require -9 to be a quadratic residue modulo p . This forces $p = 3$ or $p \equiv 1 \pmod{4}$. It means we need to find all n such that all prime divisors of $2^n - 1$ are of these forms.

Solution. Firstly, suppose n has an odd prime divisor q . Then we have $2^q - 1 \mid 2^n - 1$ so that $2^q - 1 \mid m^2 + 9$. Note that $2^q - 1 \equiv 3 \pmod{4}$. Therefore, $2^q - 1$ has a prime divisor p of the form $4k + 3$. This gives $p \mid m^2 + 9$ so that -9 is a quadratic residue modulo p . Note that $p \neq 3$ since the odd powers of 2 are congruent to 2 modulo 3. Thus, we find that

$$\left(\frac{-9}{p}\right) = \left(\frac{9}{p}\right) \left(\frac{-1}{p}\right) = \left(\frac{-1}{p}\right) = -1.$$

This is a contradiction.

Therefore, it suffices to consider $n = 2^\alpha$ for some $\alpha \geq 0$. We claim that all such n 's are solutions. Clearly, $\alpha = 0$ is possible as $2^0 - 1 = 1$ where we can take any $m \in \mathbb{N}$. For $\alpha \geq 1$, n is even so that $3 \mid 2^n - 1$. We write $2^n - 1 = 3^\beta p_1^{\beta_1} p_2^{\beta_2} \cdots p_s^{\beta_s}$. Note that $\beta = 1$ since $2^{2^\alpha} - 1 \pmod{9}$ follows the pattern $3, 6, 3, 6, \dots$ for $\alpha = 1, 2, 3, 4, \dots$. We can also show that $p_j \equiv 1 \pmod{4}$ for each $1 \leq j \leq s$ as follows.

Indeed, from $2^{2^\alpha} \equiv 1 \pmod{p_j}$, we have $2^{2^{\alpha-1}} \equiv \pm 1 \pmod{p_j}$. If $2^{2^{\alpha-1}} \equiv 1 \pmod{p_j}$, then we have $2^{2^{\alpha-2}} \equiv \pm 1 \pmod{p_j}$. Inductively, since we cannot have $2 \equiv 1 \pmod{p_j}$, there must be an exponent γ with $0 \leq \gamma \leq \alpha - 1$ such that $2^{2^\gamma} \equiv -1 \pmod{p_j}$.

If $\gamma \geq 1$, then 2^γ is even, and hence -1 is a quadratic residue modulo p_j . This yields $p_j \equiv 1 \pmod{4}$ by [proposition 5.1\(e\)](#). If $\gamma = 0$, then $2 \equiv -1 \pmod{p_j}$, which forces $p_j = 3$. This is rejected since the prime divisor 3 has been singled out.

Now, for each $1 \leq j \leq s$, the equation $m^2 \equiv -9 \pmod{p_j}$ is solvable, as $\left(\frac{-9}{p_j}\right) = 1$ for $p_j \equiv 1 \pmod{4}$. Then, by Hensel's lifting lemma, the equation $m^2 \equiv -9 \pmod{p_j^{\beta_j}}$ is solvable since $2m \not\equiv 0 \pmod{p_j}$. We pick any such solution m_j . By the Chinese remainder theorem, we can find $m \in \mathbb{N}$ such that $m \equiv m_j \pmod{p_j^{\beta_j}}$ for each j and $3 \mid m$. For such an m , we have $2^n - 1 \mid m^2 + 9$.

Therefore, the answer is $n = 2^\alpha$ for any integer $\alpha \geq 0$. □

The entire section is about quadratic residues. One should understand this concept and various ways to compute or simplify the Legendre symbols, including the quadratic reciprocity law.

6 Orders and Primitive Roots

The purpose of this section is to go through some facts related to powers of integers modulo a certain integer. As in the solution of [problem 5.5](#), we find that there are some patterns of $a^k \pmod{n}$ for fixed a, n when k runs through the positive integers. We have a more concrete concept on this topic.

Definition 6.1. Let n be a positive integer and let a be an integer with $(a, n) = 1$. Then the *order* (指數) of a modulo n is defined to be the smallest positive integer d such that $a^d \equiv 1 \pmod{n}$.

By the Euler-Fermat theorem, we have $a^{\varphi(n)} \equiv 1 \pmod{n}$. Hence, such an integer d exists and we must have $d \leq \varphi(n)$. Indeed, we have more than that.

Proposition 6.1. Let n be a positive integer and let a be an integer with $(a, n) = 1$. Let d be the order of a modulo n . Then $a^k \equiv a^m \pmod{n}$ iff $k \equiv m \pmod{d}$ for any $k, m \in \mathbb{N}_0$.

Proof. WLOG assume $k \geq m$. Write $k - m = qd + r$ where $0 \leq r < d$ by the division algorithm. As $(a, n) = 1$, we have $a^k \equiv a^m \pmod{n}$ iff $a^{k-m} \equiv 1 \pmod{n}$. This holds iff $a^{qd+r} \equiv 1 \pmod{n}$. Note that

$$a^{qd+r} = (a^d)^q a^r \equiv a^r \pmod{n}.$$

From this, if $k \equiv m \pmod{d}$, then $r = 0$ so that $a^{qd+r} \equiv 1 \pmod{n}$. It follows that $a^k \equiv a^m \pmod{n}$.

Conversely, if $a^k \equiv a^m \pmod{n}$, then $a^r \equiv 1 \pmod{n}$. By the minimality of d in the definition of order, we can only have $r = 0$. This implies $k \equiv m \pmod{d}$.

Thus, the equivalence is established. □

In particular, we have $a^k \equiv 1 \pmod{n}$ iff $d \mid k$. This shows the order d must divide $\varphi(n)$. Also, the above proposition shows that the powers of a can only be congruent to $1, a, a^2, \dots, a^{d-1}$ modulo n , and these d numbers are pairwise incongruent. Note that [proposition 6.1](#) holds even for $k, m \in \mathbb{Z}$, with the convention of negative powers mentioned before.

There is no simple way to find the order of a modulo n . But since we know that the order d must divide $\varphi(n)$, it suffices to test the positive divisors of $\varphi(n)$. We can first test whether $a^{\frac{\varphi(n)}{p}} \equiv 1 \pmod{n}$ for each prime divisor p of $\varphi(n)$. If none of these holds, then it means $a^b \not\equiv 1 \pmod{n}$ for every proper divisor b of $\varphi(n)$. Thus, we conclude that the order is $\varphi(n)$. If one of these holds, say $a^{\frac{\varphi(n)}{p}} \equiv 1 \pmod{n}$, then

d must divide $\frac{\varphi(n)}{p}$ and we can repeat the same procedure. Eventually, we must obtain the order we want.

It would be nice if we have $d = \varphi(n)$, since in that case the powers of a run through all residue classes modulo n that are coprime to n . This motivates the following definition.

Definition 6.2. Let n be a positive integer and let a be an integer with $(a, n) = 1$. Then a is said to be a *primitive root* (原根) modulo n if the order of a modulo n is $\varphi(n)$.

As mentioned, if a is a primitive root modulo n , then $1, a, a^2, \dots, a^{\varphi(n)-1}$ are pairwise incongruent modulo n . Thus, they correspond to the $\varphi(n)$ residue classes modulo n which are relatively prime to n . However, such a primitive root may not exist. The following existence theorem of primitive roots is the most important fact in this topic.

Theorem 6.1. Let n be a positive integer. Then there exists a primitive root modulo n iff $n = 1, 2, 4, p^\alpha, 2p^\alpha$ for some odd prime p and positive integer α .

Proof. We have several cases to consider. Firstly, we may assume $n > 1$ since the case $n = 1$ is trivial.

Case 1. $n = 2^\alpha$ where $\alpha \geq 1$

When $\alpha = 1$, we can take $g = 1$ so that the order of g modulo 2 is $\varphi(2) = 1$.

When $\alpha = 2$, we can take $g = 3$ so that the order of g modulo 4 is $\varphi(4) = 2$. This is because $3 \not\equiv 1 \pmod{4}$ but $3^2 \equiv 1 \pmod{4}$.

When $\alpha = 3$, the only residue classes modulo 8 coprime to 8 are $a \equiv 1, 3, 5, 7$. Since $a^2 \equiv 1 \pmod{8}$ for each of these cases, and $\varphi(8) = 4$, there is no primitive root by definition.

Suppose $n = 2^\alpha$ has a primitive root for some $\alpha \geq 3$. We can pick the smallest α for which this happens. Then $\alpha \geq 4$. This means there exists $g \in \mathbb{Z}$ such that $1, g, g^2, \dots, g^{2^{\alpha-1}-1}$ run through all the odd residue classes modulo 2^α . In particular, the powers of g also run through all the odd residue classes modulo $2^{\alpha-1}$. This shows g is a primitive root modulo $2^{\alpha-1}$ where $\alpha - 1 \geq 3$. This contradicts the minimality of α . Hence, n has no primitive root when $\alpha \geq 3$.

Case 2. $n = p^\alpha$ where p is an odd prime and $\alpha \geq 1$

We first suppose $\alpha = 1$ so that $n = p$ is a prime. Consider any divisor d of $p - 1$. We have

$$a^{p-1} - 1 = (a^d - 1)(a^{p-1-d} + a^{p-1-2d} + \dots + 1).$$

The equation $a^{p-1} \equiv 1 \pmod{p}$ has exactly $p-1$ solutions by the Fermat little theorem. The equation $a^{p-1-d} + a^{p-1-2d} + \dots + 1 \equiv 0 \pmod{p}$ has at most $p-1-d$ solutions by [proposition 4.1](#). It follows that there are at least $(p-1) - (p-1-d) = d$ solutions to $a^d \equiv 1 \pmod{p}$. Again, by [proposition 4.1](#), there are at most d solutions. Combining these, we know that there are exactly d solutions to $a^d \equiv 1 \pmod{p}$. Note that the solutions are exactly those a whose orders divide d .

For each divisor d of $p-1$, let $N(d)$ be the number of $a \pmod{p}$ with order d . From above, we have $\sum_{d'|d} N(d') = d$ for every d , where the summation runs through all positive divisors of d . We shall prove that $N(d) = \varphi(d)$ for every d . Suppose on the contrary that $N(d) \neq \varphi(d)$ for some $d \mid p-1$. We pick the smallest such d . Note that $d \neq 1$ since we have $N(1) = 1 = \varphi(1)$. We claim that

$$\sum_{d'|d} \varphi(d') = d. \quad (1)$$

For every integer ℓ with $1 \leq \ell \leq d$, define $\ell' = \frac{\ell}{d'}$ where $d' = (\ell, d)$. Note that $\left(\ell', \frac{d}{d'}\right) = \frac{(\ell, d)}{d'} = 1$. Conversely, for each divisor d' of d , and for each of the $\varphi\left(\frac{d}{d'}\right)$ choices of ℓ' such that $1 \leq \ell' \leq \frac{d}{d'}$ and $\left(\ell', \frac{d}{d'}\right) = 1$, we can define $\ell = \ell'd'$ so that $1 \leq \ell \leq d$ and $(\ell, d) = d'$. Thus, by double counting, we get

$$d = \sum_{d'|d} \varphi\left(\frac{d}{d'}\right).$$

Using the change of variable $d'' = \frac{d}{d'}$, this is equal to $\sum_{d''|d} \varphi(d'')$ as claimed.

From (1) and the minimality of d , we find that

$$d = \sum_{d'|d} N(d') = \sum_{\substack{d'|d \\ d' < d}} \varphi(d') + N(d) = (d - \varphi(d)) + N(d).$$

This yields the contradiction $N(d) = \varphi(d)$. Therefore, $N(d) = \varphi(d)$ should hold for any $d \mid p-1$. In particular, $N(p-1) = \varphi(p-1)$. This shows there are exactly $\varphi(p-1)$ primitive roots modulo p .

We now move on to the general case $n = p^\alpha$. We pick a primitive root g modulo p . Suppose first that $g^{p-1} \not\equiv 1 \pmod{p^2}$. Let d be the order of g modulo p^2 . Then $d \mid \varphi(p^2) = p(p-1)$. Also, since $g^d \equiv 1 \pmod{p^2}$ implies $g^d \equiv 1 \pmod{p}$, we must have $p-1 \mid d$ by [proposition 6.1](#). Therefore, we can only have $d = p(p-1)$ or $p-1$. The latter case is rejected since we have assumed $g^{p-1} \not\equiv 1 \pmod{p^2}$. Thus, g is a primitive root modulo p^2 .

If $g^{p-1} \equiv 1 \pmod{p^2}$, then we consider $g+p$. It is again a primitive root modulo p

since $g + p \equiv g \pmod{p}$. We check that

$$(g + p)^{p-1} \equiv g^{p-1} + (p-1)g^{p-2}p \equiv 1 - pg^{p-2} \pmod{p^2}.$$

Since $(g, p) = 1$, this is not congruent to 1 modulo p^2 . Thus, we can apply the same argument to $g + p$ and show that it is a primitive root modulo p^2 .

Suppose g is a primitive root modulo p such that $g^{p-1} \not\equiv 1 \pmod{p^2}$ (we can replace g by $g + p$ if necessary). We shall prove inductively on α that g is a primitive root modulo p^α for any $\alpha \geq 2$. Indeed, as in the base case $\alpha = 2$, it suffices to verify $g^{p^{\alpha-2}(p-1)} \not\equiv 1 \pmod{p^\alpha}$ for $\alpha \geq 2$.

Suppose $g^{p^{t-2}(p-1)} \not\equiv 1 \pmod{p^t}$ for some $t \geq 2$. As g is a primitive root modulo p^{t-1} , we have $g^{p^{t-2}(p-1)} = 1 + kp^{t-1}$ where $p \nmid k$ from the assumption. We check that

$$g^{p^{t-1}(p-1)} = (1 + kp^{t-1})^p \equiv 1 + kp^t \not\equiv 1 \pmod{p^{t+1}}.$$

Hence, the case $\alpha = t + 1$ is established. By induction, g is a primitive root modulo p^α for any $\alpha \geq 1$.

Case 3. $n = 2p^\alpha$ where p is an odd prime and $\alpha \geq 1$

Let g be an odd primitive root modulo p^α (if g is even, then we pick $g + p^\alpha$). We claim that g is a primitive root modulo $2p^\alpha$. Indeed, let d be the order of g modulo $2p^\alpha$. As above, we have $\varphi(p^\alpha) \mid d$ and $d \mid \varphi(2p^\alpha)$. However, since $\varphi(p^\alpha) = p^{\alpha-1}(p-1) = \varphi(2p^\alpha)$, we get $d = \varphi(2p^\alpha)$ so that the claim is true.

Case 4. $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_s^{\alpha_s}$ where $s \geq 2$ and $n \neq 2p^\alpha$

Consider any integer a . For each $1 \leq j \leq s$, we have $a^{\varphi(p_j^{\alpha_j})} \equiv 1 \pmod{p_j^{\alpha_j}}$ by the Euler-Fermat theorem. Then any multiple c of $\varphi(p_j^{\alpha_j})$ satisfies $a^c \equiv 1 \pmod{p_j^{\alpha_j}}$. In particular, consider the least common multiple ℓ of $\varphi(p_1^{\alpha_1}), \varphi(p_2^{\alpha_2}), \dots, \varphi(p_s^{\alpha_s})$. Then $a^\ell \equiv 1 \pmod{p_j^{\alpha_j}}$ for every j . Hence, $a^\ell \equiv 1 \pmod{n}$. This shows the order of a modulo n is at most ℓ .

Note that $\varphi(n) = \varphi(p_1^{\alpha_1})\varphi(p_2^{\alpha_2})\cdots\varphi(p_s^{\alpha_s})$. As the integers $\varphi(p_1^{\alpha_1}), \varphi(p_2^{\alpha_2}), \dots, \varphi(p_s^{\alpha_s})$ are not pairwise relatively prime (indeed at least two of them are even since only $\varphi(2)$ is odd), their least common multiple is less than their product. This shows the order of a modulo n is less than $\varphi(n)$. Thus, n cannot have any primitive root in this case.

The proof is complete by combining the 4 cases. □

As in the case of order, there is no simple way to find out a primitive root of n when n has one of the forms described in the existence theorem of primitive roots. However, in most cases we only need to use the existence of such a primitive root.

We now go through the corresponding result of [proposition 1.12](#) when $q \mid p-1$, as an application of primitive roots.

Proposition 6.2. Let p and q be odd primes such that $q \mid p-1$. Then there are exactly $\frac{p-1}{q}$ nonzero residue classes a modulo p such that the congruence equation

$$x^q \equiv a \pmod{p}$$

in x is solvable. In that case, there are exactly q solutions.

Proof. Let g be a primitive root modulo p . Since $(a, p) = 1$, x cannot be divisible by p . Let $a \equiv g^\alpha \pmod{p}$ and $x \equiv g^y \pmod{p}$. Then the equation $x^q \equiv a \pmod{p}$ holds iff $g^{qy} \equiv g^\alpha \pmod{p}$, which holds iff $qy \equiv \alpha \pmod{p-1}$ since g is a primitive root. This equation in y is solvable iff $q \mid \alpha$ since $q \mid p-1$. Thus, α can only be $q, 2q, \dots, p-1$. There are $\frac{p-1}{q}$ possible values of α , which corresponds to $\frac{p-1}{q}$ possibilities of a .

Suppose $\alpha = \beta q$ where $1 \leq \beta \leq \frac{p-1}{q}$. Then we need to solve $y \equiv \beta \pmod{\frac{p-1}{q}}$. There is a unique solution modulo $\frac{p-1}{q}$, and hence there are q solutions modulo $p-1$. These correspond to q distinct solutions of x modulo p , as desired. $\square$

Problem 6.1.

- (a) Find the order of 2 modulo 23.
- (b) Find a primitive root of 23.

Analysis. As mentioned, there is no fast method other than trial and error.

Solution.

- (a) Let d be the order. Then d divides $\varphi(23) = 22$. Firstly, as $2^2 = 4 \not\equiv 1 \pmod{23}$, we cannot have $d = 1$ or $d = 2$. Next, we check that

$$2^{11} = 2048 \equiv 1 \pmod{23}.$$

Therefore, the order of 2 modulo 23 is 11.

- (b) From (a), a simple way is to write 2 as a square modulo 23. Indeed, we easily find that $2 \equiv 5^2 \pmod{23}$. We claim that 5 is a primitive root of 23. Clearly, $5^2 \equiv 2 \not\equiv 1 \pmod{23}$. Also, we have

$$5^{11} = 5 \times (5^2)^5 \equiv 5 \times 2^5 = 160 \equiv -1 \pmod{23}.$$

Thus, the order of 5 modulo 23 is 22, and hence 5 is a primitive root of 23. $\square$

Problem 6.2. Show that if n is an integer greater than 1, then n does not divide $2^n - 1$.

Analysis. Suppose $n \mid 2^n - 1$. Then $2^n \equiv 1 \pmod{n}$. This congruence is related to the concept of order. It is usually simpler to consider modulo a prime.

Solution. Suppose $n \mid 2^n - 1$ for some $n > 1$. Let p be any prime divisor of n . Then we get $2^n \equiv 1 \pmod{p}$. Let d be the order of 2 modulo p . Then $d \mid n$ and $d \mid p - 1$ by [proposition 6.1](#) and the Fermat little theorem. If p is chosen to be the smallest prime divisor of n , then $(n, p - 1) = 1$ so that d has to be 1. This gives the contradiction $2 \equiv 1 \pmod{p}$. Hence, n does not divide $2^n - 1$. $\square$

Remarks. Considering the smallest prime divisor of a positive integer greater than 1 is a useful trick.

Problem 6.3. (USA TST 2003 Problem 3) Find all ordered triples of primes (p, q, r) such that

$$p \mid q^r + 1, \quad q \mid r^p + 1, \quad r \mid p^q + 1.$$

Analysis. A similar trick as in [problem 6.2](#) can be used, although more have to be done since there are more divisibilities.

Solution. We first suppose p, q, r are odd.

Consider $p \mid q^r + 1$. This gives $q^r \equiv -1 \pmod{p}$ and hence $q^{2r} \equiv 1 \pmod{p}$. Let d be the order of q modulo p . Then $d \mid p - 1$ and $d \mid 2r$ but $d \nmid r$ by [proposition 6.1](#) and the Fermat little theorem (we have $d \nmid r$ since $-1 \not\equiv 1 \pmod{p}$ for odd p).

WLOG assume p is the smallest prime among p, q, r . Then $d \mid (p - 1, 2r)$ implies $d \mid 2$. As $d \nmid r$, we must have $d = 2$. This yields $q^2 \equiv 1 \pmod{p}$ and $q \not\equiv 1 \pmod{p}$. It follows that $q \equiv -1 \pmod{p}$.

Similarly, $q \mid r^p + 1$ gives $r^p \equiv -1 \pmod{q}$ and $r^{2p} \equiv 1 \pmod{q}$. Let f be the order of r modulo q . As above, we have $f \mid (q - 1, 2p)$ so that $f \mid 2$ since $(q - 1, p) = (-2, p) = 1$. Also, as $f \nmid p$, we have $f = 2$. This gives $q \mid r + 1$ as above.

For exactly the same reasons, we must have $r \mid p + 1$ from $r \mid p^q + 1$. From $p \mid q + 1$, $q \mid r + 1$, $r \mid p + 1$ and the assumption that p, q, r are odd, they are distinct primes and hence $p < q < r < p$. This yields a contradiction.

It remains to consider the case when one of p, q, r is even, say $p = 2$. From $r \mid 2^q + 1$, the order h of 2 modulo r satisfies $h \mid 2q$ and $h \mid r - 1$. Since $q \mid r^2 + 1$, we must have $(q, r - 1) = 1$ (otherwise $q \mid 2$, which contradicts $2 \nmid q^r + 1$). Thus, we get $h \mid 2$. Again, we cannot have $h = 1$. Thus, $h = 2$, and hence $r \mid 2^2 - 1 = 3$. This gives $r = 3$. Then $q \mid 10$ and $2 \mid q^3 + 1$, which imply $q = 5$.

We check that $(p, q, r) = (2, 5, 3), (5, 3, 2), (3, 2, 5)$ are solutions. $\square$

Remarks. For congruences of the form $a^b \equiv -1 \pmod{p}$, we always square both sides to obtain an equation related to the concept of order.

Problem 6.4. (CMO 2009 Problem 2) Find all pairs (p, q) of prime numbers such that $pq \mid 5^p + 5^q$.

Analysis. The special case when one of p, q is divisible by 5 should be treated separately. If $5 \nmid p, q$, we can remove the powers of 5 to get a congruence for which the concept of order can be applied.

Solution. If both p, q are 5, then $5^2 \mid 5^5 + 5^5$ clearly holds.

If exactly one of p, q is 5, say $p = 5$, then the divisibility is equivalent to $q \mid 5^4 + 5^{q-1}$. Since $q \neq 5$, we have $5^{q-1} \equiv 1 \pmod{q}$ by the Fermat little theorem. This means the divisibility holds iff $q \mid 5^4 + 1 = 626$. Thus, we have $q = 2$ or 313 . Note that the steps are reversible.

Suppose $p, q \neq 5$. From $p \mid 5^p + 5^q$, by applying the Fermat little theorem, we get $p \mid 5 + 5^q$. As $p \neq 5$, this implies $p \mid 1 + 5^{q-1}$. Rewrite this as $5^{q-1} \equiv -1 \pmod{p}$. Hence we have $5^{2(q-1)} \equiv 1 \pmod{p}$.

Let d be the order of 5 modulo p . Then we have $d \mid 2(q-1)$ and $d \mid p-1$. As in the previous problem, we have $d \nmid q-1$ if $p \neq 2$. Therefore, we should better consider the special case first.

If one of p, q is even, say $p = 2$, then $q \mid 5^2 + 5^q$ and hence $q \mid 5^2 + 5 = 30$ by the Fermat little theorem. Since we have assumed $q \neq 5$, we can only have $q = 2, 3$. The case $q = 2$ is rejected since $2 \times 2 = 4 \nmid 50 = 5^2 + 5^2$, while $(p, q) = (2, 3)$ is a solution by checking $2 \times 3 = 6 \mid 150 = 5^2 + 5^3$.

Suppose p, q are odd. From $d \mid 2(q-1)$ and $d \nmid q-1$, we may try to consider the 2-adic valuations of the terms. Indeed, we let $2^\alpha \parallel p-1$ and $2^\beta \parallel q-1$. Since $d \mid 2(q-1)$ and $d \nmid q-1$, we must have $2^{\beta+1} \parallel d$. Note that we also have $d \mid p-1$. Thus, we get $2^{\beta+1} \mid p-1$ and hence $\beta+1 \leq \alpha$. However, by symmetry, we also have $\alpha+1 \leq \beta$. This yields the contradiction $\beta+1 \leq \alpha \leq \beta-1$.

Therefore, the solutions are $(p, q) = (2, 3), (3, 2), (2, 5), (5, 2), (5, 5), (5, 313), (313, 5)$. $\square$

Remarks. As can be seen in these problems, the concept of order and the Fermat little theorem are the most important tools that we use to solve this kind of problems. Problems involving congruences of the form $a^n \equiv \pm 1 \pmod{p}$ can always be solved using order. In particular, if we have a congruence $a^n \equiv -1 \pmod{p}$, then we know some information about the 2-adic valuations of some terms.

Problem 6.5. Let p be a prime number. Prove that $p \mid 1^k + 2^k + \dots + (p-1)^k$ for any positive integer k such that $p-1 \nmid k$.

Analysis. There is a formula for $1^k + 2^k + \dots + n^k$ for any $k \in \mathbb{N}$. However, one probably does not know the general formula for large k . Here, we shall see an application of primitive roots.

Solution. Let g be a primitive root of p . Then the residues $1, 2, \dots, p-1$ are congruent to $g, g^2, \dots, g^{p-1}$ modulo p in some order. Thus,

$$1^k + 2^k + \dots + (p-1)^k \equiv g^k + g^{2k} + \dots + g^{(p-1)k} \pmod{p}.$$

Note that we have transformed the sum into a geometric sequence. We have

$$g^k + g^{2k} + \dots + g^{(p-1)k} = \frac{g^{pk} - g^k}{g^k - 1}.$$

Note that the numerator $g^{pk} - g^k \equiv g^k - g^k = 0 \pmod{p}$ by the Fermat little theorem. Also, since $p-1 \nmid k$ and g is a primitive root modulo p , we have $g^k - 1 \not\equiv 0 \pmod{p}$. This shows the denominator is coprime to p . Hence, $1^k + 2^k + \dots + (p-1)^k$ is divisible by p . $\square$

Remarks. As can be seen, using the primitive roots, we can convert everything to powers of g . This often converts between sums and products like how the logarithm function works.

For $p-1 \mid k$, we can simply apply the Fermat little theorem to get

$$1^k + 2^k + \dots + (p-1)^k \equiv 1 + 1 + \dots + 1 \equiv -1 \pmod{p}.$$

Problem 6.6. (IMO Shortlist 1997) An infinite arithmetic progression whose terms are positive integers contains the square of an integer and the cube of an integer. Show that it contains the sixth power of an integer.

Analysis. In terms of congruences, it actually says a certain integer a is both a square and a cube modulo a positive integer d . It asks us to show that a is a sixth power modulo d . This sounds to be obvious, but we need a more careful proof for that. Using the concept of primitive root would make things simpler.

Solution. Suppose the terms are $a + nd$ where $a, d \in \mathbb{N}_0$ are fixed. If $d = 0$, then there is only 1 distinct term, which is both a square and a cube. Thus, it is a sixth power. The result is also obvious if $d = 1$, since all large integers are there in that case.

Suppose $d = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_s^{\alpha_s}$ is the prime factorization. From the conditions, the equations $x^2 \equiv a \pmod{d}$ and $y^3 \equiv a \pmod{d}$ are both solvable. It suffices to show that $z^6 \equiv a \pmod{d}$ is solvable, since then $(z + kd)^6$ is a term in the arithmetic progression for sufficiently large $k \in \mathbb{N}$.

Fix $1 \leq j \leq s$ and write for simplicity $p = p_j$ and $\alpha = \alpha_j$. If $(p, a) = 1$, let g be a primitive root of p . Let $a \equiv g^t \pmod{p^\alpha}$ and $x \equiv g^u \pmod{p^\alpha}$. Then the congruence $x^2 \equiv a \pmod{p^\alpha}$ holds iff $g^{2u} \equiv g^t \pmod{p^\alpha}$ iff $2u \equiv t \pmod{p^{\alpha-1}(p-1)}$. Since this

is solvable, we have $(2, p^{\alpha-1}(p-1)) \mid t$. Similarly, by considering the cubic equation, we have $(3, p^{\alpha-1}(p-1)) \mid t$. These combine to give $(6, p^{\alpha-1}(p-1)) \mid t$. Note that by reversing similar arguments, this shows $z^6 \equiv a \pmod{p^\alpha}$ is solvable.

If $p^\alpha \mid a$, then 0 is a solution of $z^6 \equiv a \pmod{p^\alpha}$. If $p^\beta \parallel a$ where $\beta < \alpha$, then since $x^2 \equiv a \pmod{p^\alpha}$, we must have $p^\beta \parallel x^2$. This shows β is even. Similarly, from the cubic equation, we know that β is divisible by 3. Then $6 \mid \beta$. Let $\beta = 6\beta'$. We have the following equivalences

$$\begin{aligned} x^2 \equiv a \pmod{p^\alpha} &\Leftrightarrow \left(\frac{x}{p^{3\beta'}}\right)^2 \equiv \frac{a}{p^{6\beta'}} \pmod{p^{\alpha-6\beta'}}, \\ y^3 \equiv a \pmod{p^\alpha} &\Leftrightarrow \left(\frac{y}{p^{2\beta'}}\right)^3 \equiv \frac{a}{p^{6\beta'}} \pmod{p^{\alpha-6\beta'}}, \\ z^6 \equiv a \pmod{p^\alpha} &\Leftrightarrow \left(\frac{z}{p^{\beta'}}\right)^6 \equiv \frac{a}{p^{6\beta'}} \pmod{p^{\alpha-6\beta'}}. \end{aligned}$$

The solvability of the equations on the left-hand sides is the same as the solvability of the equations on the right-hand sides, which is indeed the case we have considered since $\frac{a}{p^{6\beta'}}$ is coprime to p , and $\alpha - 6\beta' > 0$. Therefore, the equation $z^6 \equiv a \pmod{p^\alpha}$ is solvable in this case.

Now, let z_j be a solution to $z^6 \equiv a \pmod{p_j^{\alpha_j}}$ for $j = 1, 2, \dots, s$. Then $z^6 \equiv a \pmod{d}$ is solvable as we can take $z \equiv z_j \pmod{p_j^{\alpha_j}}$. Such a $z \in \mathbb{Z}$ exists by the Chinese remainder theorem. This completes the proof. $\square$

Problem 6.7. (IMO Shortlist 2001 N4) Let $p \geq 5$ be a prime number. Prove that there exists an integer a with $1 \leq a \leq p-2$ such that neither $a^{p-1} - 1$ nor $(a+1)^{p-1} - 1$ is divisible by p^2 .

Analysis. Note that order is not so useful since the base is not fixed. However, primitive root would be quite useful. Indeed, we are considering congruences of the form $k^{p-1} \equiv 1 \pmod{p^2}$. Such a congruence equation can be easily solved in terms of a primitive root. We just need to see whether it is possible that there are many solutions in the range $1 \leq k \leq p-1$.

Solution. Suppose on the contrary that for any $a = 1, 2, \dots, p-2$, at least one of the congruences $a^{p-1} \equiv 1 \pmod{p^2}$ and $(a+1)^{p-1} \equiv 1 \pmod{p^2}$ holds.

Let g be a primitive root modulo p^2 . For any b which is relatively prime to p , if we write $b \equiv g^\alpha \pmod{p^2}$, then

$$\begin{aligned} &b^{p-1} \equiv 1 \pmod{p^2} \\ \Leftrightarrow &g^{(p-1)\alpha} \equiv 1 \pmod{p^2} \\ \Leftrightarrow &\varphi(p^2) = p(p-1) \mid (p-1)\alpha \\ \Leftrightarrow &p \mid \alpha. \end{aligned}$$

In the following, we say b a good number if $b^{p-1} \equiv 1 \pmod{p^2}$.

We count the number of good numbers b in the range $1 \leq b \leq p^2 - 1$. Note that we must have $(b, p) = 1$ if b is good. From above, b is congruent to one of $g^p, g^{2p}, \dots, g^{p(p-1)}$ modulo p^2 . Since g is a primitive roots, these are pairwise incongruent. This shows there are exactly $p - 1$ good numbers.

Next, there is a good number in each of the pairs $(1, 2), (3, 4), \dots, (p-2, p-1)$ from the assumption. Thus, there are at least $\frac{p-1}{2}$ good numbers b in the range $1 \leq b \leq p-1$.

As b is good iff $-b$ is good from $b^{p-1} = (-b)^{p-1}$, there are at least $\frac{p-1}{2}$ good numbers b in the range $p^2 - p + 1 \leq b \leq p^2 - 1$. From this double counting, we cannot have any other good numbers in the range $p \leq b \leq p^2 - p$. We shall derive a contradiction from this.

Indeed, we pick b_1 from $\{p-3, p-2\}$ and b_2 from $\{p-2, p-1\}$ which are good. Then $b_1^{p-1} \equiv 1 \pmod{p^2}$ and $b_2^{p-1} \equiv 1 \pmod{p^2}$, and hence $(b_1 b_2)^{p-1} \equiv 1 \pmod{p^2}$. So $b_1 b_2$ is good. Note that

$$p \leq (p-3)(p-2) \leq b_1 b_2 \leq (p-2)(p-1) \leq p^2 - p$$

since $p \geq 5$. Thus, we get another good number $b_1 b_2$ in the range from p to $p^2 - p$. This is a contradiction.

Hence, there exists $a \in \mathbb{Z}$ with $1 \leq a \leq p-2$ such that neither $a^{p-1} - 1$ nor $(a+1)^{p-1} - 1$ is divisible by p^2 . $\square$

Problem 6.8. Find all positive integers n such that $m^{n+1} \equiv m \pmod{n}$ for every integer m .

Analysis. The congruence equation is similar to $m^n \equiv 1 \pmod{n}$ in some sense. So we may think about the tools covered in this section. Since m is arbitrary, we can choose some m with special properties, like primitive roots of some primes.

Solution. Clearly, $n = 1$ is possible. If $p^2 \mid n$ for some prime p , then we take $m = p$ so that $p^2 \mid p^{n+1} - p$. This yields $p^2 \mid p$, which is a contradiction. Therefore, n is squarefree.

Let p be a prime divisor of n . Let g be a primitive root of p . We take $m = g$. Then $g^{n+1} \equiv g \pmod{p}$ implies $g^n \equiv 1 \pmod{p}$. This gives $p-1 \mid n$.

Let $n = p_1 p_2 \cdots p_s$ where $p_1 < p_2 < \cdots < p_s$ are distinct primes. From above, since $p_1 - 1 \mid n$ and p_1 is the smallest prime divisor of n , we must have $p_1 = 2$.

Next, if $s \geq 2$, as $p_2 - 1 \mid n$, we can only have $p_2 - 1 = 2$ so that $p_2 = 3$.

Similarly, if $s \geq 3$, then $p_3 - 1 \mid n$ implies $p_3 - 1$ equals one of $1, 2, 3, 6$. As $p_3 > 3$ and p_3 is a prime, the only possible choice is $p_3 = 7$.

If $s \geq 4$, then $p_4 - 1 \mid n$ and $p_4 > 7$ imply $p_4 - 1$ equals one of 7, 14, 21, 42. The only possible case is $p_4 = 43$.

If $s \geq 5$, then $p_5 - 1 \mid n$ and $p_5 > 43$ imply $p_5 - 1$ equals one of 43, 86, 129, 258, 301, 602, 903, 1806. There is no solution for p_5 to be a prime.

Thus, the only possible solutions are $n = 1, 2, 6, 42, 1806$. We show that each of these is a solution. It suffices to show $m^{n+1} \equiv m \pmod{p}$ for each prime divisor p of n since n is squarefree. This clearly holds for $p \mid m$. For $p \nmid m$, using the same primitive root, we let $m \equiv g^k \pmod{p}$. Then

$$\begin{aligned} & m^{n+1} \equiv m \pmod{p} \\ \Leftrightarrow & g^{k(n+1)} \equiv g^k \pmod{p} \\ \Leftrightarrow & k(n+1) \equiv k \pmod{p-1} \\ \Leftrightarrow & kn \equiv 0 \pmod{p-1}. \end{aligned}$$

Note that $p-1$ does divide n from the above construction. Therefore, the congruence does hold. $\square$

Here are the common usages of orders and primitive roots introduced in this section.

- order
 - solving congruences of the form $a^x \equiv 1 \pmod{p}$ ([problem 6.2](#))
 - solving congruences of the form $a^x \equiv -1 \pmod{p}$ ([problems 6.3, 6.4](#))
 - Fermat little theorem ([problems 6.2, 6.3, 6.4](#))
- primitive root
 - expressing residue classes as powers of a primitive root ([problem 6.5, 6.6, 6.7, 6.8](#))
 - solving congruences of the form $x^k \equiv a \pmod{p}$ ([problems 6.6, 6.7, 6.8](#))

7 More on Diophantine Equations

In this section, we introduce a few more advanced theorems on Diophantine equations. We would also use the more advanced tools introduced in previous sections to solve Diophantine equations, such as the lifting the exponent lemma, orders and primitive roots.

Theorem 7.1. (Sum of 2 squares theorem(兩平方和定理)) A positive integer n is a sum of 2 squares iff the exponent of every prime divisor of n of the form $4k + 3$ is even.

Proof. Suppose there exists a prime $p \equiv 3 \pmod{4}$ such that $p^{2\alpha+1} \parallel n$. Let $n = a^2 + b^2$. Then $a^2 \equiv -b^2 \pmod{p}$. Note that we must have $p \nmid b$. Otherwise, let b^{-1} be the inverse of b . Then $(ab^{-1})^2 \equiv -1 \pmod{p}$. However, -1 is a quadratic nonresidue by proposition 5.1(e). This is a contradiction. Thus, $p \mid b$ and hence $p \mid a$. We can divide both sides of $n = a^2 + b^2$ by p^2 . Inductively, we can divide both sides by $p^{2\alpha}$. The same argument still applies, which shows $p^{2\alpha+2}$ divides n . This is a contradiction.

Next, we show that every prime $p \equiv 1 \pmod{4}$ is a sum of 2 squares. Let d be an integer such that $d^2 \equiv -1 \pmod{p}$. We shall prove the following result, which is known as Thue's lemma(圖厄引理) (or its special case).

Claim. There exist $a, b \in \mathbb{Z}$ with $0 < |a|, |b| < \sqrt{p}$ satisfying $da \equiv b \pmod{p}$.

Indeed, let $k = \lfloor \sqrt{p} \rfloor + 1$. Consider integers x, y in the range $0 \leq x, y \leq k - 1$. There are k^2 pairs of x, y . Since $k^2 > p$, there must be 2 pairs $(x_1, y_1) \neq (x_2, y_2)$ such that $dx_1 - y_1 \equiv dx_2 - y_2 \pmod{p}$ by the pigeonhole principle. This implies

$$d(x_1 - x_2) \equiv y_1 - y_2 \pmod{p}.$$

This suggests we can pick $a = x_1 - x_2$ and $b = y_1 - y_2$. Note that $0 \leq |a|, |b| < \sqrt{p}$, and we cannot have $a = 0$ or $b = 0$ (since otherwise we have $a = b = 0$ and hence $(x_1, y_1) = (x_2, y_2)$). This proves Thue's lemma.

Now, for this pair of (a, b) , we have

$$a^2 + b^2 \equiv a^2 + (da)^2 \equiv a^2 - a^2 = 0 \pmod{p}$$

and $0 < a^2 + b^2 < p + p = 2p$. These yield $a^2 + b^2 = p$.

The next step is to show that if both n_1 and n_2 are sums of 2 squares, then so is $n_1 n_2$. Indeed, let $n_1 = a_1^2 + b_1^2$ and $n_2 = a_2^2 + b_2^2$. Then

$$n_1 n_2 = (a_1 a_2 + b_1 b_2)^2 + (a_1 b_2 - a_2 b_1)^2.$$

Since every prime $p \equiv 1 \pmod{4}$, every $m^2 = 0^2 + m^2$ and the number $2 = 1^2 + 1^2$ are sums of 2 squares, we know that n is a sum of 2 squares if n is a product of these numbers by using the above fact repeatedly. Also, the integer 1 is clearly a sum of 2 squares. This completes the proof. $\square$

For example, $343 = 7^3$ is not a sum of 2 squares, while $245 = 5 \times 7^2$ is. Indeed, we have $245 = 7^2 + 14^2$.

Theorem 7.2. (Sum of 3 squares theorem(三平方和定理)) A positive integer n is a sum of 3 squares iff it is not of the form $4^\alpha(8k+7)$ where α, k are nonnegative integers.

The ‘only if’ part is simple by considering modulo 8, while the ‘if’ part is much harder and we are not going to give a proof here.

For example, $4028 = 4 \times 1007$ is not a sum of 3 squares, while $1080 = 4 \times 270$ is. Indeed, we have $1080 = 6^2 + 12^2 + 30^2$.

Theorem 7.3. (Sum of 4 squares theorem(四平方和定理)) Every positive integer n is a sum of 4 squares.

Proof. We first show that if both n_1 and n_2 are sums of 4 squares, then so is $n_1 n_2$. Indeed, let

$$\begin{aligned} n_1 &= a_1^2 + b_1^2 + c_1^2 + d_1^2, \\ n_2 &= a_2^2 + b_2^2 + c_2^2 + d_2^2. \end{aligned}$$

Then

$$\begin{aligned} n_1 n_2 &= (a_1 a_2 + b_1 b_2 + c_1 c_2 + d_1 d_2)^2 + (a_1 b_2 - b_1 a_2 + c_1 d_2 - d_1 c_2)^2 \\ &\quad + (a_1 c_2 - b_1 d_2 - c_1 a_2 + d_1 b_2)^2 + (a_1 d_2 + b_1 c_2 - c_1 b_2 - d_1 a_2)^2. \end{aligned} \quad (1)$$

Since 1 and 2 are clearly sums of 4 squares, it suffices to show p is a sum of 4 squares for every odd prime p . By [problem 4.3](#), there exist $a, b \in \mathbb{Z}$ such that $p \mid a^2 + b^2 + 1$.

We can pick a, b such that $0 \leq |a|, |b| \leq \frac{p-1}{2}$. Then

$$pm = a^2 + b^2 + 1^2 + 0^2$$

is a sum of 4 squares for some integer m . Note that $pm \leq \frac{(p-1)^2}{2} + 1 < p^2$ so that $m < p$. We pick the smallest positive integer r for which

$$pr = a_1^2 + b_1^2 + c_1^2 + d_1^2. \quad (2)$$

If $r = 1$, then we are done. Suppose $r > 1$. We shall obtain a contradiction by finding another positive integer smaller than r having the same property. We pick the unique a_2, b_2, c_2, d_2 which are congruent to a_1, b_1, c_1, d_1 modulo r respectively and satisfy $-\frac{r-1}{2} \leq a_2, b_2, c_2, d_2 \leq \frac{r}{2}$. Then $r \mid a_2^2 + b_2^2 + c_2^2 + d_2^2$ so that

$$rs = a_2^2 + b_2^2 + c_2^2 + d_2^2 \quad (3)$$

for some $s \in \mathbb{N}_0$. Note that $rs \leq 4\left(\frac{r}{2}\right)^2 = r^2$ so that $s \leq r$. If $s = r$, then equality holds and we must have $a_2 = b_2 = c_2 = d_2 = \frac{r}{2}$. Let $r = 2r'$. Then a_1, b_1, c_1, d_1 are all congruent to r' modulo $2r'$. From (2), we get

$$2p = r' \left(\left(\frac{a_1}{r'} \right)^2 + \left(\frac{b_1}{r'} \right)^2 + \left(\frac{c_1}{r'} \right)^2 + \left(\frac{d_1}{r'} \right)^2 \right).$$

Note that the bracket on the right-hand side is a multiple of 4 since each square is congruent to 1 modulo 4 (as $\frac{a_1}{r'}$ is odd, etc.). This contradicts p being an odd prime. Similarly, if $s = 0$, then $a_2 = b_2 = c_2 = d_2 = 0$ and hence $r \mid a_1, b_1, c_1, d_1$. This gives

$$p = r \left(\left(\frac{a_1}{r} \right)^2 + \left(\frac{b_1}{r} \right)^2 + \left(\frac{c_1}{r} \right)^2 + \left(\frac{d_1}{r} \right)^2 \right).$$

In order that p is an odd prime, we can only have $p = r = a_1$ and $b_1 = c_1 = d_1 = 0$ up to permutation. This contradicts $r \leq m < p$.

Thus, we have $1 \leq s < r$. Using (2) and (3), we have

$$pr^2s = (pr)(rs) = a_3^2 + b_3^2 + c_3^2 + d_3^2 \quad (4)$$

where a_3, b_3, c_3, d_3 are determined by equation (1). Since a_2, b_2, c_2, d_2 are congruent to a_1, b_1, c_1, d_1 modulo r respectively, we easily find that $r \mid b_3, c_3, d_3$ using equation (1). Also,

$$a_3 \equiv a_1^2 + b_1^2 + c_1^2 + d_1^2 \equiv 0 \pmod{r}$$

from (2). Thus, we can divide both sides of (4) by r^2 to get

$$ps = a_4^2 + b_4^2 + c_4^2 + d_4^2.$$

Thus, we have found a positive integer s smaller than r such that ps is a sum of 4 squares. This is a contradiction. Therefore, every odd prime is a sum of 4 squares, and the theorem is proved. $\square$

In this proof, we have used the well-known and useful trick, called the *method of infinite descent* (無窮遞降法). To prove that something cannot happen, we assume it does happen and pick the smallest case. If we can find a further smaller case from this having the same property, then we obtain a contradiction. This is indeed another way to express the idea of induction. It is useful in solving some Diophantine equations.

The next few theorems are very difficult to prove (and not too useful). Therefore, we only quote their results here.

Theorem 7.4. (Fermat last theorem (費馬大定理)) Let $n \geq 3$ be a positive integer. Then there is no solution to $a^n + b^n = c^n$ where a, b, c are nonzero integers.

Recall that there are infinitely many nontrivial solutions when $n = 2$ given by Pythagorean triples.

Theorem 7.5. ([Catalan's conjecture/Mihăilescu's theorem](#)(卡塔蘭猜想/米哈伊列斯庫定理)) The only positive integer solution to $a^m - b^n = 1$ with $m, n \geq 2$ is $(a, b, m, n) = (3, 2, 2, 3)$.

Theorem 7.6. ([Skolem's theorem](#)(斯科倫定理)) Let d be a nonzero integer. Then there is at most one integer solution to $m^3 - dn^3 = 1$ with $n \neq 0$.

Let us combine all techniques we have learned to solve some problems in Diophantine equations (or problems which can be rephrased as Diophantine equations).

Problem 7.1. (IMO Prelim 2015 Problem 11) Let n be a positive integer. If the two numbers $(n+1)(2n+15)$ and $n(n+5)$ have exactly the same prime factors, find the greatest possible value of n .

Analysis. By examining the common divisors of the linear factors, we can find all possible prime divisors of the 2 numbers. This gives us a Diophantine equation. We can use the concept of order to solve Diophantine equations involving exponents.

Solution. Let p be any prime divisor of $n+1$. Then $p \mid (n+1)(2n+15)$ so that $p \mid n(n+5)$. Since $n \equiv -1 \pmod{p}$, we have $n(n+5) \equiv (-1) \cdot 4 = -4 \pmod{p}$. This gives $p \mid 4$ so that p can only be 2.

Similarly, let p be any prime divisor of $n+5$. Then $p \mid n(n+5)$ so that we obtain $p \mid (n+1)(2n+15)$. As $n \equiv -5 \pmod{p}$, we get

$$(n+1)(2n+15) \equiv (-4) \cdot 5 = -20 \pmod{p}.$$

This gives $p \mid 20$ so that p can be 2 or 5.

Let $n+1 = 2^a$ and $n+5 = 2^b \cdot 5^c$. Then we have

$$2^a + 4 = 2^b \cdot 5^c. \tag{1}$$

We should check the 2-adic valuations. Firstly, we deal with the small cases.

- If $a = 0$, then $2^a + 4 = 5$ and we get $(a, b, c) = (0, 0, 1)$. But then $n = 0$ is not a positive integer.
- If $a = 1$, then $2^a + 4 = 6$ is divisible by 3.
- If $a = 2$, then $2^a + 4 = 8$ and we get $(a, b, c) = (2, 3, 0)$. This implies $n = 3$. One checks that $(n+1)(2n+15) = 84 = 2^2 \times 3 \times 7$ and $n(n+5) = 24 = 2^3 \times 3$ do not have the same prime divisors.

- If $a \geq 3$, then the left-hand side of (1) is congruent to 4 modulo 8. Therefore, $b = 2$ and the equation is reduced to

$$2^{a-2} + 1 = 5^c. \quad (2)$$

This kind of equation can be solved by the concept of order. We first find that $a = 3$ gives no solution, while $a = 4$ gives $c = 1$. In that case, we have $n = 15$. Note that $(n+1)(2n+15) = 16 \times 45 = 2^4 \times 3^2 \times 5$ and $n(n+5) = 15 \times 20 = 2^2 \times 3 \times 5^2$. Therefore, $n = 15$ is a solution.

By testing other small values, one naturally guesses there is no other solution to (2). Indeed, for $a \geq 5$, the main difference is that the left-hand side of (2) is congruent to 1 modulo 8. Since the order of 5 modulo 8 is 2, [proposition 6.1](#) implies $2 \mid c$.

Let $c = 2d$. Then the equation becomes $2^{a-2} + 1 = 25^d$. Considering modulo 3, we get

$$2^{a-2} + 1 = 25^d \equiv 1^d = 1 \pmod{3}.$$

This shows 2^{a-2} is divisible by 3, which is impossible.

Thus, the only solution is $n = 15$. □

Remarks. Since there is an obvious pattern for the powers of an odd number modulo 8, using order is not necessary. However, one should note that looking at the pattern is not a reliable and rigorous method.

One can rewrite (2) as $5^c - 2^{a-2} = 1$. By Catalan's conjecture, we must have $c = 1$ or $a - 2 = 1$. Thus, one can also use this deep theorem to find the solutions easily.

Problem 7.2. (IMO Shortlist 1991) Find all positive integer solutions x, y, z of the equation

$$3^x + 4^y = 5^z. \quad (1)$$

Analysis. A usual way is to first consider congruences with small moduli. This provides many information to us about the exponents.

Solution. By considering modulo 3, we get $1 \equiv (-1)^z \pmod{3}$. This clearly implies $2 \mid z$. Let $z = 2w$. The equation becomes

$$3^x + 4^y = 25^w. \quad (2)$$

If one continues to consider modulo 4, then one would find that x is even. Though we do not need this information in the following proof. Indeed, we use another useful technique when solving Diophantine equations, i.e. factorization.

Equation (2) can be rewritten as

$$3^x = 25^w - 4^y = (5^w)^2 - (2^y)^2 = (5^w - 2^y)(5^w + 2^y).$$

Note that $(5^w - 2^y, 5^w + 2^y) = (5^w - 2^y, 2 \times 5^w) = 1$. In order that their product is a power of 3, we must have $5^w - 2^y = 1$ and $5^w + 2^y = 3^x$. The equation $5^w - 2^y = 1$ is again the one that we have considered in the previous problem. We use Catalan's conjecture to conclude that the only solution is $(y, w) = (2, 1)$. In that case $x = 2$.

We check that the only solution is $(x, y, z) = (2, 2, 2)$. $\square$

Problem 7.3. (CMO 2005 Problem 6) Find all nonnegative integer solutions (x, y, z, w) of the equation

$$2^x \cdot 3^y - 5^z \cdot 7^w = 1. \quad (1)$$

Analysis. This can be viewed as a variation of equations of the form given in Catalan's conjecture. A similar trick to that given in [problem 7.1](#) can be used. The steps used in the following solution are standard in solving this kind of problems.

Solution. Considering the parity, we find that $x \geq 1$.

Case 1. $x = 1$

For $y = 0, 1$, we easily get 2 solutions $(y, z, w) = (0, 0, 0), (1, 1, 0)$. There seems to be no solutions for $y \geq 2$. Suppose $y \geq 2$. If $z, w \geq 1$, then we consider modulo 5 and modulo 7.

Firstly, $2 \cdot 3^y \equiv 1 \pmod{5}$ gives $3^y \equiv 3 \pmod{5}$, so that $3^{y-1} \equiv 1 \pmod{5}$. As the order of 3 modulo 5 is 4, this implies $4 \mid y - 1$.

Secondly, $2 \cdot 3^y \equiv 1 \pmod{7}$ gives $3^y \equiv 4 \pmod{7}$. As $3^4 \equiv 4 \pmod{7}$ and the order of 3 modulo 7 is 6, this implies $y \equiv 4 \pmod{6}$. However, $4 \mid y - 1$ and $6 \mid y - 4$ clearly contradict each other.

Thus, one of z, w is 0. If $z = 0$, we have $-7^w \equiv 1 \pmod{9}$ since $y \geq 2$. This cannot hold as $7^w \equiv 7, 4, 1 \pmod{9}$. Similarly, if $w = 0$, we have $-5^z \equiv 1 \pmod{9}$. This leads to $z \equiv 3 \pmod{6}$ by order. In particular, z is divisible by 3. Then

$$126 = 5^3 + 1 \mid 5^z + 1 = 2 \cdot 3^y.$$

This is a contradiction.

Case 2. $x \geq 2$ and $y = 0$

Equation (1) becomes $2^x - 5^z \cdot 7^w = 1$. There is no solution for $x = 2$. When $x = 3$, we get a solution $(x, z, w) = (3, 0, 1)$. We claim that there is no solution for $x \geq 4$.

If $z \geq 1$, then $2^x \equiv 1 \pmod{5}$ so that $4 \mid x$. Thus, $15 = 2^4 - 1 \mid 2^x - 1 = 5^z \cdot 7^w$. This is a contradiction.

If $z = 0$, then there is no solution to $2^x - 7^w = 1$ since $x \geq 4$ due to Catalan's conjecture.

Case 3. $x \geq 2$ and $y \geq 1$

In this case, $-(-1)^w \equiv 1 \pmod{4}$ implies w is odd. Similarly, $-(-1)^z \equiv 1 \pmod{3}$ implies z is odd. Therefore, $5^z \cdot 7^w \equiv 5 \cdot 7 \equiv 3 \pmod{8}$. This means $2^x \cdot 3^y \equiv 4 \pmod{8}$, and hence $x = 2$.

Since z is odd, it is at least 1. Then $4 \cdot 3^y \equiv 1 \pmod{5}$. This implies $3^y \equiv 4 \pmod{5}$, and hence $y \equiv 2 \pmod{4}$. In particular, $y = 2y_1$ for some $y_1 \in \mathbb{N}$. Then (1) can be rewritten as

$$5^z \cdot 7^w = 4 \cdot 9^{y_1} - 1 = (2 \cdot 3^{y_1})^2 - 1 = (2 \cdot 3^{y_1} - 1)(2 \cdot 3^{y_1} + 1).$$

As $(2 \cdot 3^{y_1} - 1, 2 \cdot 3^{y_1} + 1) = 1$ and $2 \cdot 3^{y_1} \pm 1 > 1$, there are only 2 possibilities.

- If $2 \cdot 3^{y_1} - 1 = 5^z$ and $2 \cdot 3^{y_1} + 1 = 7^w$, we clearly get 1 solution $(y_1, z, w) = (1, 1, 1)$. If $y_1 \geq 2$, we have $1 \equiv 7^w \pmod{9}$ so that $3 \mid w$. Then $342 = 7^3 - 1 \mid 7^w - 1 = 2 \cdot 3^{y_1}$. This is impossible.
- If $2 \cdot 3^{y_1} - 1 = 7^w$ and $2 \cdot 3^{y_1} + 1 = 5^z$, $y_1 = 1$ is not a solution, while for $y_1 \geq 2$, we have $-1 \equiv 7^w \pmod{9}$. This cannot hold.

To conclude, $(x, y, z, w) = (1, 0, 0, 0), (1, 1, 1, 0), (3, 0, 0, 1), (2, 2, 1, 1)$ are the only solutions. $\square$

Problem 7.4. (Balkan MO 1993 Problem 4) Let p be a prime and $m \geq 2$ be an integer. Prove that the equation

$$\frac{x^p + y^p}{2} = \left(\frac{x + y}{2} \right)^m \quad (1)$$

has a positive integer solution $(x, y) \neq (1, 1)$ if and only if $m = p$.

Analysis. Note that we cannot directly apply Zsigmondy's theorem here to conclude that there exists a prime divisor of $x^p + y^p$ not dividing $x + y$ apart from some exceptional cases, since x and y need not be relatively prime. Indeed, a solution in the case $m = p$ is exactly $x = y$, where the variables share the same prime divisors. So a better way to start is to take out the greatest common divisor of the variables.

Next, the key observation is that $x + y$ divides $x^p + y^p$ in general. However, the exponent of each prime divisor of the right-hand side of (1) is large. This is impossible by the lifting the exponent lemma.

Solution. Suppose $(x, y) \neq (1, 1)$. By the power mean inequality, we have

$$\frac{x^p + y^p}{2} \geq \left(\frac{x + y}{2} \right)^p.$$

This implies $m \geq p$ since $\frac{x + y}{2} > 1$. Let $d = (x, y)$, $x = da$ and $y = db$ so that $(a, b) = 1$. Equation (1) can be rewritten as

$$2^{m-1}(a^p + b^p) = d^{m-p}(a + b)^m. \quad (2)$$

Case 1. $p \geq 3$

Since p is a prime, it must be odd in this case. Consider any odd prime divisor q of $a + b$, if exists. Note that $q \nmid a, b$ since otherwise $q \mid a, b$, contradicting $(a, b) = 1$. By the lifting the exponent lemma, we have

$$v_q(a^p + b^p) = v_q(a + b) + v_q(p) \leq v_q(a + b) + 1.$$

However, by equation (2), we need $v_q(a^p + b^p) = v_q(d^{m-p}(a + b)^m) \geq mv_q(a + b)$. Combining these, it follows that

$$1 \geq (m - 1)v_q(a + b) \geq m - 1 \geq p - 1 \geq 2,$$

contradiction.

Therefore, $a + b$ must be a power of 2. As $(a, b) = 1$, both a, b are odd. Again by the lifting the exponent lemma, we have

$$v_2(a^p + b^p) = v_2(a + b).$$

This time by equation (2), we need

$$(m - 1) + v_2(a^p + b^p) = v_2(d^{m-p}(a + b)^m) \geq mv_2(a + b).$$

This implies $m - 1 \geq (m - 1)v_2(a + b)$, and hence $v_2(a + b) \leq 1$ (note that $m - 1 \geq p - 1 > 1$). As $a + b$ is a power of 2, the only possibility is $a + b = 2$, i.e. $a = b = 1$. This shows $x = y$, which only holds when $m = p$.

Case 2. $p = 2$

A similar argument works in this case. If $q \mid a + b$ for some odd prime q , then $q \mid a^2 + b^2$ by equation (2). This implies $q \mid 2a^2$, and hence $q \mid a, b$. This contradicts $(a, b) = 1$.

Now, $a + b$ must be a power of 2, and a, b are odd. Thus, $a^2 + b^2 \equiv 1 + 1 = 2 \pmod{4}$, and so

$$m = (m - 1) + 1 = v_2(2^{m-1}(a^2 + b^2)) = v_2(d^{m-2}(a + b)^m) \geq mv_2(a + b).$$

This implies $v_2(a + b) \leq 1$, and hence $a + b = 2$. Again, this means $a = b = 1$ and $x = y$, which only holds when $m = p = 2$.

Thus, we have seen that there is no solution $(x, y) \neq (1, 1)$ when $m \neq p$, and there are solutions $x = y$ when $m = p$. $\square$

Problem 7.5. (IMO Shortlist 2021 N5) Prove that there are only finitely many quadruples (a, b, c, n) of positive integers such that

$$n! = a^{n-1} + b^{n-1} + c^{n-1}. \quad (1)$$

Analysis. A factorial is a product of many small integers. Therefore, by considering congruences modulo any small numbers, the left-hand side of (1) becomes 0. This is often one idea in solving such Diophantine equations. By considering modulo 4, we can deduce n is even (or a, b, c are all even, which can be easily rejected).

One natural idea is to consider the congruence modulo a prime divisor p of c , for example. It is not hard to show that $c < n$. So p must divide

$$a^{n-1} + b^{n-1} = (a+b)(a^{n-2} - a^{n-3}b + \dots + b^{n-2}).$$

However, this does not seem to be too useful as it is possible that p divides the clumsy term $a^{n-2} + a^{n-3}b + \dots + b^{n-2}$, in which case we do not have much information. Rather than this natural idea, we can indeed do the opposite. We can first start with a prime divisor of $a+b$ and see what happens. For this purpose, we need a tighter bound for $a+b$.

Solution. Note that there are finitely many solutions for each fixed n . So it suffices to prove there is no solution for all large n . We first prove that $a, b, c < \frac{n-1}{2}$ if n is sufficiently large. Indeed, by the AM-GM inequality, we have

$$\begin{aligned} n! &= 2(n-1)(n(n-2))(n-3)(3 \times 4 \times \dots \times (n-4)) \\ &\leq 2(n-1)(n-1)^2(n-1) \left(\frac{3+4+\dots+(n-4)}{n-6} \right)^{n-6} \\ &= 2(n-1)^4 \left(\frac{n-1}{2} \right)^{n-6} \\ &< \left(\frac{n-1}{2} \right)^{n-1}. \end{aligned}$$

As $n! = a^{n-1} + b^{n-1} + c^{n-1} > a^{n-1}$, we must have $a < \frac{n-1}{2}$, etc.

Case 1. n is odd

Note that the exponent $n-1$ is even. Therefore, $a^{n-1}, b^{n-1}, c^{n-1} \equiv 0, 1 \pmod{4}$. From (1), since $4 \mid n!$ for large n , we must have $a^{n-1} \equiv b^{n-1} \equiv c^{n-1} \equiv 0 \pmod{4}$. This means a, b, c are even. Then we see that

$$v_2(a^{n-1} + b^{n-1} + c^{n-1}) \geq n-1.$$

If one is familiar with the factorial function, one should know that $v_2(n!) \geq n-1$ cannot hold. Indeed, let s be the sum of digits of the base 2 representation of n . Since n is odd and $n > 1$, we must have $s \geq 2$. By Legendre's formula, we deduce

$$v_2(n!) = \frac{n-s}{2-1} \leq n-2.$$

This contradicts $v_2(n!) = v_2(a^{n-1} + b^{n-1} + c^{n-1}) \geq n-1$.

Case 2. n is even

Consider any odd prime divisor p of $a+b$, if exists. Since $a+b < n-1$ from above, we know that $p \mid n!$. Together with $p \mid a+b \mid a^{n-1} + b^{n-1}$, we see from equation (1) that $p \mid c^{n-1}$, which implies $p \mid c$.

The next idea is that we now know that a large power of p divides c^{n-1} , so in view of [proposition 2.2](#), the p -adic valuations of $n!$ and $a^{n-1} + b^{n-1}$ should be the same. This is not quite possible. Here are the details.

Firstly, if p divides one of a, b , it must divide both of a, b . Again, this yields

$$n-1 \leq v_p(a^{n-1} + b^{n-1} + c^{n-1}) = v_p(n!) < \frac{n}{p-1} \leq \frac{n}{2},$$

which is impossible as $n \geq 2$. Therefore, we now have $p \mid a+b$ and $p \nmid a, b$. By the lifting the exponent lemma, we have

$$v_p(a^{n-1} + b^{n-1}) = v_p(a+b) + v_p(n-1). \quad (2)$$

Also, we have

$$v_p(n!) = v_p(1) + v_p(2) + \cdots + v_p(n), \quad (3)$$

$$v_p(c^{n-1}) \geq n-1. \quad (4)$$

By [proposition 2.2](#), the smallest two among (2), (3), (4) are equal. Since we have shown that $v_p(n!) < \frac{n}{2} \leq n-1$, we must have

$$v_p(a+b) + v_p(n-1) = v_p(1) + v_p(2) + \cdots + v_p(n).$$

As $a+b < n-1$, the two terms on the left are distinct and both appear on the right. This means the only multiples of p from 1 to n are $a+b$ and $n-1$. Thus, $a+b = p$ and $n-1 = 2p$, contradicting $2 \mid n$.

Therefore, we may assume $a+b$ has no odd prime divisor. In other words, $a+b$ is a power of 2. Similarly, assume $b+c$ and $c+a$ are powers of 2. There are several cases to consider.

- If all of a, b, c are divisible by 4, then $v_2(a^{n-1} + b^{n-1} + c^{n-1}) \geq 2(n-1)$. We easily deduce a contradiction as before since $v_2(n!) \leq n-1$.
- If all of a, b, c are odd, then $a^{n-1} + b^{n-1} + c^{n-1}$ is odd, while $n!$ is even, contradiction.
- If $2 \parallel a$ (or other similar cases), then $2a = (a+b) + (c+a) - (b+c)$ is divisible by 4 but not 8. As it is clear that $a+b, b+c, c+a > 2$, one of them must be 4. The case $1+3$ is rejected since otherwise the third number should also be odd. WLOG assume $a = b = 2$. Then $c = 2^k - 2$ for some $k \geq 2$. As $c < n$, we have

$$c \mid n! - c^{n-1} = 2^{n-1} + 2^{n-1} = 2^n.$$

This implies $c \mid 2^n$, and it is not hard to see that $c = 2$. But then $n! = 3 \cdot 2^{n-1}$ does not hold since $5 \mid n!$ but $5 \nmid 3 \cdot 2^{n-1}$ for large n .

Finally, all these conclude that there is no solution to (1) when n is large, so we are done. $\square$

Problem 7.6. (IMO Shortlist 2006 N5) Find all integer solutions of the equation

$$\frac{x^7 - 1}{x - 1} = y^5 - 1. \quad (1)$$

Analysis. The trick used in the first few problems cannot be applied, since we now have fixed exponents. In view of the factorizations

$$\begin{aligned} x^7 - 1 &= (x - 1)(x^6 + x^5 + \cdots + 1), \\ y^5 - 1 &= (y - 1)(y^4 + y^3 + \cdots + 1), \end{aligned}$$

We may consider the possible prime divisors of both sides first. If p is such a prime, then we obtain congruences of the form $x^7 \equiv 1 \pmod{p}$ and $y^5 \equiv 1 \pmod{p}$. Thus, it may be a good idea to use primitive roots.

Solution. Let p be any prime divisor of $\frac{x^7 - 1}{x - 1}$. Then $x^7 \equiv 1 \pmod{p}$. Let g be a primitive root of p and let $x \equiv g^\alpha \pmod{p}$. Then $g^{7\alpha} \equiv 1 \pmod{p}$, and hence $p - 1 \mid 7\alpha$. If $(p - 1, 7) = 1$, then $p - 1 \mid \alpha$. In other words, we have $x \equiv 1 \pmod{p}$. Note that $p \mid \frac{x^7 - 1}{x - 1} = x^6 + x^5 + \cdots + 1$. Thus,

$$0 \equiv x^6 + x^5 + \cdots + 1 \equiv 1 + 1 + \cdots + 1 = 7 \pmod{p}.$$

Therefore, p is either 7 or of the form $7k + 1$ (which corresponds to $(p - 1, 7) > 1$).

From (1) and above, every prime divisor of $y^5 - 1$ is congruent to 0 or 1 modulo 7. In particular, as

$$y^5 - 1 = (y - 1)(y^4 + y^3 + y^2 + y + 1),$$

the same holds for the integers $y - 1$ and $y^4 + y^3 + y^2 + y + 1$. Thus, both of them are congruent to 0 or 1 modulo 7. From $y - 1 \equiv 0, 1 \pmod{7}$, we get $y \equiv 1, 2 \pmod{7}$. This implies

$$y^4 + y^3 + y^2 + y + 1 \equiv 5, 3 \pmod{7}.$$

This is a contradiction. Hence, there is no solution to (1). $\square$

Problem 7.7. (CWMO 2010 Problem 8) Find all integers k such that there exist positive integers a and b satisfying $\frac{b+1}{a} + \frac{a+1}{b} = k$.

Analysis. In this solution, we shall introduce the method of *Vieta jumping*(韋達跳躍), which is indeed a kind of infinite descent method with the use of Vieta's theorem.

Solution. We first consider the case $a = b$. In that case,

$$\frac{b+1}{a} + \frac{a+1}{b} = \frac{2(a+1)}{a} = 2 + \frac{2}{a}.$$

When $a = b = 1$, $k = 4$. When $a = b = 2$, $k = 3$.

Suppose $k \neq 3, 4$. We claim that there are no positive integer solutions to

$$a^2 + b^2 + a + b = kab. \quad (1)$$

Suppose this is solvable. Choose a positive integer solution to (1) for which $a + b$ is minimized. WLOG assume $a > b$. This can be rewritten as

$$a^2 + (1 - kb)a + (b^2 + b) = 0.$$

As a quadratic equation in a , there is another solution a' . By Vieta's theorem, we have

$$\begin{cases} a + a' = kb - 1, \\ aa' = b^2 + b. \end{cases}$$

From the first equation, a' is an integer. From the second equation, a' is positive, and we have

$$a' = \frac{b^2 + b}{a} \leq \frac{(a-1)^2 + (a-1)}{a} = a - 1.$$

This shows $a' < a$. This gives a pair (a', b) of positive integers with $\frac{b+1}{a'} + \frac{a'+1}{b} = k$ and $a' + b < a + b$. This is a contradiction. Therefore, k can only be 3 or 4. $\square$

Remarks. The Vieta jumping can be used to solve this kind of quadratic Diophantine equations which have no solutions (perhaps under some assumption like $a \neq b$). The idea is to pick the smallest solution (a, b) (usually a solution with minimum $a + b$), then consider another root a' of the quadratic equation in a , and show that (a', b) is yet a smaller solution. This gives a contradiction, which means there is no solution.

Problem 7.8. (IMO 2007 Problem 5) Let a and b be positive integers. Show that if $4ab - 1$ divides $(4a^2 - 1)^2$, then $a = b$.

Analysis. This contest problem is a simplification of N6 in the shortlist. It can also be solved using the Vieta jumping.

Solution. Suppose on the contrary that $a \neq b$. Note that $4ab - 1 \mid (4a^2 - 1)^2$ iff $4ab - 1 \mid (4a^2 - 4ab)^2$. As $(4ab - 1, 4a) = 1$, this is equivalent to $4ab - 1 \mid (a - b)^2$. This yields

$$(a - b)^2 = k(4ab - 1) \quad (1)$$

for some $k \in \mathbb{N}$. The merit is that we now obtain a quadratic equation which is symmetric in a, b . We pick the solution to (1) for which $a + b$ is minimized. WLOG assume $a > b$. This equation can be rewritten as

$$a^2 - 2b(2k + 1)a + (b^2 + k) = 0.$$

Let a' be the other solution to this quadratic equation. Note that

$$\begin{cases} a + a' = 2b(2k + 1), \\ aa' = b^2 + k. \end{cases}$$

These show a' is a positive integer. By the minimal choice, we must have $a' \geq a$. Thus,

$$k = aa' - b^2 \geq a^2 - b^2.$$

Then (1) implies $(a - b)^2 \geq (a^2 - b^2)(4ab - 1)$. This gives

$$a - b \geq (a + b)(4ab - 1).$$

This is clearly impossible as $a - b < a + b$. As $a \neq b$ leads to a contradiction, we must have $a = b$. $\square$

Problem 7.9. Solve the equation

$$abc^k = a^k + b^k \tag{1}$$

in integers, where $\gcd(a, b, c) = 1$ and k is nonnegative.

Analysis. It looks like a variation of the Fermat last theorem. Therefore, we guess nontrivial solutions only exist for small k . For the general case, note that the equation implies a, b have the same prime divisors. This recalls the trick we have used in [problem 2.9](#).

Solution. We first handle the cases when k is small.

Case 1. $k = 0$

Equation (1) becomes $ab = 2$. Note that we cannot have $c = 0$ or otherwise c^k is undefined. The solutions are $(a, b, c) = (\pm 1, \pm 2, t), (\pm 2, \pm 1, t)$ where $t \neq 0$.

Case 2. $k = 1$

If one of a, b is zero, then so is the other. Since $\gcd(a, b, c) = 1$, c cannot have any prime divisor. Thus, $c = \pm 1$.

Suppose $a, b \neq 0$. Equation (1) becomes $abc = a + b$. Note that we have $a \mid b$ and $b \mid a$. Therefore, $a = \pm b$.

If $a = b$, then $ac = 2$. The solutions are $(a, b, c) = (\pm 1, \pm 1, \pm 2), (\pm 2, \pm 2, \pm 1)$.

If $a = -b$, then we must have $c = 0$ since $a, b \neq 0$. Again, from $\gcd(a, b, c) = 1$, a, b cannot have any prime divisors. The only solutions are $(a, b, c) = (\pm 1, \mp 1, 0)$.

Case 3. $k = 2$

As above, $(a, b, c) = (0, 0, \pm 1)$ is a solution. Suppose $a, b \neq 0$. Equation (1) becomes $abc^2 = a^2 + b^2$. From $a \mid b^2$ and $b \mid a^2$, we know that a, b have the same prime divisors.

Let p be any common prime divisor of a, b . Let $\alpha = v_p(a)$ and $\beta = v_p(b)$. Since $\gcd(a, b, c) = 1$, we have $p \nmid c$. By proposition 2.2, the smallest 2 among $\alpha + \beta$, 2α and 2β are equal. In any case, we must have $\alpha = \beta$. Since this holds for every prime p , we have $a = \pm b$. The equation becomes $c^2 = \pm 2$, which has no solution in integers.

Case 4. $k \geq 3$

As above, $(a, b, c) = (0, 0, \pm 1)$ is a solution. Suppose $a, b \neq 0$. Similar to case 3, we find that a, b have the same prime divisors.

Let p be any common prime divisor of a, b . Let $\alpha = v_p(a)$ and $\beta = v_p(b)$. Since $\gcd(a, b, c) = 1$, we have $p \nmid c$. By proposition 2.2, the smallest 2 among $\alpha + \beta$, $k\alpha$ and $k\beta$ are equal. Since $k \geq 3$, $\alpha + \beta$ must be one of the smallest 2. We either have $\alpha + \beta = k\alpha$ or $\alpha + \beta = k\beta$. This means $\beta = (k-1)\alpha$ or $\alpha = (k-1)\beta$.

Thus, we can partition the prime divisors of a, b into 2 groups. There exist coprime positive integers P, Q such that $a = \mu P Q^{k-1}$ and $b = \nu P^{k-1} Q$ where $\mu, \nu = \pm 1$. The equation becomes

$$\mu\nu P^k Q^k c^k = (\mu P Q^{k-1})^k + (\nu P^{k-1} Q)^k.$$

This can be simplified to give

$$\mu\nu c^k = \mu^k (Q^{k-2})^k + \nu^k (P^{k-2})^k. \quad (2)$$

This has the form of the Fermat last theorem as each term is a k th power multiplied to ± 1 . We can study each case in detail.

- If $2 \mid k$ with $\mu = \nu$, or $2 \nmid k$ with $\mu = \nu = 1$, then (2) becomes

$$c^k = (Q^{k-2})^k + (P^{k-2})^k.$$

By the Fermat last theorem, as $P, Q > 0$, there is no solution.

- If $2 \mid k$ with $\mu = -\nu$, then (2) becomes

$$-c^k = (Q^{k-2})^k + (P^{k-2})^k.$$

This has no solution since $-c^k \leq 0$ while the right-hand side is positive.

- If $2 \nmid k$ with $\mu = \nu = -1$, then (2) becomes

$$c^k = -(Q^{k-2})^k - (P^{k-2})^k.$$

This has no solution since $-c^k \geq 0$ while the right-hand side is negative.

- If $2 \nmid k$ with $\mu = 1$ and $\nu = -1$, then (2) becomes

$$-c^k = (Q^{k-2})^k - (P^{k-2})^k.$$

By the Fermat last theorem, the only solution is $c = 0$ and $P = Q$. As $(P, Q) = 1$ and $P, Q > 0$, this implies $P = Q = 1$. One checks that $(a, b, c) = (1, -1, 0)$ is a solution.

- If $2 \nmid k$ with $\nu = -1$ and $\mu = 1$, then the case is analogous to the case we have just handled. This gives another solution $(a, b, c) = (-1, 1, 0)$.

To conclude, we can list out the following solutions:

$$\begin{aligned} (a, b, c, k) = & (\pm 1, \pm 2, t, 0), (\pm 2, \pm 1, t, 0) \quad \text{where } t \neq 0, \\ & (0, 0, \pm 1, u) \quad \text{where } u > 0, \\ & (\pm 1, \pm 1, \pm 2, 1), (\pm 2, \pm 2, \pm 1, 1), (\pm 1, \mp 1, 0, v) \quad \text{where } v \text{ is odd.} \end{aligned}$$

□

Problem 7.10. Solve the equation

$$p^q - q^k = (p + q)^{k-q} \tag{1}$$

where p, q are prime numbers and k is a nonnegative integer.

Analysis. We shall combine various tricks that we have learnt together to solve this Diophantine equation. These include the concept of order, the binomial expansion and establishing algebraic bounds.

Solution. Since the left-hand side of (1) is an integer, so is the right-hand side, and we have $k \geq q$. Note that $k = q$ yields $p^q - q^q = 1$, which has no solution as there is no pair of consecutive positive q th powers with $q \geq 2$. Therefore, we have $k > q$.

By applying weak bounds to (1), we get

$$p^q = (p + q)^{k-q} + q^k > p^{k-q}.$$

This gives $q > k - q$ so that $2q > k$.

Also, from (1) and $k > q$, we have

$$p^q = (p + q)^{k-q} + q^k > q^k > q^q.$$

This implies $p > q$. We have obtained several simple relations between the variables. The next natural step is to consider modulo p or q . This motivates the consideration of orders.

Consider modulo p . Equation (1) gives $-q^k \equiv q^{k-q} \pmod{p}$. As $p \neq q$, this implies $q^q \equiv -1 \pmod{p}$. Let d be the order of q modulo p . Note that $p > q \geq 2$. Then we have $d \mid 2q$ but $d \nmid q$. Also, we have $d \mid p - 1$. As q is a prime, we can only have $d = 2$ or $d = 2q$.

Case 1. $d = 2$

This shows $q^2 \equiv 1 \pmod{p}$ and $q \not\equiv 1 \pmod{p}$. Thus, $q \equiv -1 \pmod{p}$. As $p \mid q + 1$ and $p > q$, we can only have $(p, q) = (3, 2)$. Then equation (1) becomes $9 - 2^k = 5^{k-2}$. There is no solution.

Case 2. $d = 2q$

Then $2q \mid p - 1$ so that $p \equiv 1 \pmod{q}$. Let $p = aq + 1$. Equation (1) becomes

$$(aq + 1)^q - q^k = (aq + 1 + q)^{k-q}. \quad (2)$$

Since we have considered modulo p , this time we should consider modulo powers of q . There is no result obtained from modulo q . We consider modulo q^2 . By the binomial expansion, as $k > q \geq 2$, equation (2) implies

$$1 + q(aq) \equiv (k - q)(a + 1)q + 1 \pmod{q^2},$$

i.e. $q \mid (k - q)(a + 1)$. As $2q > k > q$ and q is a prime, we have $(k - q, q) = 1$ so that $q \mid a + 1$. Let $a = bq - 1$. Then $p = aq + 1 = bq^2 - q + 1$. Equation (2) becomes

$$(bq^2 - q + 1)^q - q^k = (bq^2 + 1)^{k-q}. \quad (3)$$

In view of Hensel's lifting lemma, there is no hope to obtain contradiction by considering congruences modulo higher powers of q . However, equation (3) is sufficient to yield a contradiction with some algebraic bounds for the general case. Indeed, we note that $(bq^2 - q + 1)^q$ dominates $(bq^2 + 1)^{k-q}$ since $q > k - q$. Also, $(bq^2 - q + 1)^q$ is larger than q^{2q} if $b > 1$, and this dominates q^k . We can write down these things more mathematically.

We shall prove that

$$(bq^2 - q + 1)^q > q^{2q-1} + (bq^2 + 1)^{q-1} \quad (4)$$

for $b > 1$. If this is true, then we get

$$q^k + (bq^2 + 1)^{k-q} = (bq^2 - q + 1)^q > q^{2q-1} + (bq^2 + 1)^{q-1},$$

which is a contradiction as $k \leq 2q - 1$ and $k - q \leq q - 1$.

If $q = 2$, then (4) is the same as

$$(4b - 1)^2 > 8 + (4b + 1).$$

This is equivalent to $4b^2 - 3b - 2 > 0$. This clearly holds for $b \geq 2$.

Suppose q is an odd prime. Note that the terms $bq^2 - q + 1$ and $bq^2 + 1$ only differ by a q . Thus, we apply the binomial expansion to $((bq^2 + 1) - q)^q$. We get

$$(bq^2 + 1 - q)^q = \sum_{r=0}^q \binom{q}{r} (bq^2 + 1)^r (-q)^{q-r}. \quad (5)$$

Only the terms with $r = q - 1$ and q are important. We first show that the remaining terms have a positive sum. It suffices to show that the sum of terms with $r = q - 2m$ and $r = q - 2m - 1$ is positive where $1 \leq m \leq \frac{q-1}{2}$. Indeed,

$$\begin{aligned}
& \binom{q}{q-2m} (bq^2 + 1)^{q-2m} \cdot q^{2m} > \binom{q}{q-2m-1} (bq^2 + 1)^{q-2m-1} \cdot q^{2m+1} \\
\Leftrightarrow & \frac{q!}{(q-2m)!(2m)!} \cdot (bq^2 + 1) > \frac{q!}{(q-2m-1)!(2m+1)!} \cdot q \\
\Leftrightarrow & (2m+1)(bq^2 + 1) > (q-2m)q \\
\Leftrightarrow & 2m(bq^2 + q + 1) + (b-1)q^2 + 1 > 0.
\end{aligned}$$

This is true as $b > 1$. Therefore, from (5), we get

$$(bq^2 + 1 - q)^q > (bq^2 + 1)^q - (bq^2 + 1)^{q-1} q^2.$$

Comparing with our goal (4), it suffices to show

$$(bq^2 + 1)^q - (bq^2 + 1)^{q-1} q^2 \geq q^{2q-1} + (bq^2 + 1)^{q-1}.$$

Note that

$$\begin{aligned}
& (bq^2 + 1)^q - (bq^2 + 1)^{q-1} q^2 \geq q^{2q-1} + (bq^2 + 1)^{q-1} \\
\Leftrightarrow & (b-1)(bq^2 + 1)^{q-1} q^2 \geq q^{2q-1} \\
\Leftrightarrow & (b-1)(bq^2 + 1)^{q-1} \geq q^{2q-3}.
\end{aligned}$$

This is true since $(b-1)(bq^2 + 1)^{q-1} > (q^2)^{q-1} > q^{2q-3}$ for $b > 1$. Therefore, we have established (4), which leads to a contradiction as explained.

Thus, it remains to consider the case $b = 1$. Equation (3) becomes

$$(q^2 - q + 1)^q - q^k = (q^2 + 1)^{k-q}, \quad (6)$$

where $p = q^2 - q + 1$ is a prime. One may note that there is a solution when $q = 3$ by trial and error. Indeed, in that case the equation gives $7^3 - 3^k = 10^{k-3}$, which has a unique solution $k = 5$. Also, $p = 3^2 - 3 + 1 = 7$ is a prime.

Suppose $q \neq 3$. We guess there is no solution. In order to use the condition $q \neq 3$, naturally we should consider modulo 3. As q is a prime, we must have $3 \nmid q$.

- If $q \equiv 1 \pmod{3}$, then (6) yields $1 - 1 \equiv 2^{k-q} \pmod{3}$, which cannot hold.
- If $q \equiv 2 \pmod{3}$, then $p = q^2 - q + 1 \equiv 0 \pmod{3}$. This implies $p = 3$ since p is a prime. Then $q = 2$, which has been discussed in case 1.

To sum up, the only solution is $(p, q, k) = (7, 3, 5)$. □

Problem 7.11. Solve the system of equations

$$\begin{cases} p^{2\alpha} + q^{2\beta} + (pq)^{2\gamma} = (pq + 1)^{2\delta}, \\ p^\alpha + q^\beta + \gamma^3 = (pq + 1)^\delta, \end{cases}$$

where p, q are prime numbers and $\alpha, \beta, \gamma, \delta$ are positive integers.

Analysis. The tools for solving this problem is very similar to the same kind of problems we have come across. But this time we have a simultaneous equation, which makes things more complicated. By checking the parity, it is easy to conclude one of p, q is 2. Another possible observation is that the first equation is close to the square of the second equation. Therefore, the first step in the following solution is natural.

Solution. We label

$$p^{2\alpha} + q^{2\beta} + (pq)^{2\gamma} = (pq + 1)^{2\delta} \quad (1)$$

and

$$p^\alpha + q^\beta + \gamma^3 = (pq + 1)^\delta. \quad (2)$$

If p, q have the same parity, then (1) cannot hold. By symmetry, we may assume $p = 2$ and $q > 2$.

Consider (1)–(2)². This gives

$$(2q)^{2\gamma} = \gamma^6 + 2^{\alpha+1}q^\beta + 2^{\alpha+1}\gamma^3 + 2q^\beta\gamma^3. \quad (3)$$

Since this equation contains many powers of 2, we should consider the 2-adic valuations. Let $\gamma = 2^m t$ where t is odd. We see from (2) that γ is even. Thus, $m \geq 1$.

The 2-adic valuations of the terms in (3) are 2γ , $6m$, $\alpha + 1$, $\alpha + 1 + 3m$ and $3m + 1$ respectively. We now apply [proposition 2.2](#). There are 2 possibilities.

Case 1. $m = t = 1$ so that $\gamma = 2$

Equation (3) becomes $16q^4 = 64 + 2^{\alpha+1}q^\beta + 2^{\alpha+4} + 16q^\beta$. Since $\alpha + 1$ cannot be the unique smallest exponent of 2, the two sides must be divisible by 16. This gives

$$q^4 = 4 + 2^{\alpha-3}q^\beta + 2^\alpha + q^\beta. \quad (4)$$

Now, we should consider modulo q . We get $2^\alpha \equiv -4 \pmod{q}$. However, from (2), we have $2^\alpha + q^\beta + 8 = (2q + 1)^\delta$ so that $2^\alpha \equiv -7 \pmod{q}$. These yield $-4 \equiv -7 \pmod{q}$. We can only have $q = 3$, in which case (4) becomes $77 = 2^{\alpha-3}3^\beta + 2^\alpha + 3^\beta$. Clearly, $\beta \leq 3$. We check that only $\beta = 2$ gives a solution $\alpha = 5$ and $\delta = 2$.

Case 2. $(m, t) \neq (1, 1)$

Note that $6m > 3m + 1$ and $\alpha + 1 + 3m > \alpha + 1$. Also, we can prove in this case that $2\gamma = 2^{m+1}t > 3m + 1$ by induction. Therefore, the smallest 2 terms among 2γ , $6m$,

$\alpha + 1$, $\alpha + 1 + 3m$ and $3m + 1$ must be $\alpha + 1$ and $3m + 1$. Thus, $\alpha = 3m$. The system becomes

$$\begin{cases} 2^{6m} + q^{2\beta} + (2q)^{2^{m+1}t} = (2q + 1)^{2\delta}, \\ 2^{3m} + q^\beta + 2^{3m}t^3 = (2q + 1)^\delta. \end{cases} \quad (5)$$

Again, we would like to consider modulo a certain power of q . But first of all, we need some comparison between the exponents 2β and $2^{m+1}t$.

The first equation of (5) gives $(2q + 1)^{2\delta} > (2q)^{2^{m+1}t}$. Note that $(2q)^{4\delta} > (2q + 1)^{2\delta}$. These combine to give $\delta > 2^{m-1}t$.

Consider the second equation of (5). We find that

$$2^{3m} + q^\beta + 2^{3m}t^3 = (2q + 1)^\delta > (2q + 1)^{2^{m-1}t} \geq 7^{2^{m-1}t}. \quad (6)$$

We claim that

$$7^{2^{m-1}t} > 2(2^{3m} + 2^{3m}t^3) \quad (7)$$

for $m \geq 3$ and $t \geq 1$. When $(m, t) = (3, 1)$, we have $7^4 = 2401 > 2048 = 2(512 + 512)$. Suppose (7) holds for some $m \geq 3$ and $t \geq 1$. Then we have

$$\begin{aligned} 7^{2^mt} &= (7^{2^{m-1}t})^2 > 4(2^{3m} + 2^{3m}t^3)^2 \\ &= 4(2^{6m} + 2^{6m}t^6 + 2^{6m+1}t^3) > 2(2^{3(m+1)} + 2^{3(m+1)}t^3). \end{aligned}$$

This shows the case $(m + 1, t)$ holds. Also, we have

$$7^{2^{m-1}(t+1)} > 7^{2^{m-1}} \cdot 2(2^{3m} + 2^{3m}t^3) > 2(2^{3m} + 2^{3m}(2t)^3) > 2(2^{3m} + 2^{3m}(t + 1)^3).$$

This shows the case $(m, t + 1)$ holds. By induction on both m and t , we establish (7) for $m \geq 3$.

Suppose $m \geq 3$. From (6) and (7), we have $q^\beta > 2^{3m} + 2^{3m}t^3$. Using this, from (6), we get

$$q^\beta > \frac{1}{2}(2q + 1)^\delta > \frac{1}{2}(2q + 1)^{2^{m-1}t}.$$

Then $q^{\beta+1} > (2q + 1)^{2^{m-1}t} > q^{2^{m-1}t}$ so that $\beta \geq 2^{m-1}t$.

We can now consider modulo $q^{2^{m-1}t}$ for the system (5). This gives

$$\begin{cases} 2^{6m} \equiv (2q + 1)^{2\delta} \pmod{q^{2^{m-1}t}}, \\ 2^{3m} + 2^{3m}t^3 \equiv (2q + 1)^\delta \pmod{q^{2^{m-1}t}}. \end{cases}$$

Equating these congruences, we get $(1 + t^3)^2 \equiv 1 \pmod{q^{2^{m-1}t}}$. This is the same as $q^{2^{m-1}t} \mid t^3(t^3 + 2)$. As $(t^3, t^3 + 2) = (t^3, 2)$ is either 1 or 2, only one of t^3 and $t^3 + 2$ is divisible by q . Then $q^{2^{m-1}t}$ divides t^3 or $t^3 + 2$. In both cases, we have

$$t^3 + 2 \geq q^{2^{m-1}t} \geq 3^{4t}. \quad (8)$$

When $t = 1$, we have $t^3 + 2 = 3 < 81 = 3^{4t}$. If $t^3 + 2 < 3^{4t}$ for some $t \geq 1$, then

$$(t + 1)^3 + 2 \leq (2t)^3 + 2 < 81(t^3 + 2) < 3^{4(t+1)}.$$

Therefore, (8) cannot hold by induction.

It remains to consider cases $m = 1, 2$.

- When $m = 1$, equation (1) gives $2^6 \equiv 1 \pmod{q}$. Therefore, $q = 3, 7$.

If $q = 3$, equation (1) becomes $2^6 + 3^{2\beta} + 6^{4t} = 7^{2\delta}$. This gives $3^{2\beta} + 2 \equiv 0 \pmod{7}$. However, $3^{2\beta} \equiv 2, 4, 1 \pmod{7}$. There is no solution.

If $q = 7$, equation (1) becomes $2^6 + 7^{2\beta} + 14^{4t} = 15^{2\delta}$. This gives $7^{2\beta} \equiv 0 \pmod{5}$, which cannot hold.

- When $m = 2$, equation (1) gives $2^{12} \equiv 1 \pmod{q}$. Therefore, $q = 3, 5, 7, 13$.

If $q = 3$, equation (1) becomes $2^{12} + 3^{2\beta} + 6^{8t} = 7^{2\delta}$. This gives $3^{2\beta} + 2 \equiv 0 \pmod{7}$. Again, there is no solution.

If $q = 5$, equation (1) becomes $2^{12} + 5^{2\beta} + 10^{8t} = 11^{2\delta}$. From this, we obtain $5^{2\beta} + 5 \equiv 0 \pmod{11}$. Note that the Legendre symbol $\left(\frac{-5}{11}\right) = -\left(\frac{11}{5}\right) = -1$. Thus, -5 cannot be a quadratic residue modulo 11. There is no solution.

If $q = 7$, equation (1) becomes $2^{12} + 7^{2\beta} + 14^{8t} = 15^{2\delta}$. From this, we obtain $2^{2\beta} + 2 \equiv 0 \pmod{5}$. However, $2^{2\beta} \equiv 4, 1 \pmod{5}$. There is no solution.

If $q = 13$, equation (1) becomes $2^{12} + 13^{2\beta} + 26^{8t} = 27^{2\delta}$. From this, we obtain $5^{8t} + 2 \equiv 1 \pmod{7}$. This cannot hold since -1 is a quadratic nonresidue modulo 7.

Therefore, the only solution is $(p, q, \alpha, \beta, \gamma, \delta) = (2, 3, 5, 2, 2, 2)$ or $(3, 2, 5, 2, 2, 2)$. We check that this is a solution as $2^{10} + 3^4 + 6^4 = 2401 = 7^4$ and $2^5 + 3^2 + 2^3 = 7^2$. $\square$

Problem 7.12. (Brazil MO 2010 Problem 6) Find all pairs (a, b) of positive integers such that

$$3^a = 2b^2 + 1. \quad (1)$$

Analysis. Equation (1) looks simple. However, it turns out that this is a difficult problem. It again combines a lot of techniques that we have introduced.

Solution.

Case 1. a is odd

It seems that the case $2 \mid a$ should be simpler, since 3^a would then be a square. However, we find that we shall need the method of infinite descent in that case. Therefore, it is better to solve the odd case completely first.

Let $c = 3^{\frac{a-1}{2}}$. Equation (1) becomes $3c^2 = 2b^2 + 1$. This is similar to Pell's equation, but with some coefficients. One usual way is to use substitutions to cancel one of the coefficients. For example, we can consider $c = x + 2y$ and $b = x + 3y$. We get Pell's equation $x^2 - 6y^2 = 1$. Note that $9c^2 > 3(2b^2) > 4b^2$ so that $x = 3c - 2b > 0$. Also, $3c^2 \leq 3b^2$ so that $y = b - c \geq 0$.

The smallest positive solution is $(x_1, y_1) = (5, 2)$. Thus, all positive solutions are given by (x_n, y_n) where $x_n + \sqrt{6}y_n = (5 + 2\sqrt{6})^n$ for $n \geq 1$. From

$$\begin{aligned} x_n + \sqrt{6}y_n &= (5 + 2\sqrt{6})^{n-1}(5 + 2\sqrt{6}) = (x_{n-1} + \sqrt{6}y_{n-1})(5 + 2\sqrt{6}) \\ &= (5x_{n-1} + 12y_{n-1}) + (2x_{n-1} + 5y_{n-1})\sqrt{6}, \end{aligned}$$

we get the recurrence relation $x_n = 5x_{n-1} + 12y_{n-1}$ and $y_n = 2x_{n-1} + 5y_{n-1}$. The second relation can be rewritten as

$$x_{n-1} = \frac{1}{2}y_n - \frac{5}{2}y_{n-1}.$$

Applying this relation to $x_n = 5x_{n-1} + 12y_{n-1}$, we get

$$\frac{1}{2}y_{n+1} - \frac{5}{2}y_n = \frac{5}{2}y_n - \frac{25}{2}y_{n-1} + 12y_{n-1},$$

and hence

$$y_{n+1} = 10y_n - y_{n-1}.$$

Similarly, we also have $x_{n+1} = 10x_n - x_{n-1}$. For each solution x_n, y_n , we define c_n to be the corresponding value of c , i.e. $c_n = x_n + 2y_n$. Then we get $c_{n+1} = 10c_n - c_{n-1}$. From the small solutions $(x_1, y_1) = (5, 2)$ and $(x_2, y_2) = (49, 20)$, we find $c_1 = 9$ and $c_2 = 89$. We can also define $c_0 = 1$ corresponding to the solution $(x_0, y_0) = (1, 0)$. This gives us the sequence $\{c_n\}$, which are the possible candidates of c . Recall that $c = 3^{\frac{a-1}{2}}$ is a power of 3. We should find those c_n which are powers of 3.

Note that $c_0 = 1$ gives $(a, b) = (1, 1)$ while $c_1 = 9$ gives $(a, b) = (5, 11)$. Suppose c_n is a power of 3 greater than 9. Taking modulo 27, we have

$$c_0, c_1, \dots \equiv 1, 9, 8, 17, 0, 10, 19, 18, 26, 26, 18, 19, 10, 0, 17, 8, 9, 1 \pmod{27},$$

where the pattern repeats again every 18 terms. Thus, $27 \mid c_n$ iff $n \equiv 4, 13 \pmod{18}$ from the above pattern. In particular, note that $c_4 = 8721 = 27 \times 17 \times 19$. Therefore, we may further consider modulo 17 or 19. Indeed, we have

$$c_0, c_1, \dots \equiv 1, 9, 4, 14, 0, 3, 13, 8, 16, 16, 8, 13, 3, 0, 14, 4, 9, 1 \pmod{17}.$$

Again, the pattern repeats every 18 terms. Whenever $n \equiv 4, 13 \pmod{18}$, we always have $17 \mid c_n$. Therefore, c_n cannot be a power of 3 greater than 9.

Case 2. a is even

Let $a = 2d$. Then equation (1) yields

$$2b^2 = 3^{2d} - 1 = (3^d - 1)(3^d + 1).$$

Note that $(3^d - 1, 3^d + 1) = (3^d - 1, 2) = 2$. Analogous to [proposition 2.3](#), we must have $3^d - 1 = 2u^2$ and $3^d + 1 = 4v^2$ for some coprime u, v , or $3^d - 1 = 4u^2$ and $3^d + 1 = 2v^2$. The latter case is rejected since we have the contradiction

$$-1 \equiv 3^d - 1 = 4u^2 \equiv 1 \pmod{3}.$$

For the former case, we get the relation $3^d - 1 = 2u^2$. This is the same as equation [\(1\)](#), with $d < a$ (indeed $a = 2d$). Therefore, every solution with $2 \mid a$ is generated by a solution with smaller exponent of 3. Thus, we can use the descent method to find all solutions.

Firstly, from the solution $(d, u) = (1, 1)$, we have $a = 2$ and hence $b = 2$. If we further consider $(d, u) = (2, 2)$, we get $a = 4$ but there is no solution for b .

Secondly, from the solution $(d, u) = (5, 11)$, we have $a = 10$. There is no solution for b .

Hence, the only solutions are $(a, b) = (1, 1), (2, 2), (5, 11)$. □

We have used many tricks in this section to solve various Diophantine equations, as listed below.

- factorization ([problems 7.2, 7.3, 7.5, 7.6, 7.12](#))
- deriving bounds ([problems 7.5, 7.7, 7.8, 7.10, 7.11](#))
- considering congruences ([problems 7.1, 7.2, 7.3, 7.5, 7.6, 7.10, 7.11, 7.12](#))
- writing $x = da$ and $y = db$ where $d = (x, y)$ ([problem 7.4](#))
- considering p -adic valuations ([problems 7.1, 7.4, 7.5, 7.9, 7.11](#))
- lifting the exponent lemma ([problems 7.4, 7.5](#))
- orders ([problems 7.1, 7.2, 7.3, 7.10, 7.11](#))
- primitive roots ([problem 7.6](#))
- quadratic residues ([problem 7.11](#))
- infinite descent, Vieta jumping ([problems 7.7, 7.8, 7.12](#))

8 Prime Numbers

Lastly, we use this short section to give a few more advanced theorems and conjectures related to primes, mainly for self-interest.

Theorem 8.1. ([Dirichlet's theorem](#)(狄利克雷定理)) Let a and d be relatively prime integers where $d > 0$. Then there are infinitely many primes of the form $a + kd$.

This is the only useful theorem for our purpose. This can be used to construct primes lying in certain residue classes. In addition, it is known that there exist positive constants c and L (independent of a and d) such that the smallest prime in the arithmetic progression $a + kd$ with $1 \leq a \leq d-1$ and $(a, d) = 1$ is less than cd^L . The exponent L is called [Linnik's constant](#)(林尼克常數). It is known that $L = 5$ works, and the conjecture is $L = 2$.

Theorem 8.2. ([Bertrand's postulate](#)(伯特蘭-切比雪夫定理)) Let $n > 1$ be a positive integer. Then there exists a prime number strictly between n and $2n$.

Theorem 8.3. ([Green-Tao theorem](#)(格林-陶定理)) Let k be a positive integer. Then there exists an arithmetic progression of prime numbers containing k terms.

For example, when $k = 6$, we may choose 7, 37, 67, 97, 127, 157. Note that we cannot specify the common difference in this theorem. There is a generalization of the Green-Tao theorem given below.

Let $P_1(n), P_2(n), \dots, P_k(n)$ be any polynomials with integer coefficients and vanishing constant terms (i.e. $P_1(0) = P_2(0) = \dots = P_k(0) = 0$). Then there exist infinitely many pairs (a, b) of integers such that $P_j(a) + b$ is a prime for $j = 1, 2, \dots, k$.

Theorem 8.4. There exist infinitely many pairs of primes whose differences are at most 246.

This means there are infinitely many pairs of primes with bounded gaps.

The following give some well-known conjectures related to prime numbers, which have not been proven yet. Therefore, one should not use these or try to prove these when working on MO problems!

Conjecture 1. ([Twin prime conjecture](#)(孿生質數猜想)) There exist infinitely many pairs of primes whose differences are 2.

If this is true, then it serves as an improved result of [theorem 8.4](#).

Conjecture 2. ([Goldbach conjecture](#)(哥德巴赫猜想)) Let $n > 2$ be an even integer. Then n is a sum of 2 prime numbers.

Although this has not been proven, the weak Goldbach problem has been established, i.e. every odd integer greater than 5 is a sum of 3 prime numbers.

Conjecture 3. ([Legendre's conjecture](#)(勒讓德猜想)) Let n be a positive integer. Then there exists a prime number between n^2 and $(n+1)^2$.

Conjecture 4. There are infinitely many positive integers n for which $n^2 + 1$ is a prime number.

The above 4 conjectures together are known as [Landau's problems](#)(蘭道猜想).

Conjecture 5. Let n be a sufficiently large integer of the form $24k + 4$. Then n is a sum of 4 squares of prime numbers.

If this is true, then it can be viewed as an improved result of the sum of 4 squares theorem. Note that the condition that $n \equiv 4 \pmod{24}$ is necessary since any prime p greater than 3 satisfies $p^2 \equiv 1 \pmod{24}$. On the other hand, it has been proven that any sufficiently large integer of the form $24k + 5$ is the sum of 5 squares of primes.

Problem 8.1. Let n be a positive integer. If n is a quadratic residue modulo every prime p , prove that n is a square number.

Analysis. Suppose n is not a square number. It suffices to find a prime p for which $\left(\frac{n}{p}\right) = -1$. Note that $\left(\frac{n}{p}\right)$ can be expressed as a product of terms $\left(\frac{q}{p}\right)$ where q is a prime. Each of these terms equals $\left(\frac{p}{q}\right)$ if p is of the form $4k+1$. Then we can control the value of $\left(\frac{p}{q}\right)$ by choosing p to lie in a certain residue class modulo q . This is how we construct the desired p .

Solution. Suppose on the contrary that n is not a square. Let $n = m^2 p_1 p_2 \cdots p_s$ where $p_1 < p_2 < \cdots < p_s$ are primes. By the assumption, we have $s \geq 1$.

Case 1. $p_1 > 2$

Let a be a quadratic nonresidue modulo p_1 . We choose a prime p such that

$$\begin{cases} p \equiv 1 \pmod{4p_2 \cdots p_s}, \\ p \equiv a \pmod{p_1}. \end{cases}$$

Indeed, this restricts p to lie in a certain residue class modulo $4p_1p_2 \cdots p_s$ by the Chinese remainder theorem. Since $(a, p_1) = 1$, this residue class must be relatively prime to $4p_1p_2 \cdots p_s$. Therefore, such a prime exists by Dirichlet's theorem on arithmetic progression.

For this prime p , since $p \equiv 1 \pmod{4}$, by the quadratic reciprocity law, we have

$$\begin{aligned} \left(\frac{n}{p}\right) &= \left(\frac{m^2}{p}\right) \left(\frac{p_1}{p}\right) \left(\frac{p_2}{p}\right) \cdots \left(\frac{p_s}{p}\right) \\ &= \left(\frac{p}{p_1}\right) \left(\frac{p}{p_2}\right) \cdots \left(\frac{p}{p_s}\right) \\ &= \left(\frac{a}{p_1}\right) \left(\frac{1}{p_2}\right) \cdots \left(\frac{1}{p_s}\right) \\ &= -1. \end{aligned}$$

This contradicts the fact that n is a quadratic residue modulo p .

Case 2. $p_1 = 2$

The above proof fails only because we cannot apply the quadratic reciprocity law for the prime 2. But we can use [proposition 5.1\(f\)](#) to replace this. We choose a prime p such that

$$\begin{cases} p \equiv 5 \pmod{8}, \\ p \equiv 1 \pmod{p_2p_3 \cdots p_s}. \end{cases}$$

Then

$$\left(\frac{n}{p}\right) = \left(\frac{m^2}{p}\right) \left(\frac{2}{p}\right) \left(\frac{p_2}{p}\right) \cdots \left(\frac{p_s}{p}\right) = \left(\frac{2}{p}\right) \left(\frac{p}{p_2}\right) \cdots \left(\frac{p}{p_s}\right) = -1.$$

This again yields a contradiction.

Hence, n must be a square number. □

Problem 8.2. (IMO Shortlist 2008 N6/Problem 3) Prove that there exist infinitely many positive integers n such that $n^2 + 1$ has a prime divisor which is greater than $2n + \sqrt{2n}$.

Analysis. In view of [conjecture 4](#), we probably do not want to show that $n^2 + 1$ is a prime for infinitely many n . Note that for a given n , we cannot say much about the prime divisors of $n^2 + 1$, since the polynomial is irreducible. Also, if this can be done,

then it should not be hard to prove or disprove the conjecture. With this insight, we may think in another way.

Indeed, we may try to fix a prime p and find those n 's with $p \mid n^2 + 1$. The goal is to show that there exists an n which is not too large. As -1 is a quadratic residue, we should focus on primes of the form $4k + 1$.

Solution. Take any prime of the form $4k + 1$. There are exactly 2 solutions to the congruence $m^2 \equiv -1 \pmod{p}$ with $1 \leq m \leq p - 1$. Note that m is a solution iff $p - m$ is a solution. Therefore, there is a solution in the range $1 \leq m \leq \frac{p-1}{2}$. We choose this small solution so that p is large when compared to m . However, for this solution m , we have only shown that $m^2 + 1$ has a prime divisor p with $p \geq 2m + 1$. This is slightly weaker than the desired bound $2m + \sqrt{2m}$.

The bound that we obtain is not strong enough because we have only used the constraint $m \leq \frac{p-1}{2}$. However, this has nothing to do with $m^2 \equiv -1 \pmod{p}$. Indeed, one can show that $m = \frac{p-1}{2}$ cannot be a solution for $p > 5$. Thus, we should modify our argument to see how large can m be.

Let $m = \frac{p-1}{2} - a$ where $a \geq 0$. Our goal is to find a nontrivial lower bound for a . Now,

$$\begin{aligned} & \left(\frac{p-1}{2} - a \right)^2 \equiv -1 \pmod{p} \\ \Leftrightarrow & (2a+1)^2 \equiv -4 \pmod{p} \\ \Leftrightarrow & 4a^2 + 4a + 5 \equiv 0 \pmod{p}. \end{aligned}$$

This implies

$$p \leq 4a^2 + 4a + 5, \tag{1}$$

which indeed provides a lower bound for a . This is sufficient for our purpose. Now, as $p > 2m$, we have

$$\begin{aligned} & p > 2m + \sqrt{2m} \\ \Leftrightarrow & (p - 2m)^2 > 2m \\ \Leftrightarrow & (2a+1)^2 > p - 1 - 2a \\ \Leftrightarrow & 4a^2 + 6a + 2 > p. \end{aligned}$$

This is true from (1), as long as $a \geq 2$. Indeed, if we choose $p \geq 37$, then (1) implies $4a^2 + 4a + 5 \geq 37$ so that $a > 2$.

Therefore, we can pick any prime $p_1 \geq 37$ with $p_1 \equiv 1 \pmod{4}$. Then there exists n_1 such that p_1 is a prime divisor of $n_1^2 + 1$ and $p_1 > 2n_1 + \sqrt{2n_1}$. Suppose we have chosen $n_1, n_2, \dots, n_s$ with the same property. We pick a prime $p_{s+1} \geq 37$ of the form $4k + 1$

such that it does not divide any $n_j^2 + 1$. Such a prime exists by Dirichlet's theorem. For this prime p_{s+1} , we again find an integer n_{s+1} such that p_{s+1} is a prime divisor of $n_{s+1}^2 + 1$ and $p_{s+1} > 2n_{s+1} + \sqrt{2n_{s+1}}$. Then n_{s+1} is distinct from $n_1, n_2, \dots, n_s$ since p_{s+1} only divides $n_{s+1}^2 + 1$. Recursively, we can find infinitely many integers n with the desired property. $\square$

Remarks. Sometimes if we find something which is weaker than what we want, it does not mean this information is useless. A slight modification may give completely different results.

Problem 8.3. (All-Russian MO 2011 Grade 11 Problem 7) Let $P(a)$ be the largest prime positive divisor of $a^2 + 1$. Prove that there exist infinitely many positive integers a, b, c such that $P(a) = P(b) = P(c)$.

Analysis. This is another problem that studies the polynomial $n^2 + 1$. Similar to the previous problem, we fix p and determine a, b, c for which $P(a) = P(b) = P(c) = p$. This problem is slightly different since we not only need p to be a prime divisor of $a^2 + 1$, but also the largest one.

Solution. Let p be any prime of the form $4k + 1$. There are 2 solutions a and b to $m^2 \equiv -1 \pmod{p}$ such that $1 \leq a \leq \frac{p-1}{2}$ and $b = p - a$. Since $b^2 + 1 \leq (p-1)^2 + 1 < p^2$, We have $p > \sqrt{b^2 + 1}$ so that $P(b) = p$. For the same reason, we have $P(a) = p$.

Next, we should consider the small numbers $p + a$, $2p - a$, $2p + a$ and $3p - a$, etc. In these cases, p needs not be the largest prime divisor. One way is to show that there is another small prime divisor, say 5 (note that it cannot be 3), so that p is still the largest one. For this reason, it is better to fix p to lie in a certain residue class modulo 5 first. For example, we can restrict p to be a prime of the form $20k + 17$ (which means $p \equiv 2 \pmod{5}$) by Dirichlet's theorem (indeed, the choice $p \equiv 3 \pmod{5}$ also works).

Case 1. $a \equiv 0, 1 \pmod{5}$

Note that $(p + a)^2 \equiv -1 \pmod{5}$ in this case. Thus, $(p + a)^2 + 1$ equals $5pd$ for some $d \in \mathbb{N}$. As

$$d = \frac{(p + a)^2 + 1}{5p} < \frac{(2p)^2}{5p} < p,$$

we have $P(p + a) = p$. This yields $P(a) = P(b) = P(p + a)$.

Case 2. $a \equiv 2 \pmod{5}$

Note that $(2p - a)^2 \equiv -1 \pmod{5}$ in this case. Thus, $(2p - a)^2 + 1$ equals $5pd$ for some $d \in \mathbb{N}$. As

$$d = \frac{(2p - a)^2 + 1}{5p} < \frac{(2p)^2}{5p} < p,$$

we have $P(2p - a) = p$. This yields $P(a) = P(b) = P(2p - a)$.

Case 3. $a \equiv 3, 4 \pmod{5}$

This time only the extra divisor 5 is not sufficient. We also need another small prime divisor, say 2. Note that $(2p+a)^2 \equiv (3p-a)^2 \equiv -1 \pmod{5}$ in this case. Also, one of $2p+a$ and $3p-a$ is odd. We choose the one which is odd, say ℓ . Thus, $\ell^2 + 1$ is even and it equals $10pd$ for some $d \in \mathbb{N}$. As

$$d = \frac{\ell^2 + 1}{10p} < \frac{(3p)^2}{10p} < p,$$

we have $P(\ell) = p$. This yields $P(a) = P(b) = P(\ell)$.

In any case, we can find distinct a, b, c with $P(a) = P(b) = P(c) = p$. Since there are infinitely many primes p of the form $20k + 17$, we can find infinitely many such triples. $\square$

Problem 8.4. Let n be a positive integer. Prove that there exist n positive integers such that the arithmetic mean of any k of them is a prime number where $k = 1, 2, \dots, n$.

Analysis. Arithmetic means are closely related to arithmetic sequences. Therefore, we should think of the Green-Tao theorem. Using this theorem, the problem is indeed trivial.

Solution. By the Green-Tao theorem, there exists an arithmetic sequence of primes $a + jd$ where $a \in \mathbb{Z}$, $d \in \mathbb{N}$ and $j = 1, 2, \dots, s$ with $s = (n+1)!$.

Define $n_j = a + n! \cdot jd$ for $1 \leq j \leq n$. Then for any k of them, say $n_{i_1}, \dots, n_{i_k}$, we have

$$\frac{n_{i_1} + \dots + n_{i_k}}{k} = a + \frac{n!}{k} \cdot (i_1 + \dots + i_k)d.$$

Note that $\frac{n!}{k}$ is a positive integer, and

$$\frac{n!}{k} \cdot (i_1 + \dots + i_k) \leq n! \cdot \max\{i_1, \dots, i_k\} < s.$$

Thus, the average of $n_{i_1}, \dots, n_{i_k}$ is of the form $a + jd$ with $1 \leq j \leq s$. Hence, it is a prime number from the construction. Since this holds for any k and any $i_1, \dots, i_k$, the proof is complete. $\square$

Problem 8.5. (IMO Shortlist 2019 N7) Prove that there is a constant $c > 0$ and infinitely many positive integers n with the following property: there are infinitely many positive integers that cannot be expressed as the sum of fewer than $cn \log(n)$ pairwise coprime n th powers.

Analysis. Suppose $m = a_1^n + a_2^n + \cdots + a_s^n$ where $a_1, a_2, \dots, a_s$ are relatively prime. We want to show that $s \geq cn \log n$ for infinitely many m . The relatively prime condition is the main condition which forces s to be large in some cases. The first idea is that, if a_j^n only belongs to a few residue classes modulo some prime p , then $a_1^n + a_2^n + \cdots + a_s^n$ cannot cover all residue classes modulo p when s is small. It follows that there is no solution for infinitely many m (in that residue class).

For example, we may take any prime p and take n to be a multiple of $p-1$, or just $n = p-1$ as a trial. Since at most one of the a_j 's is divisible by p using the relatively prime condition, by the Fermat little theorem, we must have $a_j^n \equiv 1 \pmod{p}$ for all but at most one j . If $m \equiv p-1 \pmod{p}$, then we need $s \geq p-1 = n$. This already gives us a certain bound for s , though it is slightly weaker than $cn \log n$. Therefore, probably we are on the right track, and the next idea is just to work on more than one prime.

If we want to apply a similar argument to the case of two distinct primes p and q , we want n to be a multiple of both $p-1$ and $q-1$. If we simply take $n = (p-1)(q-1)$, as s is still of the size pq , this does not really help much. However, we can indeed take $n = [p-1, q-1]$, which is much smaller than pq if $p-1$ and $q-1$ share some large common divisor.

Solution. We would like to choose distinct primes p and q such that $p, q \equiv 1 \pmod{d}$ and d is large. Indeed, we first fix a large d . By Dirichlet's theorem and Linnik's theorem, we can choose a prime p in the arithmetic progression $1 + kd$ such that $p \ll d^5$ (which means $p \leq cd^5$ for some constant c). Next, one of the arithmetic progressions $1 + k(2d)$ and $(1+d) + k(2d)$ does not contain p . We can choose another prime q in that arithmetic progression such that $q \ll (2d)^5 \ll d^5$. Then we must have $p \neq q$, $p, q \equiv 1 \pmod{d}$ and $p, q \ll d^5$.

Now, consider $n = \frac{(p-1)(q-1)}{d} \in \mathbb{N}$ and fix a constant S . Suppose

$$m = a_1^n + a_2^n + \cdots + a_s^n \quad (1)$$

where $a_1, a_2, \dots, a_s$ are relatively prime and $s \leq S$. In particular, at most one of a_j 's is a multiple of p , and at most one is a multiple of q . By the Fermat little theorem, since $p-1, q-1 \mid n$, we have $a_i^n \equiv 1 \pmod{p}$ and $a_j^n \equiv 1 \pmod{q}$ for all but at most one i and one j . This implies

$$\begin{aligned} m &\equiv s-1, s \pmod{p}, \\ m &\equiv s-1, s \pmod{q}. \end{aligned}$$

By the Chinese remainder theorem, m only belongs to 4 different residue classes modulo pq for each fixed s . Therefore, in order that (1) is solvable for m in all residue classes modulo pq (with different s), we must have $S \geq \frac{pq}{4}$. It remains to show that this is much larger than $cn \log n$.

Indeed, from $n = \frac{(p-1)(q-1)}{d}$ and $p, q \ll d^5$, we have

$$n \log n \ll \frac{pq}{d} \log \frac{pq}{d} \ll \frac{pq}{d} \log d^9 \ll pq \frac{\log d}{d} \ll pq.$$

This implies there exists a constant c such that $cn \log n \leq \frac{pq}{4} \leq S$. From above, there are infinitely many m such that (1) is not solvable for every $s < cn \log n$. This completes the proof. $\square$

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- [Bounded gaps of prime numbers](#)
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- [Cauchy-Davenport theorem](#)
- [Chinese remainder theorem](#)
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- Sum of 2 squares theorem
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Terminologies and Notations

- $\mathbb{N}$
- $\mathbb{N}_0$
- $\mathbb{Z}$
- $a \nmid b$
- $d(n)$
- $\tau(n)$
- $\sigma(n)$
- $a \not\equiv b \pmod{n}$
- $a \parallel b$
- common divisor
- common multiple
- composite number
- congruent $a \equiv b \pmod{n}$
- coprime
- Diophantine equation
- divide $a \mid b$
- divisible
- divisor
- divisor function $\sigma_\alpha(n)$
- Euler function $\varphi(n)$
- Euler's totient function $\varphi(n)$
- Goldbach conjecture
- greatest common divisor $\gcd(a, b)$, (a, b)
- infinite descent method
- Landau's problems
- least common multiple $\text{lcm}(a, b)$, $[a, b]$
- Legendre symbol $\left(\frac{a}{p}\right)$
- Legendre's conjecture
- Linnik's constant

- multiple
- multiplicative function
- order
- p -adic valuation $v_p(n)$
- prime number
- primitive root
- quadratic nonresidue
- quadratic residue
- quotient
- relatively prime
- remainder
- squarefree
- twin prime conjecture
- Vieta jumping

References

- Lifting the Exponent Lemma (LTE) (by Amir Hossein Parvardi)
(Available from: <http://www.artofproblemsolving.com/community/c6h393335>)

「要漂亮人生再變繽紛，要努力才能有運。」

《醉凡塵》